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# ==== run.py ====
# -*- coding: utf-8 -*-
"""系统启动入口。
运行：
    python run.py
然后在浏览器打开 http://127.0.0.1:5000
"""
from app import create_app
from app.models import db
import sys
if hasattr(sys.stdout, "reconfigure"):
    sys.stdout.reconfigure(encoding="utf-8")    # 控制台中文正常显示
app = create_app()
if __name__ == "__main__":
    with app.app_context():
        db.create_all()    # 首次运行时自动建表
        from app.utils.schema import ensure_schema
        added = ensure_schema(db)    # 旧库补齐新增字段
        if added:
            print("已补充字段：%s" % "、".join(added))
    print("=" * 58)
    print("  历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1.0")
    print("=" * 58)
    print("  访问地址：http://127.0.0.1:5000")
    print("  演示账号：admin    密码：admin123")
    print("  停止运行：按 Ctrl+C")
    print("=" * 58)
    app.run(host="127.0.0.1", port=5000, debug=True)
# ==== app/__init__.py ====
# -*- coding: utf-8 -*-
"""Flask 应用工厂。"""
from flask import Flask
from flask_login import LoginManager
from app.config import Config
from app.models import db, User
login_manager = LoginManager()
login_manager.login_view = "auth.login"
login_manager.login_message = "请先登录"
@login_manager.user_loader
def load_user(user_id):
    return db.session.get(User, int(user_id))
def create_app(config_class=Config):
    app = Flask(__name__)
    app.config.from_object(config_class)
    db.init_app(app)
    login_manager.init_app(app)
    # 注册视图（各模块路由）
    from app.views import auth, project, unit, standard,
    constraint, \
        ahp, indicator, evaluate, report
```

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app.register_blueprint(auth.bp)
app.register_blueprint(project.bp)
app.register_blueprint(unit.bp)
app.register_blueprint(standard.bp)
app.register_blueprint(constraint.bp)
app.register_blueprint(ahp.bp)
app.register_blueprint(indicator.bp)
app.register_blueprint(evaluate.bp)
app.register_blueprint(report.bp)
#  全局登录保护：除登录页与静态资源外，均需登录
from flask import request, redirect, url_for
from flask_login import current_user, login_user
PUBLIC_ENDPOINTS = {"auth.login", "static"}
@app.before_request
def require_login():
    #  截图模式（仅用于生成说明书配图）：自动以管理员身份登录
    if app.config.get("SCREENSHOT_MODE"):
        if (request.endpoint not in ("auth.login",
"static")
                and not current_user.is_authenticated):
            admin = User.query.filter_by(role="admin").f
irst()
            if admin:
                login_user(admin)
        return None
    if request.endpoint in PUBLIC_ENDPOINTS or
request.endpoint is None:
        return None
    if not current_user.is_authenticated:
        return redirect(url_for("auth.login"))
    return None
#  侧边栏导航：按论文评价流程的顺序编排，分为「评价流程」与「系统管理」两组。
#  不使用模块编号，避免出现与实际流程不一致的代号。
@app.context_processor
def inject_nav():
    path = request.path or "/"
    flow = [
        ("项目与矿山档案", "/project/", path.startswith("/proje
ct")),
        ("政策与标准知识库", "/standard/",
path.startswith("/standard")),
        ("限制性指标判定", "/constraint/",
path.startswith("/constraint")),
        ("微地质单元", "/unit/", path.startswith("/unit")),
        ("指标体系与权重", "/ahp/", path.startswith("/ahp")),
        ("指标数据采集", "/indicator/",
path.startswith("/indicator")),
        ("适宜性定量评价", "/evaluate/",
path.startswith("/evaluate")),
        ("报告与结果输出", "/report/", path.startswith("/report
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        ")),
            ]
            admin = [
                ("用户与操作日志", "/users", path.startswith(("/users",
        "/logs"))),
            ]
            groups = [
                {"title": "评价流程", "links": flow},
                {"title": "系统管理", "links": admin},
            ]
            return {"nav_groups": groups, "home_active": path
        == "/"}
        # 操作审计：统一在响应阶段记录成功的写操作（M1 操作日志）
        from app.services import audit
        @app.after_request
        def audit_writes(response):
            if request.method in ("POST", "PUT", "PATCH",
        "DELETE"):
                audit.record(request.endpoint,
        response.status_code)
            return response
        # 错误页：给出可读提示，而不是默认的英文错误页
        from flask import render_template
        @app.errorhandler(403)
        def forbidden(e):
            return render_template("error.html", code=403,
                                   message="没有权限执行该操作"), 403
        @app.errorhandler(404)
        def not_found(e):
            return render_template("error.html", code=404,
                                   message="请求的页面不存在"), 404
        return app
# ==== app/config.py ====
# -*- coding: utf-8 -*-
"""应用配置。"""
import os
from datetime import timedelta
BASE_DIR = os.path.abspath(os.path.dirname(__file__))
class Config:
        SECRET_KEY = os.environ.get("SECRET_KEY",
        "mine-restoration-dev-key")
        # 单机部署，SQLite 数据库文件放在应用目录下
        SQLALCHEMY_DATABASE_URI = "sqlite:///" +
        os.path.join(BASE_DIR, "app.db")
        SQLALCHEMY_TRACK_MODIFICATIONS = False
        # 禁止跨站请求伪造（CSRF），登录与表单提交依赖此项
        WTF_CSRF_ENABLED = True
        # 「记住我」：登录时勾选则下发持久 cookie，关闭浏览器后仍可自动登录
        REMEMBER_COOKIE_DURATION = timedelta(days=30)
        REMEMBER_COOKIE_HTTPONLY = True
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    REMEMBER_COOKIE_SAMESITE = "Lax"
    # 影像与数据文件上传（M2 遥感影像挂接、M7 批量导入）
    UPLOAD_FOLDER = os.path.join(BASE_DIR, "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024          # 单次上传上限
    32 MB
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
    ALLOWED_DATA_EXT = {"csv", "xlsx", "xls"}
# ==== app/models.py ====
# -*- coding: utf-8 -*-
"""数据模型：系统全部数据表。
本文件由甲在阶段 0 一次性写完，作为两人独立开发的地基。
约定：两人只读此文件；确需改表时，由乙提需求、甲统一修改。
"""
import json
from datetime import datetime
from flask_login import UserMixin
from flask_sqlalchemy import SQLAlchemy
db = SQLAlchemy()
class User(UserMixin, db.Model):
    """用户（M1 用户与权限管理）。"""
    __tablename__ = "users"
    id = db.Column(db.Integer, primary_key=True)
    username = db.Column(db.String(64), unique=True,
    nullable=False, index=True)
    password_hash = db.Column(db.String(256),
    nullable=False)
    # 角色：管理员 admin / 评价员 evaluator / 查看者 viewer
    role = db.Column(db.String(20), nullable=False,
    default="viewer")
    active = db.Column(db.Boolean, nullable=False,
    default=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    def to_dict(self):
        return {"id": self.id, "username": self.username,
                "role": self.role, "active": self.active}
class OperationLog(db.Model):
    """操作日志（M1）。"""
    __tablename__ = "operation_logs"
    id = db.Column(db.Integer, primary_key=True)
    user_id = db.Column(db.Integer, db.ForeignKey("users.id"
    ), nullable=True)
    action = db.Column(db.String(128), nullable=False)
    # 操作类型
    target_type = db.Column(db.String(64), nullable=True)
    # 对象类型
    target_id = db.Column(db.Integer, nullable=True)
    # 对象主键
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    detail = db.Column(db.Text, nullable=True)
    # 操作详情
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow, index=True)
class Project(db.Model):
    """评价项目（M2 项目与矿山档案）。"""
    __tablename__ = "projects"
    id = db.Column(db.Integer, primary_key=True)
    name = db.Column(db.String(128), nullable=False)
    code = db.Column(db.String(64), nullable=True)
    # 项目编号
    description = db.Column(db.Text, nullable=True)
    status = db.Column(db.String(20), default="进行中")
    # 进行中 / 已完成 / 已归档
    created_by = db.Column(db.Integer,
    db.ForeignKey("users.id"), nullable=True)
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
    patches = db.relationship("MinePatch",
    backref="project", lazy="dynamic",
                              cascade="all, delete-orphan")
class MinePatch(db.Model):
    """矿山图斑（M2）。"""
    __tablename__ = "mine_patches"
    id = db.Column(db.Integer, primary_key=True)
    project_id = db.Column(db.Integer,
    db.ForeignKey("projects.id"), nullable=False)
    name = db.Column(db.String(128), nullable=False)
    admin_region = db.Column(db.String(128), nullable=True)
    # 所在行政区
    center_x = db.Column(db.Float, nullable=True)
    # 中心经度 / 横坐标
    center_y = db.Column(db.Float, nullable=True)
    # 中心纬度 / 纵坐标
    area_ha = db.Column(db.Float, nullable=True)
    # 图斑面积（公顷）
    mineral_type = db.Column(db.String(64), nullable=True)
    # 开采矿种
    mining_method = db.Column(db.String(64), nullable=True)
    # 开采方式
    stop_years = db.Column(db.Float, nullable=True)
    # 停产年限
    responsible_entity_missing = db.Column(db.Boolean,
    default=True)  # 责任主体是否缺失
    # 遥感底图 / 无人机正射影像挂接（M2）
    image_path = db.Column(db.String(256), nullable=True)
    # 相对 static 的路径
    image_note = db.Column(db.String(256), nullable=True)
    # 影像说明（如来源、时相）
    image_uploaded_at = db.Column(db.DateTime,
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    nullable=True)                                                    251
    created_at = db.Column(db.DateTime,                               252
    default=datetime.utcnow)                                          253
    units = db.relationship("MicroUnit", backref="patch",             254
    lazy="dynamic",                                                   255
                                    cascade="all, delete-orphan")     256
class ReferenceEcosystem(db.Model):                                   257
    """参照生态系统（M2）：标准化基准值，与图斑绑定。"""                  258
    __tablename__ = "reference_ecosystems"                            259
    id = db.Column(db.Integer, primary_key=True)                      260
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat          261
    ches.id"), nullable=False)                                        262
    coverage_benchmark = db.Column(db.Float, default=75.0)            263
    # 植被覆盖度基准 %                                                264
    shannon_benchmark = db.Column(db.Float, default=0.7)              265
    # Shannon-Wiener 基准                                             266
    dominant_species = db.Column(db.String(256),                      267
    nullable=True)  # 建群种                                         268
    soil_benchmark = db.Column(db.String(256),                        269
    nullable=True)     # 土壤基准                                     270
class Standard(db.Model):                                             271
    """地方标准 / 政策文件（M3 政策与标准知识库）。"""                    272
    __tablename__ = "standards"                                       273
    id = db.Column(db.Integer, primary_key=True)                      274
    province = db.Column(db.String(32), nullable=False)               275
    # 省份                                                            276
    name = db.Column(db.String(256), nullable=False)                  277
    # 名称                                                            278
    doc_number = db.Column(db.String(128), nullable=True)             279
    # 标准编号 / 文号                                                 280
    publish_date = db.Column(db.String(32), nullable=True)            281
    implement_date = db.Column(db.String(32), nullable=True)          282
    # 类别：地方标准 / 政策文件                                       283
    category = db.Column(db.String(32), nullable=False,               284
    default="地方标准")                                               285
    summary = db.Column(db.Text, nullable=True)                       286
    # 特点概述                                                        287
    rules = db.relationship("StandardRule",                           288
    backref="standard",                                               289
                                    cascade="all, delete-orphan")     290
class StandardRule(db.Model):                                         291
    """结构化规则（M3）：将标准条文拆解为可执行规则。"""                  292
    __tablename__ = "standard_rules"                                  293
    id = db.Column(db.Integer, primary_key=True)                      294
    standard_id = db.Column(db.Integer,                               295
    db.ForeignKey("standards.id"), nullable=False)                    296
    item = db.Column(db.String(128), nullable=False)                  297
    # 判定项                                                          298
    operator = db.Column(db.String(8), nullable=False)                299
    # 比较符：>  >=  <  <=  ==                                        300
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    threshold = db.Column(db.String(64), nullable=True)
    # 阈值
    condition_type = db.Column(db.String(32),
    default="必要条件")  # 必要条件 / 充分条件
    region = db.Column(db.String(64), nullable=True)
    # 适用分区（宁夏分区差异化）
    note = db.Column(db.Text, nullable=True)
class ConstraintCheck(db.Model):
    """限制性判定记录（M4）。"""
    __tablename__ = "constraint_checks"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    policy_pass = db.Column(db.Boolean, nullable=True)
    # 政策约束
    location_pass = db.Column(db.Boolean, nullable=True)
    # 区位约束
    pollution_pass = db.Column(db.Boolean, nullable=True)
    # 土壤污染
    geo_pass = db.Column(db.Boolean, nullable=True)
    # 地质灾害
    farmland_pass = db.Column(db.Boolean, nullable=True)
    # 耕地属性
    passed = db.Column(db.Boolean, nullable=True)
    # 总体是否通过
    reason = db.Column(db.Text, nullable=True)
    # 不通过原因
    checked_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class MicroUnit(db.Model):
    """微地质单元（M5）：岩质边坡 / 碎石边坡 / 平台。"""
    __tablename__ = "micro_units"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    unit_type = db.Column(db.String(32), nullable=False)
    # 单元类型
    area = db.Column(db.Float, nullable=True)
    # 面积
    ratio = db.Column(db.Float, nullable=True)
    # 面积占比（自动计算）
    description = db.Column(db.Text, nullable=True)
    # 现场描述
class Indicator(db.Model):
    """指标定义（M6）：指标树，含角色与启用单元类型。"""
    __tablename__ = "indicators"
    id = db.Column(db.Integer, primary_key=True)
    code = db.Column(db.String(32), unique=True,
    nullable=False)  # 指标编码
    name = db.Column(db.String(64), nullable=False)
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    # 指标名称
    level = db.Column(db.Integer, nullable=False)
    # 1/2/3 级
    parent_code = db.Column(db.String(32), nullable=True)
    # 父指标编码
    # 角色：土壤污染指标同时作为限制性指标与评价指标
    is_constraint = db.Column(db.Boolean, default=False)
    is_evaluative = db.Column(db.Boolean, default=True)
    unit = db.Column(db.String(32), nullable=True)
    # 单位
    ideal_value = db.Column(db.String(64), nullable=True)
    # 理想值（可配置）
    ideal_type = db.Column(db.String(32), nullable=True)
    # benefit/cost/mid/discrete
    # 启用该指标的微地质单元（逗号分隔）。岩质边坡不启用土壤指标、平台不启用地形指标
    enabled_units = db.Column(db.String(128), nullable=True)
    description = db.Column(db.Text, nullable=True)
    # 取值依据说明
    def units_list(self):
        return [s for s in (self.enabled_units or
"").split(",") if s]
class WeightScheme(db.Model):
    """权重方案（M6）：按微地质单元类型分别保存。"""
    __tablename__ = "weight_schemes"
    id = db.Column(db.Integer, primary_key=True)
    unit_type = db.Column(db.String(32), nullable=False)
    # 所属单元类型
    name = db.Column(db.String(64), nullable=True)
    # 方案名称
    level = db.Column(db.Integer, nullable=False,
    default=1)          # 1=一级指标 2=二级指标
    is_default = db.Column(db.Boolean, default=False)
    # 是否默认（论文表 3）
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class WeightItem(db.Model):
    """权重明细（M6）：方案内的各项权重。"""
    __tablename__ = "weight_items"
    id = db.Column(db.Integer, primary_key=True)
    scheme_id = db.Column(db.Integer,
    db.ForeignKey("weight_schemes.id"), nullable=False)
    indicator_code = db.Column(db.String(32),
    nullable=False)        # 指标编码
    weight = db.Column(db.Float, nullable=False)
    # 权重值
class ExpertMatrix(db.Model):
    """专家判断矩阵（M6）：多专家打分的原始矩阵。"""
    __tablename__ = "expert_matrices"
    id = db.Column(db.Integer, primary_key=True)
    scheme_id = db.Column(db.Integer,
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    db.ForeignKey("weight_schemes.id"), nullable=True)
    # 直接标注适用对象，便于无需先建权重方案即可录入专家矩阵
    unit_type = db.Column(db.String(32), nullable=True)
    # 微地质单元类型
    l1_name = db.Column(db.String(64), nullable=True)
    # 所属一级指标（二级矩阵用）
    expert_name = db.Column(db.String(64), nullable=True)
    # 专家姓名
    level = db.Column(db.Integer, nullable=False,
    default=1)          # 矩阵对应指标层级
    matrix_json = db.Column(db.Text, nullable=False)
    # 判断矩阵（JSON 二维数组）
    cr = db.Column(db.Float, nullable=True)
    # 该专家矩阵的一致性比例
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class IndicatorData(db.Model):
    """指标实测数据（M7）：按微地质单元组织。"""
    __tablename__ = "indicator_data"
    id = db.Column(db.Integer, primary_key=True)
    unit_id = db.Column(db.Integer, db.ForeignKey("micro_uni
    ts.id"), nullable=False, unique=True)
    soil_ratio = db.Column(db.Float, nullable=True)
    # 土石比
    thickness = db.Column(db.Float, nullable=True)
    # 土层厚度 cm
    ph = db.Column(db.Float, nullable=True)
    organic_matter = db.Column(db.Float, nullable=True)
    # 有机质 g/kg
    # 重金属含量，JSON 字符串：{"Hg": 0.05, ...}，未检测项缺失或为 null
    heavy_metals = db.Column(db.Text, nullable=True)
    coverage = db.Column(db.Float, nullable=True)
    # 植被覆盖度 %
    shannon = db.Column(db.Float, nullable=True)
    # Shannon-Wiener 多样性指数
    slope = db.Column(db.Float, nullable=True)
    # 坡度 °
    aspect = db.Column(db.String(16), nullable=True)
    # 坡向：阳坡/阴阳坡/阴坡
    def heavy_metals_dict(self):
        return json.loads(self.heavy_metals) if
    self.heavy_metals else {}
class EvaluationResult(db.Model):
    """评价结果（M8）。"""
    __tablename__ = "evaluation_results"
    id = db.Column(db.Integer, primary_key=True)
    patch_id = db.Column(db.Integer, db.ForeignKey("mine_pat
    ches.id"), nullable=False)
    final_score = db.Column(db.Float, nullable=True)
    # 最终评价因子 F
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    grade = db.Column(db.String(32), nullable=True)
    # 适宜性等级
    measures = db.Column(db.Text, nullable=True)
    # 推荐措施
    unit_scores = db.Column(db.Text, nullable=True)
    # 各单元得分 JSON
    created_at = db.Column(db.DateTime,
    default=datetime.utcnow)
class ScoreDetail(db.Model):
    """得分明细（M8）：每一层指标的权重、标准化值、加权贡献，供追溯与复核。"""
    __tablename__ = "score_details"
    id = db.Column(db.Integer, primary_key=True)
    result_id = db.Column(db.Integer,
    db.ForeignKey("evaluation_results.id"), nullable=False)
    unit_id = db.Column(db.Integer, nullable=True)
    l1_name = db.Column(db.String(64), nullable=True)
    # 一级指标
    l2_name = db.Column(db.String(64), nullable=True)
    # 二级指标
    raw_value = db.Column(db.String(64), nullable=True)
    # 实测值
    std_value = db.Column(db.Float, nullable=True)
    # 标准化值
    weight = db.Column(db.Float, nullable=True)
    # 权重
    contribution = db.Column(db.Float, nullable=True)
    # 加权贡献
# ==== app/forms.py ====
# -*- coding: utf-8 -*-
"""表单定义（共享，由甲统一维护）。
各模块的 Flask-WTF 表单集中于此。
"""
from flask_wtf import FlaskForm
from wtforms import SelectField, SubmitField, StringField,
    PasswordField, BooleanField
from wtforms.validators import DataRequired, Length
from app.utils.constants import UNIT_TYPES, ASPECT_TYPES
class AHPSelectForm(FlaskForm):
    """M6：选择微地质单元类型与指标层级，以确定判断矩阵规模。"""
    unit_type = SelectField("微地质单元类型",
    validators=[DataRequired()])
    level = SelectField("指标层级",
                        choices=[("1", "一级指标"), ("2",
    "二级指标")],
                        validators=[DataRequired()])
    l1 = SelectField("一级指标", validators=[],
    validate_choice=False)  # 动态填充，层级=2 时使用
    submit = SubmitField("生成判断矩阵")
    def __init__(self, *args, **kwargs):
        super().__init__(*args, **kwargs)
```

```
            self.unit_type.choices = [(t, t) for t in
        UNIT_TYPES]
class EmptyForm(FlaskForm):
    """仅用于承载 CSRF 令牌的空表单（矩阵录入等自定义表单使用）。"""
    pass
class LoginForm(FlaskForm):
    """M1：登录表单（占位，由乙完善）。"""
    username = StringField("用户名", validators=[DataRequired()
    , Length(max=64)])
    password = PasswordField("密码", validators=[DataRequired(
    )])
    remember = BooleanField("记住我")
    submit = SubmitField("登录")
#  ==== app/utils/constants.py ====
#  -*- coding: utf-8 -*-
"""评价指标常量表。
全部数值取自论文，作为系统计算的唯一基准：
- 论文 4.2.2 节：指标标准化理想极值与取值依据
- 论文 4.2.3 节 / 表 4：适宜性等级划分
- 论文表 3：按微地质单元分别确定的一级、二级指标权重
- 论文 4.2.1 节：平均随机一致性指标 RI
"""
#  ------------------------------------------------------------------
    ------------------
#  一、微地质单元类型
#  ------------------------------------------------------------------
    ------------------
#  论文 4.1 节："按照地形地貌划分，将其划分为平台、碎石边坡和岩质边坡
#  三种地质微单元类型"
UNIT_TYPES = ("岩质边坡", "碎石边坡", "平台")
#  每个单元类型启用的指标类别（论文表 3 隐含：岩质边坡无土壤指标，平台无地形指标）
#  供数据录入界面与评价引擎判断"该单元是否需要某类指标"
UNIT_ENABLED_INDICATORS = {
    "岩质边坡": ("植被指标", "地形指标"),                  #  不设土壤指标
    "碎石边坡": ("土壤指标", "植被指标", "地形指标"),
    "平台": ("土壤指标", "植被指标"),                      #  不设地形指标
}
#  ------------------------------------------------------------------
    ------------------
#  二、指标标准化理想极值（论文 4.2.2 节）
#  ------------------------------------------------------------------
    ------------------
IDEAL_SOIL_RATIO = 0.8          #  土石比理想值
IDEAL_THICKNESS = 20.0          #  土层厚度理想值 （cm）
IDEAL_PH_RANGE = (5.0, 9.0)     #  pH 理想区间（适中型指标）
IDEAL_ORGANIC = 30.0            #  有机质理想值
    （g/kg），全国第二次土壤普查养分等级划分标准二等最低值
IDEAL_COVERAGE = 75.0           #  植被覆盖度参照生态系统 （%）
IDEAL_SHANNON = 0.7             #  Shannon-Wiener 多样性指数参照生态系统
#  重金属极值 （mg/kg），《绿化种植土壤》CJ/T 340—2016 土壤重金属含量 IV 级最高值
```

```
# 论文原文：镉、汞、铅、铬、砷、镍、铜、锌分别为 2、2、530、400、45、220、600、800
HEAVY_METALS = ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu",
    "Zn")
HEAVY_METAL_LIMIT = {
    "Cd": 2.0, "Hg": 2.0, "Pb": 530.0, "Cr": 400.0,
    "As": 45.0, "Ni": 220.0, "Cu": 600.0, "Zn": 800.0,
}
# 污染指标下 8 种重金属"重要性相当，权重平均分配"（论文 4.2.1 节）
HEAVY_METAL_EQUAL_WEIGHT = 1.0 / 8.0
# 坡度分界，论文 4.2.2 节引《新一轮退耕还林还草总体方案》发改西部〔2014〕1772 号
SLOPE_BREAK = 25.0
# 坡向离散赋值，论文 4.2.2 节："阳坡、阴阳坡、阴坡分别赋值 0.2、0.6、1"
ASPECT_SCORES = {"阳坡": 0.2, "阴阳坡": 0.6, "阴坡": 1.0}
ASPECT_TYPES = ("阳坡", "阴阳坡", "阴坡")
# pH 适中型标准化的容差边界（区间外线性衰减，避免得分突变）
PH_TOL_LOW = 4.0
PH_TOL_HIGH = 10.0
# ---------------------------------------------------------------
    ----------------
# 三、适宜性等级划分（论文表 4）
# ---------------------------------------------------------------
    ----------------
GRADE_LEVELS = (
    {"grade": "不适宜", "min": None, "max": 0.3,
     "meaning": "存在限制性条件或区域生态本底较差，自然恢复影响因子严重制约自然恢复",
     "measure": "生态重建、人工辅助"},
    {"grade": "基本适宜", "min": 0.3, "max": 0.7,
     "meaning": "生态本底良好，核心自然恢复影响因子基本满足自然恢复需求",
     "measure": "自然恢复为主，可辅以封闭场地和少量补种、补植、飞播等"},
    {"grade": "适宜", "min": 0.7, "max": None,
     "meaning": "各自然恢复影响因子均能满足自然恢复需求，自然演替速度快",
     "measure": "自然恢复"},
)
def grade_of(score):
    """按论文表 4 判定适宜性等级。"""
    for g in GRADE_LEVELS:
        lo, hi = g["min"], g["max"]
        if lo is None and score < hi:
            return g
        if hi is None and score >= lo:
            return g
        if lo is not None and hi is not None and lo <=
    score < hi:
            return g
    # score 恰好等于 0.7 时落入"适宜"
    return GRADE_LEVELS[-1]
# ---------------------------------------------------------------
    ----------------
# 四、平均随机一致性指标 RI（论文 4.2.1 节给出 n=2,3,4,5，其余为常用扩展值）
# ---------------------------------------------------------------
```

```
    ------------------                                                  601
RI_TABLE = {1: 0.0, 2: 0.0, 3: 0.58, 4: 0.90, 5: 1.12, 6:              602
    1.24,                                                               603
                7: 1.32, 8: 1.41, 9: 1.45, 10: 1.49}                    604
#  一致性比例阈值（论文：当 CR<0.1 时，判断矩阵满足一致性要求）          605
CR_THRESHOLD = 0.1                                                      606
#  ------------------------------------------------------------------  607
    ------------------                                                  608
#  五、默认权重方案（论文表 3）                                        609
#  ------------------------------------------------------------------  610
    ------------------                                                  611
#  论文明确："划分岩质边坡、碎石土边坡、平台三类微地质单元开展独立评价，  612
#  分别计算评价指标权重"。权重由 10 位专家打分后经 AHP 方法确定。      613
#  下表为论文表 3 所载的归一化权重，作为系统默认方案。                 614
DEFAULT_WEIGHTS = {                                                     615
    "岩质边坡": {                                                       616
        "l1": {"植被指标": 0.45, "地形指标": 0.55},                     617
        "l2": {                                                         618
            "植被指标": {"植被覆盖度": 0.67, "生物多样性": 0.33},       619
            "地形指标": {"坡度": 0.64, "坡向": 0.36},                   620
        },                                                              621
    },                                                                  622
    "碎石边坡": {                                                       623
        "l1": {"土壤指标": 0.44, "植被指标": 0.31, "地形指标": 0.25},   624
        "l2": {                                                         625
            "土壤指标": {"土石比": 0.25, "土层厚度": 0.25, "pH": 0.17,  626
                         "有机质": 0.17, "污染指标": 0.17},             627
            "植被指标": {"植被覆盖度": 0.64, "生物多样性": 0.36},       628
            "地形指标": {"坡度": 0.67, "坡向": 0.33},                   629
        },                                                              630
    },                                                                  631
    "平台": {                                                           632
        "l1": {"土壤指标": 0.54, "植被指标": 0.46},                     633
        "l2": {                                                         634
            "土壤指标": {"土石比": 0.25, "土层厚度": 0.25, "pH": 0.17,  635
                         "有机质": 0.17, "污染指标": 0.17},             636
            "植被指标": {"植被覆盖度": 0.61, "生物多样性": 0.39},       637
        },                                                              638
    },                                                                  639
}                                                                       640
#  ------------------------------------------------------------------  641
    ------------------                                                  642
#  六、限制性指标（论文表 2 + 4.1 节流程）                             643
#  ------------------------------------------------------------------  644
    ------------------                                                  645
#  论文摘要与 4.1 节：四项限制性指标；流程段另明确"是否为耕地"为约束条件  646
CONSTRAINT_ITEMS = ("政策", "区位约束", "土壤污染", "地质灾害", "耕地属性")  647
#  ==== app/services/ahp_service.py ====                               648
#  -*- coding: utf-8 -*-                                               649
"""AHP 权重引擎（M6）。                                                650
```

```
严格按论文 4.2.1 节实现：
- 式(2)(3)(4)：特征根法求权重（行元素乘积 → n 次方根 → 归一化）
- 式(5)(6)(7)：一致性检验（λmax、CI、CR）
- 论文要求 CR < 0.1；RI 按阶数查表
"""
import numpy as np
from app.utils.constants import RI_TABLE, CR_THRESHOLD
def matrix_from_upper(upper):
    """由判断矩阵上三角构造完整互反矩阵。
    参数
    ----
    upper : list[list[float]] 或 np.ndarray，上三角元素（不含对角线）。
        例：3 阶时传入 [[a12, a13], [a23]]。
        对角线为 1，下三角取倒数（a_ji = 1/a_ij）。
    """
    upper = np.asarray(upper, dtype=float)
    k = upper.shape[0]
    n = int(round((1 + (1 + 8 * k) ** 0.5) / 2))
    A = np.ones((n, n), dtype=float)
    idx = 0
    for i in range(n):
        for j in range(i + 1, n):
            A[i, j] = upper[idx]
            A[j, i] = 1.0 / upper[idx]
            idx += 1
    return A
def weights_and_consistency(A):
    """由判断矩阵 A 计算权重与一致性指标。
    返回
    ----
    dict：{weights: np.ndarray, lambda_max, ci, cr, n,
    consistent: bool}
    """
    A = np.asarray(A, dtype=float)
    n = A.shape[0]
    # 式(2) 行元素乘积
    row_prod = np.prod(A, axis=1)
    # 式(3) 开 n 次方根
    root = row_prod ** (1.0 / n)
    # 式(4) 归一化
    weights = root / root.sum()
    # 式(5) 最大特征值 λmax = (1/n) Σ (A·W)_i / W_i
    aw = A.dot(weights)
    lambda_max = float(np.sum(aw / weights) / n)
    # 式(6) 一致性指标 CI
    ci = (lambda_max - n) / (n - 1) if n > 1 else 0.0
    # 式(7) 一致性比例 CR = CI / RI
    ri = RI_TABLE.get(n, 1.49)
    cr = ci / ri if ri else 0.0
    return {
```

```
            "weights": weights,
            "lambda_max": lambda_max,
            "ci": ci,
            "cr": cr,
            "n": n,
            "consistent": cr < CR_THRESHOLD,
        }
def find_most_inconsistent(A):
    """定位判断矩阵中最不一致的一对比较。
    依据：完全一致矩阵应满足 a_ij = w_i / w_j。
    计算每个上三角元素与其理想值的相对偏差，返回偏差最大者。
    返回
    ----
    dict：{i, j, value, expected, deviation}，偏差最大的 (i, j)
    及对应取值。
    """
    A = np.asarray(A, dtype=float)
    r = weights_and_consistency(A)
    w = r["weights"]
    n = A.shape[0]
    best = None
    for i in range(n):
        for j in range(i + 1, n):
            expected = w[i] / w[j]
            if expected == 0:
                continue
            deviation = abs(A[i, j] - expected) / expected
            if best is None or deviation >
    best["deviation"]:
                best = {"i": i, "j": j, "value": A[i, j],
                        "expected": expected, "deviation":
    deviation}
    return best
def aggregate_matrices(matrices):
    """多专家判断矩阵聚合：几何平均（保持互反性）。
    参数
    ----
    matrices : list[np.ndarray]，多个同阶判断矩阵。
    """
    stack = np.stack([np.asarray(m, dtype=float) for m in
    matrices])
    return np.prod(stack, axis=0) ** (1.0 / len(stack))
def judge_from_weights(weights):
    """由权重向量构造完全一致判断矩阵并检验。
    用途：验证给定权重向量能否作为某个完全一致判断矩阵的主特征向量
    （A_ij = w_i / w_j 时 λmax = n、CR = 0）。
    """
    w = np.asarray(weights, dtype=float)
    n = w.shape[0]
    A = np.outer(w, 1.0 / w)
```

```
    r = weights_and_consistency(A)
    r["matrix"] = A
    return r
# ==== app/services/standardization.py ====
# -*- coding: utf-8 -*-
"""指标标准化（M8 的一部分）。
按论文 4.2.2 节式(8) 实现：
    x' = (x - x_min) / (x_max - x_min)
按指标性质分为四类：越大越优 / 越小越优 / 适中 / 离散赋值。
所有理想极值取自 app.utils.constants，可配置。
"""
from app.utils.constants import (
    IDEAL_SOIL_RATIO, IDEAL_THICKNESS, IDEAL_PH_RANGE,
    IDEAL_ORGANIC,
    IDEAL_COVERAGE, IDEAL_SHANNON, HEAVY_METAL_LIMIT,
    HEAVY_METAL_EQUAL_WEIGHT, SLOPE_BREAK, ASPECT_SCORES,
    PH_TOL_LOW, PH_TOL_HIGH,
)
def std_benefit(value, ideal):
    """越大越优指标标准化：以理想值为上限，按比例给分，饱和于 [0,1]。"""
    if value is None:
        return None
    return max(0.0, min(1.0, value / ideal))
def std_cost(value, extreme):
    """越小越优指标标准化：以极值为参考，线性折减至 [0,1]。"""
    if value is None:
        return None
    return max(0.0, min(1.0, 1.0 - value / extreme))
def std_mid_ph(value, lo, hi, tol_lo=PH_TOL_LOW,
    tol_hi=PH_TOL_HIGH):
    """适中型指标标准化（pH）：理想区间内取 1，区间外线性衰减。"""
    if value is None:
        return None
    if lo <= value <= hi:
        return 1.0
    if value > hi:
        return max(0.0, (tol_hi - value) / (tol_hi - hi))
    return max(0.0, (value - tol_lo) / (lo - tol_lo))
def std_slope(value, break_point=SLOPE_BREAK, upper=90.0):
    """坡度标准化。
    论文 4.2.2 节：以 25° 为陡坡重要分界。25° 及以下记满分 1，
    超过 25° 线性折减，90° 记 0。
    """
    if value is None:
        return None
    if value <= break_point:
        return 1.0
    return max(0.0, (upper - value) / (upper - break_point))
def aspect_score(aspect):
    """坡向离散赋值：阳坡 0.2 / 阴阳坡 0.6 / 阴坡 1.0（论文 4.2.2 节）。"""
```

```
    if aspect is None:                                                    801
        return None                                                       802
    return ASPECT_SCORES.get(aspect)                                      803
def pollution_score(heavy_metals):                                        804
    """污染指标：8 种重金属等权（各 0.125）合成。                          805
    论文注：未检测 Cr、Zn 的，从矿山类型看未受污染，按无污染（=1）处理。   806
    heavy_metals : dict，键为金属名，值为含量 mg/kg；缺失键视为未检测。     807
    """                                                                   808
    total = 0.0                                                           809
    for metal, extreme in HEAVY_METAL_LIMIT.items():                      810
        value = heavy_metals.get(metal)                                   811
        total += std_cost(value, extreme) if value is not                 812
    None else 1.0                                                         813
    return total / len(HEAVY_METAL_LIMIT)                                 814
def standardize_unit(unit_type, data, weights=None):                      815
    """对单个微地质单元的全部指标做标准化。                               816
    参数                                                                  817
    ----                                                                  818
    unit_type : str，岩质边坡 / 碎石边坡 / 平台                           819
    data : dict，实测数据，字段见接口契约：                               820
        soil_ratio, thickness, ph, organic_matter,                        821
    heavy_metals(dict),                                                   822
        coverage, shannon, slope, aspect                                  823
    weights : dict，该单元类型的权重（app.utils.constants.DEFAULT_WEIG    824
    HTS 结构），                                                          825
                未提供时用默认值。                                        826
    返回                                                                  827
    ----                                                                  828
    dict：{ 一级指标： { 二级指标： （实测值， 标准化值， 权重， 贡献） } }  829
    标准化值可能为 None，表示该指标不适用或方法待定。                     830
    """                                                                   831
    from app.utils.constants import DEFAULT_WEIGHTS                       832
    if weights is None:                                                   833
        weights = DEFAULT_WEIGHTS[unit_type]                              834
    l1_enabled = weights["l1"].keys()                                     835
    l2w = weights["l2"]                                                   836
    result = {}                                                           837
    # ---- 土壤指标 ----                                                  838
    if "土壤指标" in l1_enabled:                                          839
        result["土壤指标"] = {                                            840
            "土石比": (data.get("soil_ratio"),                            841
                       std_benefit(data.get("soil_ratio"),                842
    IDEAL_SOIL_RATIO),                                                    843
                       l2w["土壤指标"]["土石比"]),                        844
            "土层厚度": (data.get("thickness"),                           845
                         std_benefit(data.get("thickness"),               846
    IDEAL_THICKNESS),                                                     847
                         l2w["土壤指标"]["土层厚度"]),                    848
            "pH": (data.get("ph"),                                        849
                   std_mid_ph(data.get("ph"),                             850
```

```
    *IDEAL_PH_RANGE),                                                    851
                        l2w["土壤指标"]["pH"]),                          852
            "有机质": (data.get("organic_matter"),                       853
                       std_benefit(data.get("organic_matter")            854
    , IDEAL_ORGANIC),                                                    855
                       l2w["土壤指标"]["有机质"]),                       856
            "污染指标": (data.get("heavy_metals") or {},                 857
                         pollution_score(data.get("heavy_meta            858
    ls") or {}),                                                         859
                         l2w["土壤指标"]["污染指标"]),                   860
        }                                                                861
    # ---- 植被指标 ----                                                862
    if "植被指标" in l1_enabled:                                         863
        result["植被指标"] = {                                           864
            "植被覆盖度": (data.get("coverage"),                         865
                           std_benefit(data.get("coverage"),             866
    IDEAL_COVERAGE),                                                     867
                           l2w["植被指标"]["植被覆盖度"]),               868
            "生物多样性": (data.get("shannon"),                          869
                           std_benefit(data.get("shannon"),              870
    IDEAL_SHANNON),                                                      871
                           l2w["植被指标"]["生物多样性"]),               872
        }                                                                873
    # ---- 地形指标 ----                                                874
    if "地形指标" in l1_enabled:                                         875
        result["地形指标"] = {                                           876
            "坡度": (data.get("slope"),                                  877
                     std_slope(data.get("slope")),                       878
                     l2w["地形指标"]["坡度"]),                           879
            "坡向": (data.get("aspect"),                                 880
                     aspect_score(data.get("aspect")),                   881
                     l2w["地形指标"]["坡向"]),                           882
        }                                                                883
    return result                                                        884
# ==== app/services/evaluation_service.py ====                         885
# -*- coding: utf-8 -*-                                                886
"""适宜性评价引擎（M8）。                                              887
按论文式（9） 实现：                                                   888
- 单元级：F' = Σ（二级指标权重 × 该二级指标下三级指标标准化值）        889
- 矿山级：F = Σ（Si × F'i），Si 为各微地质单元面积占比                 890
- 等级判定按论文表 4。                                                 891
"""                                                                    892
from app.utils.constants import DEFAULT_WEIGHTS, grade_of              893
from app.services import standardization as std                        894
def score_unit(unit_type, data, weights=None):                         895
    """计算单个微地质单元的评价因子 F'。                               896
    返回                                                               897
    ----                                                               898
    dict：{ l1_scores: {一级指标: 得分}, unit_score: F', details:      899
    [...] }                                                            900
```

```
    若某一级指标下存在"不适用"（标准化值为 None）的二级指标，
    则按其余项权重归一化后计算，保证单元得分始终有定义。
    """
    if weights is None:
        weights = DEFAULT_WEIGHTS[unit_type]
    std_result = std.standardize_unit(unit_type, data,
    weights)
    l1_weights = weights["l1"]
    l1_scores = {}
    details = []
    for l1_name, items in std_result.items():
        wsum = 0.0
        acc = 0.0
        for l2_name, (raw, std_value, w2) in items.items():
            if std_value is None:
                # 不适用项：不计入
                details.append({
                    "l1": l1_name, "l2": l2_name, "raw":
    raw,
                    "std_value": None, "weight": w2,
    "contribution": None,
                })
                continue
            contribution = std_value * w2
            acc += contribution
            wsum += w2
            details.append({
                "l1": l1_name, "l2": l2_name, "raw": raw,
                "std_value": std_value, "weight": w2,
                "contribution": contribution,
            })
        l1_scores[l1_name] = acc / wsum if wsum > 0 else 0.0
    unit_score = sum(l1_weights[k] * l1_scores[k] for k in
    l1_scores)
    return {"l1_scores": l1_scores, "unit_score":
    unit_score, "details": details}
def score_mine(units):
    """计算整个矿山的评价因子 F 与适宜性等级。
    参数
    ----
    units : list[dict]，每个元素：
        {unit_type, ratio, data}
        ratio 为该单元面积占比（0~1），data 为实测数据 dict。
    返回
    ----
    dict:{ final_score, grade, grade_info, unit_results }
    """
    total = 0.0
    unit_results = []
    for u in units:
```

```
            r = score_unit(u["unit_type"], u["data"])
            weighted = u["ratio"] * r["unit_score"]
            total += weighted
            unit_results.append({
                "unit_type": u["unit_type"],
                "ratio": u["ratio"],
                "unit_score": r["unit_score"],
                "weighted": weighted,
                "l1_scores": r["l1_scores"],
                "details": r["details"],
            })
        grade_info = grade_of(total)
        return {
            "final_score": total,
            "grade": grade_info["grade"],
            "grade_info": grade_info,
            "unit_results": unit_results,
        }
# ==== app/services/report_service.py ====
# -*- coding: utf-8 -*-
"""报告生成（M9）：生成 Word 评价报告。
依赖 python-docx。报告结构：
标题 → 图斑信息 → 评价结论 → 各微地质单元得分 → 完整得分明细 → 生成时间。
"""
import io
from datetime import datetime
from docx import Document
from docx.shared import Pt, Cm
from docx.oxml.ns import qn
from docx.enum.text import WD_ALIGN_PARAGRAPH
def _set_run(run, text=None, size=12, bold=False):
    if text is not None:
        run.text = text
    run.font.name = "Times New Roman"
    run.font.size = Pt(size)
    run.font.bold = bold
    # 中文字体
    run._element.rPr.rFonts.set(qn("w:eastAsia"), "宋体")
    return run
def _para(doc, text, size=12, bold=False, align=None,
    space_after=6):
    p = doc.add_paragraph()
    if align is not None:
        p.alignment = align
    p.paragraph_format.space_after = Pt(space_after)
    _set_run(p.add_run(), text, size, bold)
    return p
def _heading(doc, text, level=1):
    h = doc.add_heading(text, level=level)
    for run in h.runs:
```

```
        run.font.name = "Times New Roman"                                1001
        run._element.rPr.rFonts.set(qn("w:eastAsia"), "黑体")            1002
    return h                                                             1003
def _table(doc, headers, rows, widths=None):                             1004
    t = doc.add_table(rows=1, cols=len(headers))                         1005
    t.style = "Table Grid"                                               1006
    hdr = t.rows[0].cells                                                1007
    for i, htext in enumerate(headers):                                  1008
        _set_run(hdr[i].paragraphs[0].add_run(), htext,                  1009
    10.5, bold=True)                                                     1010
    for row in rows:                                                     1011
        cells = t.add_row().cells                                        1012
        for i, val in enumerate(row):                                    1013
            _set_run(cells[i].paragraphs[0].add_run(),                   1014
    str(val), 10.5)                                                      1015
    return t                                                             1016
def build_report(patch, mine_result):                                    1017
    """生成评价报告 Word 文档，返回 BytesIO。                              1018
    参数                                                                 1019
    ----                                                                 1020
    patch : MinePatch，图斑对象                                          1021
    mine_result : evaluation_service.score_mine 的返回 dict              1022
    """                                                                  1023
    doc = Document()                                                     1024
    # 页边距                                                             1025
    for section in doc.sections:                                         1026
        section.top_margin = Cm(2.5)                                     1027
        section.bottom_margin = Cm(2.5)                                  1028
        section.left_margin = Cm(2.5)                                    1029
        section.right_margin = Cm(2.5)                                   1030
    _heading(doc, "历史遗留废弃矿山自然恢复适宜性评价报告", 0)           1031
    _para(doc, "软件：历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ1.0",     1032
          size=10.5, align=WD_ALIGN_PARAGRAPH.CENTER)                    1033
    _para(doc, "生成时间：%s" % datetime.now().strftime("%Y-%m-%d        1034
    %H：%M：%S"),                                                        1035
          size=10.5, align=WD_ALIGN_PARAGRAPH.CENTER,                    1036
    space_after=12)                                                      1037
    # 图斑信息                                                           1038
    _heading(doc, "一、图斑基本信息", 1)                                 1039
    info = [                                                             1040
        ("图斑名称", patch.name),                                        1041
        ("所在行政区", patch.admin_region or "—"),                       1042
        ("图斑面积（公顷）", str(patch.area_ha) if patch.area_ha is      1043
    not None else "—"),                                                  1044
        ("开采矿种", patch.mineral_type or "—"),                         1045
        ("开采方式", patch.mining_method or "—"),                        1046
    ]                                                                    1047
    _table(doc, ["项目", "内容"], info)                                  1048
    # 评价结论                                                           1049
    _heading(doc, "二、评价结论", 1)                                     1050
```

```
    gi = mine_result["grade_info"]                                    1051
    concl = [                                                         1052
        ("最终评价因子 F", "%.4f" % mine_result["final_score"]),      1053
        ("适宜性等级", mine_result["grade"]),                         1054
        ("判定依据", gi["meaning"]),                                  1055
        ("推荐措施", gi["measure"]),                                  1056
    ]                                                                 1057
    _table(doc, ["项目", "结果"], concl)                              1058
    # 各单元得分                                                      1059
    _heading(doc, "三、各微地质单元得分", 1)                          1060
    rows = []                                                         1061
    for u in mine_result["unit_results"]:                             1062
        rows.append((u["unit_type"], "%.2f%%" % (u["ratio"]           1063
* 100),                                                              1064
                     "%.4f" % u["unit_score"], "%.4f" %               1065
u["weighted"]))                                                       1066
    _table(doc, ["微地质单元", "面积占比", "单元评价因子 F'", "加权贡献"],  1067
rows)                                                                 1068
    # 完整得分明细                                                    1069
    _heading(doc, "四、得分明细", 1)                                  1070
    for u in mine_result["unit_results"]:                             1071
        _para(doc, "%s（占比 %.2f%%）" % (u["unit_type"],             1072
u["ratio"] * 100),                                                    1073
              size=11, bold=True, space_after=4)                      1074
        detail_rows = []                                              1075
        for d in u["details"]:                                        1076
            raw = "—" if d["raw"] is None else d["raw"]               1077
            std = "—" if d["std_value"] is None else "%.4f"           1078
% d["std_value"]                                                      1079
            w = "%.4f" % d["weight"]                                  1080
            c = "—" if d["contribution"] is None else                 1081
"%.4f" % d["contribution"]                                            1082
            detail_rows.append((d["l1"], d["l2"], raw, std,           1083
w, c))                                                                1084
        _table(doc, ["一级指标", "二级指标", "实测值", "标准化值", "权重",  1085
"贡献"],                                                              1086
               detail_rows)                                           1087
        # 一级指标得分                                                1088
        l1_row = []                                                   1089
        for k, v in u["l1_scores"].items():                           1090
            l1_row.append("%s=%.4f" % (k, v))                         1091
        _para(doc, "一级指标得分：" + ",".join(l1_row), size=10.5,    1092
space_after=10)                                                       1093
    buf = io.BytesIO()                                                1094
    doc.save(buf)                                                     1095
    buf.seek(0)                                                       1096
    return buf                                                        1097
# ==== app/views/auth.py ====                                         1098
# -*- coding: utf-8 -*-                                               1099
"""M1 用户与权限管理 —— 登录、注销、用户维护、操作日志。"""            1100
```

```
from flask import (Blueprint, render_template, redirect,
    url_for, request,
                   flash, abort)
from flask_login import login_user, logout_user,
    login_required, current_user
from werkzeug.security import check_password_hash,
    generate_password_hash
from app.forms import LoginForm, EmptyForm
from app.models import db, User, OperationLog
bp = Blueprint("auth", __name__)
ROLE_LABELS = {"admin": "管理员", "evaluator": "评价员",
    "viewer": "查看者"}
def require_admin():
    """非管理员直接 403，避免误操作基础数据。"""
    if current_user.role != "admin":
        abort(403)
@bp.route("/")
def index():
    if current_user.is_authenticated:
        return render_template("auth/home.html")
    return redirect(url_for("auth.login"))
@bp.route("/login", methods=["GET", "POST"])
def login():
    if current_user.is_authenticated:
        return redirect(url_for("auth.index"))
    form = LoginForm()
    if form.validate_on_submit():
        user = User.query.filter_by(username=form.username.d
    ata).first()
        if user and check_password_hash(user.password_hash,
    form.password.data):
            if not user.active:
                flash("该账号已停用，请联系管理员", "error")
                return render_template("auth/login.html",
    form=form)
            login_user(user, remember=form.remember.data)
            flash("登录成功", "ok")
            return redirect(url_for("auth.index"))
        flash("用户名或密码错误", "error")
    return render_template("auth/login.html", form=form)
@bp.route("/logout")
@login_required
def logout():
    logout_user()
    flash("已退出登录", "ok")
    return redirect(url_for("auth.login"))
# ---------------------------- 用户维护 ----------------------------
    ----
@bp.route("/users")
@login_required
```

```
def users():                                                    1151
    all_users = User.query.order_by(User.id).all()              1152
    return render_template("auth/users.html",                   1153
    users=all_users,                                            1154
                                role_labels=ROLE_LABELS,        1155
    form=EmptyForm())                                           1156
@bp.route("/users/add", methods=["GET", "POST"])                1157
@login_required                                                 1158
def add_user():                                                 1159
    require_admin()                                             1160
    if request.method == "POST":                                1161
        username = request.form.get("username", "").strip()     1162
        password = request.form.get("password", "")             1163
        role = request.form.get("role", "viewer")               1164
        if not username or not password:                        1165
            flash("用户名和密码不能为空", "error")                1166
            return redirect(url_for("auth.add_user"))           1167
        if User.query.filter_by(username=username).first():     1168
            flash("用户名已存在", "error")                        1169
            return redirect(url_for("auth.add_user"))           1170
        db.session.add(User(username=username, role=role,       1171
                            password_hash=generate_password_    1172
    hash(password)))                                            1173
        db.session.commit()                                     1174
        flash("用户已添加", "ok")                                 1175
        return redirect(url_for("auth.users"))                  1176
    return render_template("auth/add_user.html")                1177
@bp.route("/users/<int:user_id>/edit", methods=["GET",          1178
    "POST"])                                                    1179
@login_required                                                 1180
def edit_user(user_id):                                         1181
    require_admin()                                             1182
    user = db.get_or_404(User, user_id)                         1183
    if request.method == "POST":                                1184
        role = request.form.get("role", user.role)              1185
        active = request.form.get("active") == "1"              1186
        #  保护：不能把最后一个管理员降级或停用，否则无人能管理系统  1187
        if user.role == "admin" and (role != "admin" or not     1188
    active):                                                    1189
            others = User.query.filter(User.role == "admin",    1190
                                       User.id != user.id,      1191
                                                                1192
    User.active.is_(True)).count()                              1193
            if others == 0:                                     1194
                flash("系统至少需保留一名启用状态的管理员", "error")  1195
                return redirect(url_for("auth.edit_user",       1196
    user_id=user.id))                                           1197
        user.role = role                                        1198
        user.active = active                                    1199
        new_password = request.form.get("password",             1200
```

```
        "").strip()                                                    1201
            if new_password:                                           1202
                user.password_hash = generate_password_hash(new_       1203
        password)                                                      1204
            db.session.commit()                                        1205
            flash("用户信息已保存", "ok")                                1206
            return redirect(url_for("auth.users"))                     1207
        return render_template("auth/edit_user.html", user=user,       1208
                               role_labels=ROLE_LABELS,                1209
        form=EmptyForm())                                              1210
@bp.route("/users/<int:user_id>/delete", methods=["POST"])             1211
@login_required                                                        1212
def delete_user(user_id):                                              1213
        require_admin()                                                1214
        user = db.get_or_404(User, user_id)                            1215
        if user.id == current_user.id:                                 1216
            flash("不能删除当前登录的账号", "error")                      1217
            return redirect(url_for("auth.users"))                     1218
        if user.role == "admin":                                       1219
            others = User.query.filter(User.role == "admin",           1220
                                       User.id != user.id,             1221
                                       User.active.is_(True)).co       1222
        unt()                                                          1223
            if others == 0:                                            1224
                flash("系统至少需保留一名启用状态的管理员", "error")          1225
                return redirect(url_for("auth.users"))                 1226
        db.session.delete(user)                                        1227
        db.session.commit()                                            1228
        flash("用户「%s」已删除" % user.username, "ok")                   1229
        return redirect(url_for("auth.users"))                         1230
# ------------------------------ 操作日志 ------------------------------1231
        ----                                                           1232
@bp.route("/logs")                                                     1233
@login_required                                                        1234
def logs():                                                            1235
        page = request.args.get("page", 1, type=int)                   1236
        per_page = 50                                                  1237
        q = OperationLog.query.order_by(OperationLog.id.desc())        1238
        total = q.count()                                              1239
        records = q.offset((page - 1) * per_page).limit(per_page       1240
        ).all()                                                        1241
        users_map = {u.id: u.username for u in User.query.all()}       1242
        pages = max(1, (total + per_page - 1) // per_page)             1243
        return render_template("auth/logs.html",                       1244
        records=records, page=page,                                    1245
                               pages=pages, total=total,               1246
        users_map=users_map)                                           1247
# ==== app/views/project.py ====                                       1248
# -*- coding: utf-8 -*-                                                1249
"""M2 项目与矿山档案 —— 项目、图斑、参照生态系统、遥感影像挂接。"""          1250
```

```
from datetime import datetime                                    1251
from flask import (Blueprint, render_template, request,          1252
    redirect, url_for,                                           1253
                      flash, current_app)                        1254
from flask_login import login_required, current_user             1255
from app.forms import EmptyForm                                  1256
from app.models import db, Project, MinePatch,                   1257
    ReferenceEcosystem                                           1258
bp = Blueprint("project", __name__, url_prefix="/project")       1259
@bp.route("/")                                                   1260
@login_required                                                  1261
def index():                                                     1262
    projects = Project.query.order_by(Project.id.desc()).all1263
    ()                                                           1264
    return render_template("project/index.html",                 1265
    projects=projects,                                           1266
                                  form=EmptyForm())              1267
@bp.route("/add", methods=["GET", "POST"])                       1268
@login_required                                                  1269
def add():                                                       1270
    form = EmptyForm()                                           1271
    if request.method == "POST" and form.validate_on_submit(1272
    ):                                                           1273
        p = Project(                                             1274
            name=request.form.get("name", "").strip(),           1275
            code=request.form.get("code", "").strip() or         1276
    None,                                                        1277
            description=request.form.get("description",          1278
    "").strip() or None,                                         1279
            created_by=current_user.id,                          1280
        )                                                        1281
        if not p.name:                                           1282
            flash("项目名称不能为空", "error")                    1283
        else:                                                    1284
            db.session.add(p)                                    1285
            db.session.commit()                                  1286
            flash("项目已创建", "ok")                            1287
            return redirect(url_for("project.index"))            1288
    return render_template("project/edit.html", form=form,       1289
    project=None)                                                1290
@bp.route("/<int:pid>/edit", methods=["GET", "POST"])            1291
@login_required                                                  1292
def edit(pid):                                                   1293
    p = db.get_or_404(Project, pid)                              1294
    form = EmptyForm()                                           1295
    if request.method == "POST" and form.validate_on_submit(1296
    ):                                                           1297
        p.name = request.form.get("name", "").strip()            1298
        p.code = request.form.get("code", "").strip() or         1299
    None                                                         1300
```

```
            p.description = request.form.get("description",                1301
        "").strip() or None                                                1302
            p.status = request.form.get("status", "进行中")                 1303
            db.session.commit()                                            1304
            flash("项目已保存", "ok")                                        1305
            return redirect(url_for("project.index"))                      1306
        return render_template("project/edit.html", form=form,             1307
        project=p)                                                         1308
@bp.route("/<int:pid>/delete", methods=["POST"])                           1309
@login_required                                                            1310
def delete(pid):                                                           1311
        p = db.get_or_404(Project, pid)                                    1312
        db.session.delete(p)                                               1313
        db.session.commit()                                                1314
        flash("项目已删除", "ok")                                            1315
        return redirect(url_for("project.index"))                          1316
# ------------------------------ 矿山图斑 ----------------------------1317
        ----                                                               1318
@bp.route("/<int:pid>/patches")                                            1319
@login_required                                                            1320
def patches(pid):                                                          1321
        p = db.get_or_404(Project, pid)                                    1322
        return render_template("project/patches.html",                     1323
        project=p,                                                         1324
                                    form=EmptyForm())                      1325
@bp.route("/patch/<int:patch_id>/delete", methods=["POST"])                1326
@login_required                                                            1327
def patch_delete(patch_id):                                                1328
        """删除矿山图斑，及其微地质单元与指标数据。"""                                  1329
        from app.models import MicroUnit, IndicatorData,                   1330
        EvaluationResult, \                                                1331
            ScoreDetail, ConstraintCheck, ReferenceEcosystem               1332
        patch = db.get_or_404(MinePatch, patch_id)                         1333
        pid = patch.project_id                                             1334
        unit_ids = [u.id for u in MicroUnit.query.filter_by(patc           1335
        h_id=patch.id).all()]                                              1336
        if unit_ids:                                                       1337
            IndicatorData.query.filter(IndicatorData.unit_id.in_           1338
        (unit_ids)).delete(                                                1339
                synchronize_session=False)                                 1340
        for r in EvaluationResult.query.filter_by(patch_id=patch           1341
        .id).all():                                                        1342
            ScoreDetail.query.filter_by(result_id=r.id).delete(            1343
                synchronize_session=False)                                 1344
        EvaluationResult.query.filter_by(patch_id=patch.id).dele           1345
        te(                                                                1346
            synchronize_session=False)                                     1347
        ConstraintCheck.query.filter_by(patch_id=patch.id).delet           1348
        e(                                                                 1349
            synchronize_session=False)                                     1350
```

```
    MicroUnit.query.filter_by(patch_id=patch.id).delete(sync
    hronize_session=False)
    ReferenceEcosystem.query.filter_by(patch_id=patch.id).de
    lete(
        synchronize_session=False)
    db.session.delete(patch)
    db.session.commit()
    flash("图斑「%s」及其数据已删除" % patch.name, "ok")
    return redirect(url_for("project.patches", pid=pid))
@bp.route("/<int:pid>/patch/add", methods=["GET", "POST"])
@login_required
def patch_add(pid):
    p = db.get_or_404(Project, pid)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        def fv(name, cast=str):
            v = request.form.get(name, "").strip()
            return cast(v) if v != "" else None
        patch = MinePatch(
            project_id=p.id,
            name=request.form.get("name", "").strip(),
            admin_region=fv("admin_region"),
            area_ha=fv("area_ha", float),
            mineral_type=fv("mineral_type"),
            mining_method=fv("mining_method"),
            stop_years=fv("stop_years", float),
            responsible_entity_missing=bool(request.form.get
    ("responsible_entity_missing")),
        )
        if not patch.name:
            flash("图斑名称不能为空", "error")
        else:
            db.session.add(patch)
            db.session.flush()
            # 参照生态系统默认取论文值
            db.session.add(ReferenceEcosystem(
                patch_id=patch.id, coverage_benchmark=75.0,
    shannon_benchmark=0.7))
            db.session.commit()
            flash("图斑已创建", "ok")
            return redirect(url_for("project.patches",
    pid=p.id))
    return render_template("project/patch_edit.html",
    form=form, project=p, patch=None)
@bp.route("/patch/<int:patch_id>/edit", methods=["GET",
    "POST"])
@login_required
def patch_edit(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
```

```
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
        def fv(name, cast=str):
            v = request.form.get(name, "").strip()
            return cast(v) if v != "" else None
        patch.name = request.form.get("name", "").strip()
        patch.admin_region = fv("admin_region")
        patch.area_ha = fv("area_ha", float)
        patch.mineral_type = fv("mineral_type")
        patch.mining_method = fv("mining_method")
        patch.stop_years = fv("stop_years", float)
        patch.responsible_entity_missing =
    bool(request.form.get("responsible_entity_missing"))
        # 参照生态系统
        ref = ReferenceEcosystem.query.filter_by(patch_id=pa
    tch.id).first()
        if ref is None:
            ref = ReferenceEcosystem(patch_id=patch.id)
            db.session.add(ref)
        ref.coverage_benchmark = fv("coverage_benchmark",
    float) or 75.0
        ref.shannon_benchmark = fv("shannon_benchmark",
    float) or 0.7
        ref.dominant_species = fv("dominant_species")
        patch.image_note = fv("image_note")
        db.session.commit()
        flash("图斑已保存", "ok")
        return redirect(url_for("project.patches",
    pid=patch.project_id))
    return render_template("project/patch_edit.html",
    form=form,
                           project=patch.project,
    patch=patch,
                           reference=ReferenceEcosystem.quer
    y.filter_by(
                               patch_id=patch.id).first())
# ---------------------------- 遥感影像挂接
    ----------------------------
@bp.route("/patch/<int:patch_id>/image", methods=["POST"])
@login_required
def upload_image(patch_id):
    """上传遥感底图 / 无人机正射影像并挂接到图斑。"""
    from app.utils import uploads as up
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if not form.validate_on_submit():
        flash("表单校验失败，请重试", "error")
        return redirect(url_for("project.patch_edit",
    patch_id=patch.id))
```

```
        rel, err = up.save_upload(request.files.get("image"),
        "patches",
                                  current_app.config["ALLOWED_IM
        AGE_EXT"],
                                  current_app.config["UPLOAD_FOL
        DER"])
        if err:
            flash(err, "error")
            return redirect(url_for("project.patch_edit",
        patch_id=patch.id))
        # 替换旧影像时清理原文件，避免残留
        if patch.image_path:
            up.remove_upload(patch.image_path,
        current_app.config["UPLOAD_FOLDER"])
        patch.image_path = rel
        patch.image_uploaded_at = datetime.utcnow()
        note = request.form.get("image_note", "").strip()
        if note:
            patch.image_note = note
        db.session.commit()
        flash("遥感影像已挂接", "ok")
        return redirect(url_for("project.patch_edit",
        patch_id=patch.id))
@bp.route("/patch/<int:patch_id>/image/delete",
    methods=["POST"])
@login_required
def delete_image(patch_id):
        from app.utils import uploads as up
        patch = db.get_or_404(MinePatch, patch_id)
        form = EmptyForm()
        if form.validate_on_submit():
            up.remove_upload(patch.image_path,
        current_app.config["UPLOAD_FOLDER"])
            patch.image_path = None
            patch.image_uploaded_at = None
            db.session.commit()
            flash("遥感影像已移除", "ok")
        return redirect(url_for("project.patch_edit",
        patch_id=patch.id))
# ==== app/views/unit.py ====
# -*- coding: utf-8 -*-
"""M5 微地质单元管理 —— 岩质边坡 / 碎石边坡 / 平台的划分与面积占比。"""
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from flask_login import login_required
from app.forms import EmptyForm
from app.models import db, MinePatch, MicroUnit,
    IndicatorData
from app.utils.constants import UNIT_TYPES
bp = Blueprint("unit", __name__, url_prefix="/unit")
```

```
def recalc_ratio(patch_id):
    """按面积重算该图斑各单元面积占比。"""
    units = MicroUnit.query.filter_by(patch_id=patch_id).all()
    total = sum(u.area or 0.0 for u in units)
    if total > 0:
        for u in units:
            u.ratio = round((u.area or 0.0) / total, 6)
    return total
@bp.route("/")
@login_required
def list_patches():
    """微地质单元入口：列出各图斑及其单元划分情况。"""
    patches = MinePatch.query.order_by(MinePatch.id.desc()).all()
    rows = []
    for p in patches:
        units = MicroUnit.query.filter_by(patch_id=p.id).all()
        rows.append({
            "patch": p,
            "count": len(units),
            "ratio_sum": sum(u.ratio or 0.0 for u in units),
        })
    return render_template("unit/list.html", rows=rows)
@bp.route("/<int:patch_id>")
@login_required
def index(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    units = MicroUnit.query.filter_by(patch_id=patch_id).all()
    total = sum(u.area or 0.0 for u in units)
    ratio_sum = sum(u.ratio or 0.0 for u in units)
    return render_template("unit/index.html", patch=patch,
    units=units,
                           total_area=total,
    ratio_sum=ratio_sum,
                           unit_types=UNIT_TYPES,
    form=EmptyForm())
@bp.route("/<int:patch_id>/add", methods=["POST"])
@login_required
def add(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    if form.validate_on_submit():
        unit_type = request.form.get("unit_type")
        area = request.form.get("area", "").strip()
        u = MicroUnit(patch_id=patch.id,
    unit_type=unit_type,
                      area=float(area) if area else None,
```

```
                                description=request.form.get("descript
    ion", "").strip() or None)
            db.session.add(u)
            db.session.flush()
            recalc_ratio(patch.id)
            db.session.commit()
            flash("微地质单元已添加", "ok")
    return redirect(url_for("unit.index",
    patch_id=patch_id))
@bp.route("/edit/<int:unit_id>", methods=["GET", "POST"])
@login_required
def edit(unit_id):
    """单个微地质单元的编辑页。"""
    u = db.get_or_404(MicroUnit, unit_id)
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit(
    ):
            u.unit_type = request.form.get("unit_type",
    u.unit_type)
            area = request.form.get("area", "").strip()
            u.area = float(area) if area else None
            u.description = request.form.get("description",
    "").strip() or None
            recalc_ratio(u.patch_id)
            db.session.commit()
            flash("微地质单元已保存", "ok")
            return redirect(url_for("unit.index",
    patch_id=u.patch_id))
    return render_template("unit/edit.html", unit=u,
    patch=u.patch,
                           unit_types=UNIT_TYPES, form=form)
@bp.route("/delete/<int:unit_id>", methods=["POST"])
@login_required
def delete(unit_id):
    u = db.get_or_404(MicroUnit, unit_id)
    pid = u.patch_id
    name = u.unit_type
    # 一并删除该单元的指标数据，避免残留
    IndicatorData.query.filter_by(unit_id=u.id).delete(synch
    ronize_session=False)
    db.session.delete(u)
    db.session.flush()
    recalc_ratio(pid)
    db.session.commit()
    flash("微地质单元「%s」已删除" % name, "ok")
    return redirect(url_for("unit.index", patch_id=pid))
# ==== app/views/standard.py ====
# -*- coding: utf-8 -*-
"""M3 政策与标准知识库 —— 标准 / 政策文件的结构化规则与省际口径比对。"""
from flask import Blueprint, render_template, request,
```

```
    redirect, url_for, flash                                          1601
from flask_login import login_required                                1602
from app.forms import EmptyForm                                       1603
from app.models import db, Standard, StandardRule                     1604
bp = Blueprint("standard", __name__, url_prefix="/standard")          1605
@bp.route("/")                                                        1606
@login_required                                                       1607
def index():                                                          1608
    standards = Standard.query.order_by(Standard.category,            1609
    Standard.province).all()                                          1610
    local = [s for s in standards if s.category == "地方标准"]          1611
    policy = [s for s in standards if s.category != "地方标准"]         1612
    return render_template("standard/index.html",                     1613
    local=local, policy=policy,                                       1614
                                form=EmptyForm())                     1615
@bp.route("/add", methods=["GET", "POST"])                            1616
@login_required                                                       1617
def add():                                                            1618
    form = EmptyForm()                                                1619
    if request.method == "POST" and form.validate_on_submit(          1620
    ):                                                                1621
        s = Standard(                                                 1622
            province=request.form.get("province",                     1623
    "").strip(),                                                      1624
            name=request.form.get("name", "").strip(),                1625
            doc_number=request.form.get("doc_number",                 1626
    "").strip() or None,                                              1627
            publish_date=request.form.get("publish_date",             1628
    "").strip() or None,                                              1629
            implement_date=request.form.get("implement_date"          1630
    , "").strip() or None,                                            1631
            category=request.form.get("category", "地方标准"),         1632
            summary=request.form.get("summary", "").strip()           1633
    or None,                                                          1634
        )                                                             1635
        if not s.province or not s.name:                              1636
            flash("省份与名称不能为空", "error")                          1637
        else:                                                         1638
            db.session.add(s)                                         1639
            db.session.commit()                                       1640
            flash("标准已添加", "ok")                                   1641
            return redirect(url_for("standard.index"))                1642
    return render_template("standard/edit.html", form=form,           1643
    standard=None)                                                    1644
@bp.route("/<int:sid>/edit", methods=["GET", "POST"])                 1645
@login_required                                                       1646
def edit(sid):                                                        1647
    s = db.get_or_404(Standard, sid)                                  1648
    form = EmptyForm()                                                1649
    if request.method == "POST" and form.validate_on_submit(          1650
```

```
    ):                                                              1651
        s.province = request.form.get("province",                   1652
    "").strip()                                                     1653
        s.name = request.form.get("name", "").strip()               1654
        s.doc_number = request.form.get("doc_number",               1655
    "").strip() or None                                             1656
        s.publish_date = request.form.get("publish_date",           1657
    "").strip() or None                                             1658
        s.implement_date = request.form.get("implement_date"        1659
    , "").strip() or None                                           1660
        s.category = request.form.get("category", "地方标准")       1661
        s.summary = request.form.get("summary", "").strip()         1662
    or None                                                         1663
        db.session.commit()                                         1664
        flash("已保存", "ok")                                       1665
        return redirect(url_for("standard.rules", sid=s.id))        1666
    return render_template("standard/edit.html", form=form,         1667
    standard=s)                                                     1668
@bp.route("/<int:sid>/delete", methods=["POST"])                    1669
@login_required                                                     1670
def delete(sid):                                                    1671
    s = db.get_or_404(Standard, sid)                                1672
    StandardRule.query.filter_by(standard_id=sid).delete()          1673
    db.session.delete(s)                                            1674
    db.session.commit()                                             1675
    flash("已删除", "ok")                                           1676
    return redirect(url_for("standard.index"))                      1677
@bp.route("/<int:sid>/rules", methods=["GET", "POST"])              1678
@login_required                                                     1679
def rules(sid):                                                     1680
    s = db.get_or_404(Standard, sid)                                1681
    form = EmptyForm()                                              1682
    if request.method == "POST" and form.validate_on_submit(        1683
    ):                                                              1684
        r = StandardRule(                                           1685
            standard_id=s.id,                                       1686
            item=request.form.get("item", "").strip(),              1687
            operator=request.form.get("operator", ">="),            1688
            threshold=request.form.get("threshold",                 1689
    "").strip() or None,                                            1690
            condition_type=request.form.get("condition_type"        1691
    , "必要条件"),                                                  1692
            region=request.form.get("region", "").strip()           1693
    or None,                                                        1694
            note=request.form.get("note", "").strip() or            1695
    None,                                                           1696
        )                                                           1697
        if not r.item:                                              1698
            flash("判定项不能为空", "error")                        1699
        else:                                                       1700
```

```
                db.session.add(r)                                                1701
                db.session.commit()                                              1702
                flash("规则已添加", "ok")                                          1703
            return redirect(url_for("standard.rules", sid=s.id))                 1704
    rs = StandardRule.query.filter_by(standard_id=s.id).all(                     1705
    )                                                                            1706
    return render_template("standard/rules.html",                                1707
    form=form, standard=s, rules=rs)                                             1708
@bp.route("/rule/<int:rid>/delete", methods=["POST"])                            1709
@login_required                                                                  1710
def rule_delete(rid):                                                            1711
    r = db.get_or_404(StandardRule, rid)                                         1712
    sid = r.standard_id                                                          1713
    db.session.delete(r)                                                         1714
    db.session.commit()                                                          1715
    flash("规则已删除", "ok")                                                      1716
    return redirect(url_for("standard.rules", sid=sid))                          1717
@bp.route("/compare")                                                            1718
@login_required                                                                  1719
def compare():                                                                   1720
    """省际口径比对：同一指标在不同省份的阈值横向对比。"""                                  1721
    rules = StandardRule.query.join(Standard).all()                              1722
    groups = {}                                                                  1723
    for r in rules:                                                              1724
        groups.setdefault(r.item, []).append(r)                                  1725
    return render_template("standard/compare.html",                              1726
    groups=groups)                                                               1727
# ==== app/views/constraint.py ====                                              1728
# -*- coding: utf-8 -*-                                                          1729
"""M4 限制性指标判定 —— 一票否决判定。                                                1730
按论文表 2 与 4.1 节流程，共 5 项：政策、区位约束、土壤污染、地质灾害、耕地属性。                      1731
任一项不通过即判定为"人工辅助或生态重建"，不再进入定量评价。                                      1732
"""                                                                              1733
from flask import Blueprint, render_template, request,                           1734
    redirect, url_for, flash                                                     1735
from flask_login import login_required                                           1736
from app.forms import EmptyForm                                                  1737
from app.models import db, MinePatch, ConstraintCheck                            1738
from app.utils.constants import CONSTRAINT_ITEMS                                 1739
bp = Blueprint("constraint", __name__,                                           1740
    url_prefix="/constraint")                                                    1741
# 字段名 -> 中文名                                                                1742
FIELDS = [                                                                       1743
    ("policy_pass", "政策约束"),                                                   1744
    ("location_pass", "区位约束"),                                                 1745
    ("pollution_pass", "土壤污染"),                                                1746
    ("geo_pass", "地质灾害"),                                                      1747
    ("farmland_pass", "耕地属性"),                                                 1748
]                                                                                1749
@bp.route("/")                                                                   1750
```

```
@login_required                                                    1751
def index():                                                       1752
    patches = MinePatch.query.order_by(MinePatch.id.desc()).       1753
    all()                                                          1754
    latest = {}                                                    1755
    for p in patches:                                              1756
        latest[p.id] = ConstraintCheck.query.filter_by(patch       1757
    _id=p.id) \                                                    1758
            .order_by(ConstraintCheck.id.desc()).first()           1759
    return render_template("constraint/index.html",                1760
    patches=patches, latest=latest)                                1761
@bp.route("/<int:patch_id>", methods=["GET", "POST"])              1762
@login_required                                                    1763
def check(patch_id):                                               1764
    patch = db.get_or_404(MinePatch, patch_id)                     1765
    form = EmptyForm()                                             1766
    record = ConstraintCheck.query.filter_by(patch_id=patch_       1767
    id) \                                                          1768
        .order_by(ConstraintCheck.id.desc()).first()               1769
    if request.method == "POST" and form.validate_on_submit(       1770
    ):                                                             1771
        if record is None:                                         1772
            record = ConstraintCheck(patch_id=patch.id)            1773
            db.session.add(record)                                 1774
        for field, _ in FIELDS:                                    1775
            setattr(record, field, request.form.get(field)         1776
    == "1")                                                        1777
        record.passed = all(getattr(record, f) for f, _ in         1778
    FIELDS)                                                        1779
        failed = [name for f, name in FIELDS if not                1780
    getattr(record, f)]                                            1781
        record.reason = ("全部通过" if record.passed               1782
                         else "未通过：" + "、".join(failed))      1783
        db.session.commit()                                        1784
        flash("判定结果已保存", "ok")                              1785
        return redirect(url_for("constraint.check",                1786
    patch_id=patch.id))                                            1787
    # 预处理各条件的当前取值，供模板直接使用（Jinja 无 getattr）  1788
    field_states = [(f, name, None if record is None else          1789
    getattr(record, f))                                            1790
                    for f, name in FIELDS]                         1791
    return render_template("constraint/check.html",                1792
    patch=patch, form=form,                                        1793
                           fields=FIELDS,                          1794
    field_states=field_states,                                     1795
                           record=record)                          1796
# ==== app/views/ahp.py ====                                       1797
# -*- coding: utf-8 -*-                                            1798
"""M6 指标体系与 AHP 权重 —— 路由与页面逻辑。"""                   1799
import json                                                        1800
```

```
from flask import Blueprint, render_template, request,
    redirect, url_for, flash
from app.forms import AHPSelectForm, EmptyForm
from app.models import db, Indicator, ExpertMatrix,
    WeightScheme
from app.utils.constants import DEFAULT_WEIGHTS, UNIT_TYPES
from app.services import ahp_service as ahp
from app.services import indicator_service as ind_svc
bp = Blueprint("ahp", __name__, url_prefix="/ahp")
def indicator_list(unit_type, level, l1=None):
    """返回 （指标名称列表， 默认权重列表）。
    level=1 时返回一级指标；level=2 时返回指定一级指标下的二级指标。
    """
    weights = DEFAULT_WEIGHTS[unit_type]
    if level == 1:
        names = list(weights["l1"].keys())
        vec = [weights["l1"][n] for n in names]
        return names, vec
    # level == 2
    l2 = weights["l2"][l1]
    names = list(l2.keys())
    vec = [l2[n] for n in names]
    return names, vec
def l1_choices(unit_type):
    """某单元类型的一级指标名称列表（用于二级指标录入时选择父指标）。"""
    return list(DEFAULT_WEIGHTS[unit_type]["l1"].keys())
@bp.route("/", methods=["GET", "POST"])
def index():
    form = AHPSelectForm()
    # 先根据请求中的单元类型填充 l1 选项，再进行表单校验
    unit = request.form.get("unit_type") or
    form.unit_type.choices[0][0]
    form.l1.choices = [(n, n) for n in l1_choices(unit)]
    if form.validate_on_submit():
        unit_type = form.unit_type.data
        level = int(form.level.data)
        if level == 1:
            return redirect(url_for("ahp.matrix",
    unit_type=unit_type, level=1))
        if not form.l1.data:
            flash("选择「二级指标」时，请同时选择所属的一级指标", "error")
            return render_template("ahp/index.html",
    form=form)
        return redirect(url_for("ahp.matrix",
    unit_type=unit_type,
                                level=2, l1=form.l1.data))
    return render_template("ahp/index.html", form=form)
@bp.route("/matrix", methods=["GET", "POST"])
def matrix():
    unit_type = request.values.get("unit_type")
```

```
    level = int(request.values.get("level", 1))
    l1 = request.values.get("l1")
    if unit_type not in DEFAULT_WEIGHTS:
        flash("无效的微地质单元类型", "error")
        return redirect(url_for("ahp.index"))
    names, vec = indicator_list(unit_type, level, l1)
    n = len(names)
    form = EmptyForm()
    # 预填默认值：论文表 3 权重对应的完全一致矩阵 a_ij = w_i / w_j
    default = [[round(vec[i] / vec[j], 4) for j in
    range(n)] for i in range(n)]
    def compute(upper):
        A = ahp.matrix_from_upper(upper)
        r = ahp.weights_and_consistency(A)
        bad = ahp.find_most_inconsistent(A)
        return {
            "weights": [round(v, 4) for v in r["weights"]],
            "lambda_max": round(r["lambda_max"], 4),
            "ci": r["ci"],
            "cr": r["cr"],
            "consistent": r["consistent"],
            "matrix": [[round(A[i, j], 4) for j in
    range(n)] for i in range(n)],
            "most_inconsistent": bad,
        }
    result = None
    if request.method == "POST" and form.validate_on_submit(
    ):
        try:
            upper = [float(request.form.get("m_%d_%d" % (i,
    j), "1"))
                     for i in range(n) for j in range(i +
    1, n)]
            result = compute(upper)
        except (ValueError, ZeroDivisionError):
            flash("矩阵输入有误：请填入正数（对角线自动为 1，下三角自动取倒数）",
    "error")
    elif request.values.get("calc") == "1":
        # 按默认（论文表 3）矩阵直接试算，便于一键查看权重与一致性
        result = compute([default[i][j] for i in range(n)
    for j in range(i + 1, n)])
    return render_template(
        "ahp/matrix.html",
        form=form, unit_type=unit_type, level=level, l1=l1,
        names=names, vec=vec, default=default,
    result=result,
    )
# ---------------------------- 指标维护 ----------------------------
    ----
@bp.route("/indicators")
```

```
def indicators():                                                    1901
    """指标定义维护：查看指标树、单位、理想值与取值依据。"""                1902
    if Indicator.query.count() == 0:                                 1903
        ind_svc.seed_indicators()                                    1904
    all_ind = Indicator.query.order_by(Indicator.level,              1905
    Indicator.id).all()                                              1906
    by_parent = {}                                                   1907
    for i in all_ind:                                                1908
        by_parent.setdefault(i.parent_code, []).append(i)            1909
    return render_template("ahp/indicators.html",                    1910
    by_parent=by_parent,                                             1911
                                all_ind=all_ind,                     1912
    unit_types=UNIT_TYPES,                                           1913
                                type_label=ind_svc.ideal_type_lab    1914
    el,                                                              1915
                                form=EmptyForm())                    1916
@bp.route("/indicators/<code>/edit", methods=["GET",                 1917
    "POST"])                                                         1918
def indicator_edit(code):                                            1919
    ind = Indicator.query.filter_by(code=code).first_or_404(         1920
    )                                                                1921
    form = EmptyForm()                                               1922
    if request.method == "POST" and form.validate_on_submit(         1923
    ):                                                               1924
        name = request.form.get("name", "").strip()                  1925
        if not name:                                                 1926
            flash("指标名称不能为空", "error")                          1927
            return redirect(url_for("ahp.indicator_edit",            1928
    code=code))                                                      1929
        ind.name = name                                              1930
        ind.unit = request.form.get("unit", "").strip() or           1931
    None                                                             1932
        ind.ideal_value = request.form.get("ideal_value",            1933
    "").strip() or None                                              1934
        ind.ideal_type = request.form.get("ideal_type") or           1935
    ind.ideal_type                                                   1936
        units = request.form.getlist("enabled_units")                1937
        ind.enabled_units = ",".join(units) if units else            1938
    None                                                             1939
        ind.is_constraint = request.form.get("is_constraint"         1940
    ) == "1"                                                         1941
        ind.description = request.form.get("description",            1942
    "").strip() or None                                              1943
        db.session.commit()                                          1944
        flash("指标「%s」已保存" % ind.name, "ok")                      1945
        return redirect(url_for("ahp.indicators"))                   1946
    return render_template("ahp/indicator_edit.html",                1947
    ind=ind,                                                         1948
                                unit_types=UNIT_TYPES,               1949
                                type_label=ind_svc.ideal_type_lab    1950
```

```
    el, form=form)                                                    1951
@bp.route("/indicators/reset", methods=["POST"])                      1952
def indicators_reset():                                               1953
    """按论文取值重置指标定义。"""                                    1954
    form = EmptyForm()                                                1955
    if form.validate_on_submit():                                     1956
        n = ind_svc.seed_indicators()                                 1957
        flash("已按论文取值重置 %d 项指标定义" % n, "ok")              1958
    return redirect(url_for("ahp.indicators"))                        1959
# ---------------------------- 多专家矩阵 ---------------------------1960
    -----                                                             1961
def expert_scope():                                                   1962
    unit_type = request.values.get("unit_type") or                    1963
    UNIT_TYPES[0]                                                     1964
    level = int(request.values.get("level", 1))                      1965
    l1 = request.values.get("l1") or None                             1966
    return unit_type, level, l1                                       1967
def expert_query(unit_type, level, l1):                               1968
    q = ExpertMatrix.query.filter_by(unit_type=unit_type,             1969
    level=level)                                                      1970
    if level == 2:                                                   1971
        q = q.filter_by(l1_name=l1)                                   1972
    else:                                                             1973
        q = q.filter(ExpertMatrix.l1_name.is_(None))                  1974
    return q.order_by(ExpertMatrix.id)                                1975
def default_scheme(unit_type, level, l1=None):                        1976
    """取（或建）该范围的权重方案，专家矩阵统一挂到它下面。"""       1977
    name = "%s · %s%s" % (unit_type,                                  1978
                          "一级指标" if level == 1 else "二级指标",   1979
                          "(%s)" % l1 if l1 else "")                  1980
    scheme = WeightScheme.query.filter_by(unit_type=unit_typ1981
    e,                                                                1982
                                          level=level,                1983
    name=name).first()                                                1984
    if scheme is None:                                                1985
        scheme = WeightScheme(unit_type=unit_type,                    1986
    level=level, name=name,                                           1987
                              is_default=False)                       1988
        db.session.add(scheme)                                        1989
        db.session.flush()                                            1990
    return scheme                                                     1991
@bp.route("/experts", methods=["GET", "POST"])                        1992
def experts():                                                        1993
    """多专家判断矩阵录入与汇总。"""                                  1994
    unit_type, level, l1 = expert_scope()                             1995
    if unit_type not in DEFAULT_WEIGHTS:                              1996
        flash("无效的微地质单元类型", "error")                        1997
        return redirect(url_for("ahp.index"))                         1998
    names, vec = indicator_list(unit_type, level, l1)                 1999
    n = len(names)                                                    2000
```

```
    form = EmptyForm()
    if request.method == "POST" and form.validate_on_submit()
    ):
        action = request.form.get("action")
        if action == "add":
            expert_name = request.form.get("expert_name",
    "").strip() or "未具名专家"
            try:
                upper = [float(request.form.get("m_%d_%d" %
    (i, j), "1")) 
                         for i in range(n) for j in range(i
    + 1, n)]
            except ValueError:
                flash("矩阵输入有误，请填入正数", "error")
                return redirect(url_for("ahp.experts",
    unit_type=unit_type,
                                        level=level, l1=l1))
            A = ahp.matrix_from_upper(upper)
            r = ahp.weights_and_consistency(A)
            scheme = default_scheme(unit_type, level, l1)
            db.session.add(ExpertMatrix(
                scheme_id=scheme.id,
                unit_type=unit_type, l1_name=(l1 if level
    == 2 else None),
                expert_name=expert_name, level=level,
                matrix_json=json.dumps([list(map(float,
    row)) for row in A]),
                cr=round(r["cr"], 6)))
            db.session.commit()
            flash("已录入专家「%s」的判断矩阵（CR=%.4f）" % (expert_name,
    r["cr"]), "ok")
        return redirect(url_for("ahp.experts",
    unit_type=unit_type,
                                level=level, l1=l1))
    rows = expert_query(unit_type, level, l1).all()
    default = [[round(vec[i] / vec[j], 4) for j in
    range(n)] for i in range(n)]
    return render_template("ahp/experts.html",
    unit_type=unit_type, level=level,
                           l1=l1, names=names, rows=rows,
    default=default,
                           n=n, form=form)
@bp.route("/experts/<int:mid>/delete", methods=["POST"])
def expert_delete(mid):
    row = db.get_or_404(ExpertMatrix, mid)
    unit_type, level, l1 = row.unit_type, row.level,
    row.l1_name
    form = EmptyForm()
    if form.validate_on_submit():
        db.session.delete(row)
```

```
            db.session.commit()                                          2051
            flash("已删除专家矩阵", "ok")                                  2052
        return redirect(url_for("ahp.experts",                           2053
        unit_type=unit_type,                                             2054
                                        level=level, l1=l1))             2055
@bp.route("/experts/aggregate")                                          2056
def experts_aggregate():                                                 2057
        """按几何平均汇总多专家矩阵，并做一致性检验。"""                          2058
        unit_type, level, l1 = expert_scope()                            2059
        if unit_type not in DEFAULT_WEIGHTS:                             2060
            flash("无效的微地质单元类型", "error")                          2061
            return redirect(url_for("ahp.index"))                        2062
        names, vec = indicator_list(unit_type, level, l1)                2063
        rows = expert_query(unit_type, level, l1).all()                  2064
        result = None                                                    2065
        if rows:                                                         2066
            mats = []                                                    2067
            for r in rows:                                               2068
                try:                                                     2069
                    mats.append([list(map(float, line))                  2070
                                 for line in                             2071
        json.loads(r.matrix_json)])                                      2072
                except Exception:                                        2073
                    continue                                             2074
            if mats:                                                     2075
                agg = ahp.aggregate_matrices(mats)                       2076
                res = ahp.weights_and_consistency(agg)                   2077
                bad = ahp.find_most_inconsistent(agg)                    2078
                result = {                                               2079
                    "count": len(mats),                                  2080
                    "weights": [round(v, 4) for v in                     2081
        res["weights"]],                                                 2082
                    "lambda_max": round(res["lambda_max"], 4),           2083
                    "ci": res["ci"], "cr": res["cr"],                    2084
                    "consistent": res["consistent"],                     2085
                    "most_inconsistent": bad,                            2086
                    "expert_crs": [r.cr for r in rows if r.cr            2087
        is not None],                                                    2088
                }                                                        2089
        return render_template("ahp/experts_aggregate.html",             2090
        unit_type=unit_type,                                             2091
                                       level=level, l1=l1, names=names,  2092
        rows=rows,                                                       2093
                                       result=result)                    2094
# ==== app/views/indicator.py ====                                       2095
# -*- coding: utf-8 -*-                                                  2096
"""M7 指标数据采集与管理 —— 路由与页面逻辑。                                  2097
按微地质单元组织录入；系统按单元类型自动决定应录入哪些指标                          2098
（岩质边坡无土壤指标、平台无地形指标）。                                      2099
"""                                                                      2100
```

```
import io                                                                2101
import json                                                              2102
from flask import (Blueprint, render_template, request,                  2103
    redirect, url_for,                                                   2104
                      flash, send_file)                                  2105
from app.forms import EmptyForm                                          2106
from app.models import db, MinePatch, MicroUnit,                         2107
    IndicatorData                                                        2108
from app.utils.constants import HEAVY_METALS, ASPECT_TYPES,              2109
    UNIT_ENABLED_INDICATORS                                              2110
from app.services import import_service as imp                           2111
bp = Blueprint("indicator", __name__,                                    2112
    url_prefix="/indicator")                                             2113
# 指标字段定义：字段名 -> (中文名, 单位, 是否可选)                                2114
SOIL_FIELDS = [                                                          2115
    ("soil_ratio", "土石比", ""),                                          2116
    ("thickness", "土层厚度", "cm"),                                       2117
    ("ph", "pH", ""),                                                    2118
    ("organic_matter", "有机质", "g/kg"),                                  2119
]                                                                        2120
VEG_FIELDS = [                                                           2121
    ("coverage", "植被覆盖度", "%"),                                       2122
    ("shannon", "Shannon-Wiener 多样性指数", ""),                          2123
]                                                                        2124
TERRAIN_FIELDS = [                                                       2125
    ("slope", "坡度", "°"),                                                2126
    ("aspect", "坡向", ""),                                                2127
]                                                                        2128
def unit_field_config(unit_type):                                        2129
    """根据单元类型返回需录入的字段分组。"""                                          2130
    enabled = UNIT_ENABLED_INDICATORS.get(unit_type, ())                 2131
    groups = []                                                          2132
    if "土壤指标" in enabled:                                                2133
        groups.append(("土壤指标", SOIL_FIELDS, True))    # True          2134
    = 含重金属                                                             2135
    if "植被指标" in enabled:                                                2136
        groups.append(("植被指标", VEG_FIELDS, False))                     2137
    if "地形指标" in enabled:                                                2138
        groups.append(("地形指标", TERRAIN_FIELDS, False))                 2139
    return groups                                                        2140
@bp.route("/")                                                           2141
def index():                                                             2142
    patches = MinePatch.query.order_by(MinePatch.id.desc()).             2143
    all()                                                                2144
    return render_template("indicator/index.html",                       2145
    patches=patches)                                                     2146
@bp.route("/<int:unit_id>", methods=["GET", "POST"])                     2147
def edit(unit_id):                                                       2148
    unit = db.get_or_404(MicroUnit, unit_id)                             2149
    ind = IndicatorData.query.filter_by(unit_id=unit.id).fir             2150
```

```
    st()                                                                2151
    form = EmptyForm()                                                  2152
    groups = unit_field_config(unit.unit_type)                          2153
    if request.method == "POST" and form.validate_on_submit(            2154
    ):                                                                  2155
        def f(name, cast=float):                                        2156
            v = request.form.get(name, "").strip()                      2157
            return cast(v) if v != "" else None                         2158
        heavy = {}                                                      2159
        for m in HEAVY_METALS:                                          2160
            v = f("hm_" + m)                                            2161
            if v is not None:                                           2162
                heavy[m] = v                                            2163
        if ind is None:                                                 2164
            ind = IndicatorData(unit_id=unit.id)                        2165
            db.session.add(ind)                                         2166
        ind.soil_ratio = f("soil_ratio")                                2167
        ind.thickness = f("thickness")                                  2168
        ind.ph = f("ph")                                                2169
        ind.organic_matter = f("organic_matter")                        2170
        ind.heavy_metals = json.dumps(heavy,                            2171
    ensure_ascii=False) if heavy else None                              2172
        ind.coverage = f("coverage")                                    2173
        ind.shannon = f("shannon")                                      2174
        ind.slope = f("slope")                                          2175
        ind.aspect = request.form.get("aspect", "").strip()             2176
    or None                                                             2177
        db.session.commit()                                             2178
        flash("指标数据已保存", "ok")                                    2179
        return redirect(url_for("indicator.index"))                     2180
    # 组装当前值供模板回显                                              2181
    current = {}                                                        2182
    if ind:                                                             2183
        current = {                                                     2184
            "soil_ratio": ind.soil_ratio, "thickness":                  2185
    ind.thickness,                                                      2186
            "ph": ind.ph, "organic_matter":                             2187
    ind.organic_matter,                                                 2188
            "coverage": ind.coverage, "shannon":                        2189
    ind.shannon,                                                        2190
            "slope": ind.slope, "aspect": ind.aspect,                   2191
        }                                                               2192
        current["heavy"] = ind.heavy_metals_dict()                      2193
    return render_template("indicator/edit.html",                       2194
                           unit=unit, form=form,                        2195
    groups=groups,                                                      2196
                           heavy_metals=HEAVY_METALS,                   2197
                           aspect_types=ASPECT_TYPES,                   2198
    current=current)                                                    2199
# ---------------------------- 批量导入 ----------------------------    2200
```

```
    ----                                                                2201
@bp.route("/import", methods=["GET", "POST"])                          2202
def import_data():                                                      2203
    """批量导入指标数据（CSV / Excel）。"""                                2204
    patches = MinePatch.query.order_by(MinePatch.id.desc()).2205
    all()                                                               2206
    form = EmptyForm()                                                  2207
    report = None                                                       2208
    if request.method == "POST" and form.validate_on_submit(2209
    ):                                                                  2210
        patch_id = request.form.get("patch_id", type=int)               2211
        patch = db.session.get(MinePatch, patch_id) if                  2212
    patch_id else None                                                  2213
        if patch is None:                                               2214
            flash("请选择矿山图斑", "error")                              2215
            return redirect(url_for("indicator.import_data")2216
    )                                                                   2217
        file = request.files.get("datafile")                            2218
        if file is None or not file.filename:                           2219
            flash("请选择要导入的文件", "error")                          2220
            return redirect(url_for("indicator.import_data")2221
    )                                                                   2222
        ext = file.filename.rsplit(".", 1)[-1].lower()                  2223
        if ext not in ("csv", "xlsx", "xls"):                           2224
            flash("仅支持 CSV 或 Excel（.xlsx/.xls）文件", "error")       2225
            return redirect(url_for("indicator.import_data")2226
    )                                                                   2227
        units = MicroUnit.query.filter_by(patch_id=patch.id)2228
    .all()                                                              2229
        unit_map = {}                                                   2230
        for u in units:                                                 2231
            unit_map.setdefault(u.unit_type, u)                         2232
        if not unit_map:                                                2233
            flash("该图斑尚未划分微地质单元，请先划分后再导入", "error")      2234
            return redirect(url_for("indicator.import_data")2235
    )                                                                   2236
        try:                                                            2237
            parsed, errors = imp.parse_upload(file.filename,2238
    file.stream, unit_map)                                              2239
        except Exception as e:                                          2240
            flash("文件解析失败：%s" % e, "error")                        2241
            return redirect(url_for("indicator.import_data")2242
    )                                                                   2243
        saved = 0                                                       2244
        for rec in parsed:                                              2245
            unit = unit_map[rec["unit_type"]]                           2246
            ind = IndicatorData.query.filter_by(unit_id=unit2247
    .id).first()                                                        2248
            if ind is None:                                             2249
                ind = IndicatorData(unit_id=unit.id)                    2250
```

```
                    db.session.add(ind)                                    2251
                f = rec["fields"]                                          2252
                for key in ("soil_ratio", "thickness", "ph",               2253
    "organic_matter",                                                      2254
                            "coverage", "shannon", "slope",                2255
    "aspect"):                                                             2256
                    if key in f:                                           2257
                        setattr(ind, key, f[key])                          2258
                if rec["heavy_metals"]:                                    2259
                    ind.heavy_metals = json.dumps(rec["heavy_met           2260
    als"],                                                                 2261
                                                                           2262
    ensure_ascii=False)                                                    2263
                saved += 1                                                 2264
            db.session.commit()                                            2265
            report = {"patch": patch.name, "saved": saved,                 2266
    "errors": errors,                                                      2267
                      "total": len(parsed) + len(                          2268
                          [e for e in errors if e.startswith("第           2269
    ")])}                                                                  2270
            if saved:                                                      2271
                flash("已导入 %d 个微地质单元的指标数据" % saved, "ok")          2272
            else:                                                          2273
                flash("没有可导入的数据行，请检查文件内容", "error")             2274
    return render_template("indicator/import.html",                        2275
    patches=patches,                                                       2276
                                   form=form, report=report,               2277
    headers=imp.HEADERS)                                                   2278
@bp.route("/import/template")                                              2279
def import_template():                                                     2280
    """下载导入模板（CSV，含一行示例）。"""                                    2281
    csv_text = imp.build_template_csv()                                    2282
    data = csv_text.encode("utf-8-sig")       # 带 BOM，便于               2283
    Excel 正确识别中文                                                       2284
    return send_file(io.BytesIO(data), as_attachment=True,                 2285
                     download_name="指标数据导入模板.csv",                   2286
                     mimetype="text/csv")                                  2287
# ==== app/views/evaluate.py ====                                          2288
# -*- coding: utf-8 -*-                                                    2289
"""M8 适宜性定量评价引擎 —— 路由与页面逻辑。                                  2290
从数据库加载某图斑的微地质单元与指标数据，调用评价引擎计算，                    2291
展示各单元得分、最终评价因子 F、适宜性等级与推荐措施。                        2292
入口处按论文 4.1 节的流程顺序检查前置条件（见 app/services/workflow.py），   2293
前置步骤未完成时不予计算，避免在条件不具备时产出评价结论。                    2294
"""                                                                        2295
import json                                                                2296
from flask import Blueprint, render_template, request,                     2297
    redirect, url_for, flash                                               2298
from app.forms import EmptyForm                                            2299
from app.models import db, MinePatch, MicroUnit,                           2300
```

```
    IndicatorData, EvaluationResult
from app.services import evaluation_service as ev
from app.services import workflow as wf
bp = Blueprint("evaluate", __name__, url_prefix="/evaluate")
def load_patch_input(patch_id):
    """从数据库组装评价引擎输入。"""
    units = MicroUnit.query.filter_by(patch_id=patch_id).all()
    result = []
    for u in units:
        ind = IndicatorData.query.filter_by(unit_id=u.id).first()
        data = {
            "soil_ratio": ind.soil_ratio if ind else None,
            "thickness": ind.thickness if ind else None,
            "ph": ind.ph if ind else None,
            "organic_matter": ind.organic_matter if ind else None,
            "heavy_metals": ind.heavy_metals_dict() if ind else {},
            "coverage": ind.coverage if ind else None,
            "shannon": ind.shannon if ind else None,
            "slope": ind.slope if ind else None,
            "aspect": ind.aspect if ind else None,
        }
        result.append({
            "unit_type": u.unit_type,
            "ratio": u.ratio if u.ratio is not None else 0.0,
            "data": data,
        })
    return result
def block_reasons(progress):
    """把未满足的前置条件转成可读的拦截原因。"""
    return ["%s：%s" % (b["label"], b["reason"]) for b in progress["blocking"]]
@bp.route("/")
def index():
    patches = MinePatch.query.order_by(MinePatch.id.desc()).all()
    latest = {}
    progress = {}
    for p in patches:
        latest[p.id] = EvaluationResult.query.filter_by(patch_id=p.id) \
            .order_by(EvaluationResult.id.desc()).first()
        progress[p.id] = wf.patch_progress(p.id)
    return render_template("evaluate/index.html",
    patches=patches,
                           latest=latest, progress=progress)
```

```
@bp.route("/<int:patch_id>", methods=["GET", "POST"])
def run(patch_id):
    patch = db.get_or_404(MinePatch, patch_id)
    form = EmptyForm()
    progress = wf.patch_progress(patch_id)
    if request.method == "POST" and form.validate_on_submit(
    ):
        if not progress["ready"]:
            for reason in block_reasons(progress):
                flash("无法执行评价 —— " + reason, "error")
            return redirect(url_for("evaluate.run",
    patch_id=patch_id))
        units = load_patch_input(patch_id)
        res = ev.score_mine(units)
        saved = EvaluationResult(
            patch_id=patch.id,
            final_score=round(res["final_score"], 4),
            grade=res["grade"],
            measures=res["grade_info"]["measure"],
            unit_scores=json.dumps(
                [{"unit_type": u["unit_type"], "ratio":
    u["ratio"],
                  "unit_score": round(u["unit_score"], 4)}
                 for u in res["unit_results"]],
                ensure_ascii=False),
        )
        db.session.add(saved)
        db.session.commit()
        return redirect(url_for("evaluate.result",
    result_id=saved.id))
    return render_template("evaluate/run.html",
    patch=patch, form=form,
                           progress=progress)
@bp.route("/result/<int:result_id>")
def result(result_id):
    r = db.get_or_404(EvaluationResult, result_id)
    patch = db.session.get(MinePatch, r.patch_id)
    unit_scores = json.loads(r.unit_scores) if
    r.unit_scores else []
    return render_template("evaluate/result.html", r=r,
    patch=patch,
                           unit_scores=unit_scores)
# ==== app/views/report.py ====
# -*- coding: utf-8 -*-
"""M9 报告与结果输出 —— 得分明细、图表、Word / PDF 导出。"""
import io
from datetime import datetime
from flask import Blueprint, render_template, send_file,
    flash, redirect, url_for
from app.models import db, MinePatch, ReferenceEcosystem
```

```
from app.services import evaluation_service as ev
from app.services import report_service as rs
from app.services import workflow as wf
from app.utils import charts
from app.utils import browser
from app.views.evaluate import load_patch_input
bp = Blueprint("report", __name__, url_prefix="/report")
def _prepare(patch_id):
    """加载图斑、参照生态系统并执行评价，供明细页与导出复用。
    同时返回流程检查结果：前置步骤未完成时，调用方应拦截并提示，
    不应直接出报告（报告本质上是评价结论的呈现，不能先于评价产生）。
    """
    patch = db.get_or_404(MinePatch, patch_id)
    progress = wf.patch_progress(patch_id)
    if not progress["ready"]:
        return patch, None, None, None, progress
    units = load_patch_input(patch_id)
    if not units:
        return patch, None, None, None, progress
    result = ev.score_mine(units)
    reference = ReferenceEcosystem.query.filter_by(patch_id=
    patch_id).first()
    return patch, reference, result, units, progress
def _deny(progress, patch_id, endpoint="report.detail"):
    """前置条件未满足时的统一处理：逐条提示并退回报告列表。"""
    for b in progress["blocking"]:
        flash("暂不能生成报告 —— %s：%s" % (b["label"],
    b["reason"]), "error")
    if progress["blocking"]:
        flash("请先完成上述步骤后再生成报告。", "error")
    return redirect(url_for("report.index"))
def _charts(result):
    """生成两张图表：单元得分对比 + 一级指标平均得分分布。"""
    units = [(u["unit_type"], u["unit_score"],
              "%.4f" % u["unit_score"]) for u in
    result["unit_results"]]
    chart_units = charts.hbars(units)
    # 各一级指标在各单元上的平均得分，反映其对评价结果的实际影响
    acc, cnt = {}, {}
    for u in result["unit_results"]:
        for k, v in (u.get("l1_scores") or {}).items():
            acc[k] = acc.get(k, 0.0) + v
            cnt[k] = cnt.get(k, 0) + 1
    items = [(k, acc[k] / cnt[k], "%.4f" % (acc[k] /
    cnt[k])) for k in acc]
    items.sort(key=lambda x: -x[1])
    chart_weight = charts.donut(items)
    return chart_units, chart_weight
@bp.route("/")
def index():
```

```
    patches = MinePatch.query.order_by(MinePatch.id.desc()).
    all()
    progress = {p.id: wf.patch_progress(p.id) for p in
    patches}
    return render_template("report/index.html",
    patches=patches,
                           progress=progress)
@bp.route("/<int:patch_id>")
def detail(patch_id):
    patch, reference, result, units, progress =
    _prepare(patch_id)
    if result is None:
        return _deny(progress, patch_id)
    chart_units, chart_weight = _charts(result)
    return render_template("report/detail.html",
    patch=patch, result=result,
                           reference=reference,
    chart_units=chart_units,
                           chart_weight=chart_weight)
@bp.route("/<int:patch_id>/word")
def word(patch_id):
    patch, reference, result, units, progress =
    _prepare(patch_id)
    if result is None:
        return _deny(progress, patch_id)
    buf = rs.build_report(patch, result)
    filename = "自然恢复适宜性评价报告_%s.docx" % patch.name
    return send_file(buf, as_attachment=True,
    download_name=filename,
                     mimetype="application/vnd.openxmlformat
    s-officedocument."
                              "wordprocessingml.document")
@bp.route("/<int:patch_id>/pdf")
def pdf(patch_id):
    """导出 PDF 报告：服务端渲染打印模板，再交浏览器无头打印。"""
    patch, reference, result, units, progress =
    _prepare(patch_id)
    if result is None:
        return _deny(progress, patch_id)
    chart_units, chart_weight = _charts(result)
    html = render_template(
        "report/print.html", patch=patch,
    reference=reference, result=result,
        chart_units=chart_units, chart_weight=chart_weight,
        generated_at=datetime.now().strftime("%Y-%m-%d
    %H:%M"))
    data, err = browser.html_to_pdf(html)
    if err:
        flash("PDF 导出失败：%s" % err, "error")
        return redirect(url_for("report.detail",
```

```
        patch_id=patch_id))                                                    2501
        return send_file(io.BytesIO(data), as_attachment=True,                 2502
                         download_name="自然恢复适宜性评价报告_%s.pdf" %       2503
        patch.name,                                                            2504
                         mimetype="application/pdf")                           2505
# ==== app/templates/base.html ====                                            2506
<!DOCTYPE html>                                                                2507
<html lang="zh-CN">                                                            2508
<head>                                                                         2509
  <meta charset="UTF-8">                                                       2510
  <meta name="viewport" content="width=device-width,                           2511
    initial-scale=1.0">                                                        2512
  <title>{% block title %}历史遗留废弃矿山自然恢复适宜性评价系统{% endblock   2513
    %}</title>                                                                 2514
  <style>                                                                      2515
    /* ---------- 设计变量：单一强调色 + 中性灰阶 ---------- */                2516
    :root {                                                                    2517
      --accent: #1f3a5f;                                                       2518
      --accent-hover: #2a4d7c;                                                 2519
      --accent-soft: #eef2f7;                                                  2520
      --danger: #b3261e;                                                       2521
      --danger-soft: #fdecea;                                                  2522
      --ok: #1e6b3a;                                                           2523
      --ok-soft: #eaf6ef;                                                      2524
      --ink: #23262b;                                                          2525
      --ink-soft: #6b7280;                                                     2526
      --line: #dfe3e8;                                                         2527
      --surface: #ffffff;                                                      2528
      --canvas: #f5f6f8;                                                       2529
      --radius: 6px;                                                           2530
      --shadow: 0 1px 2px rgba(0, 0, 0, .05);                                  2531
      --header-h: 56px;                                                        2532
      --nav-w: 208px;                                                          2533
    }                                                                          2534
    * { box-sizing: border-box; }                                              2535
    html, body { margin: 0; padding: 0; }                                      2536
    body {                                                                     2537
      font-family: "Microsoft YaHei", "PingFang SC",                           2538
    "SimSun", sans-serif;                                                      2539
      font-size: 14px;                                                         2540
      line-height: 1.6;                                                        2541
      color: var(--ink);                                                       2542
      background: var(--canvas);                                               2543
    }                                                                          2544
    /* ---------- 跳到主要内容（键盘可达性） ---------- */                     2545
    .skip-link {                                                               2546
      position: absolute;                                                      2547
      left: 12px;                                                              2548
      top: -60px;                                                              2549
      z-index: 30;                                                             2550
```

```
    padding: 8px 14px;
    background: var(--accent);
    color: #fff;
    border-radius: var(--radius);
    text-decoration: none;
    transition: top .15s ease-out;
}
.skip-link:focus { top: 12px; }
/* ---------- 顶部标题栏 ---------- */
.app-header {
    display: flex;
    align-items: center;
    gap: 12px;
    height: var(--header-h);
    padding: 0 20px;
    background: var(--accent);
    color: #fff;
}
.app-header__title {
    font-size: 16px;
    font-weight: 600;
    letter-spacing: .2px;
}
.app-header__version {
    padding: 1px 8px;
    font-size: 12px;
    border: 1px solid rgba(255, 255, 255, .45);
    border-radius: 10px;
    font-variant-numeric: tabular-nums;
}
.app-header__spacer { flex: 1; }
.app-header__user {
    display: flex;
    align-items: center;
    gap: 10px;
    font-size: 13px;
}
.app-header__user a {
    color: #d6e3f4;
    text-decoration: none;
    padding: 4px 8px;
    border-radius: 4px;
}
.app-header__user a:hover,
.app-header__user a:focus-visible {
    color: #fff;
    background: rgba(255, 255, 255, .14);
}
/* ---------- 主体：左侧导航 + 内容区 ---------- */
.app-body {
```

```
    display: flex;                                                    2601
    align-items: stretch;                                             2602
    min-height: calc(100dvh - var(--header-h));                       2603
}                                                                     2604
.app-nav {                                                            2605
    flex: 0 0 var(--nav-w);                                           2606
    width: var(--nav-w);                                              2607
    padding: 14px 10px;                                               2608
    background: var(--surface);                                       2609
    border-right: 1px solid var(--line);                              2610
}                                                                     2611
.app-nav__group {                                                     2612
    margin: 0 0 4px;                                                  2613
    padding: 10px 10px 6px;                                           2614
    font-size: 12px;                                                  2615
    color: var(--ink-soft);                                           2616
}                                                                     2617
.app-nav ul {                                                         2618
    margin: 0;                                                        2619
    padding: 0;                                                       2620
    list-style: none;                                                 2621
}                                                                     2622
.app-nav a {                                                          2623
    display: block;                                                   2624
    padding: 9px 12px;                                                2625
    margin-bottom: 2px;                                               2626
    border-radius: var(--radius);                                     2627
    color: var(--ink);                                                2628
    text-decoration: none;                                            2629
    border-left: 3px solid transparent;                               2630
}                                                                     2631
.app-nav a:hover { background: var(--accent-soft); }                  2632
.app-nav a:focus-visible {                                            2633
    outline: 2px solid var(--accent);                                 2634
    outline-offset: 1px;                                              2635
}                                                                     2636
.app-nav a[aria-current="page"] {                                     2637
    background: var(--accent-soft);                                   2638
    border-left-color: var(--accent);                                 2639
    color: var(--accent);                                             2640
    font-weight: 600;                                                 2641
}                                                                     2642
.app-main {                                                           2643
    flex: 1;                                                          2644
    min-width: 0;                                                     2645
    padding: 22px 26px 40px;                                          2646
}                                                                     2647
.container { max-width: 1040px; }                                     2648
/* ---------- 卡片 ---------- */                                      2649
.card {                                                               2650
```

```
    background: var(--surface);
    border: 1px solid var(--line);
    border-radius: var(--radius);
    box-shadow: var(--shadow);
    padding: 22px 24px;
    margin-bottom: 18px;
}
.card > h2 {
    margin: 0 0 14px;
    font-size: 16px;
    font-weight: 600;
    color: var(--accent);
    text-wrap: balance;
}
.card > h3 {
    margin: 20px 0 10px;
    font-size: 14px;
    font-weight: 600;
    color: var(--accent);
}
/* ---------- 表格 ---------- */
table {
    width: 100%;
    border-collapse: collapse;
    margin: 10px 0 16px;
    font-variant-numeric: tabular-nums;
}
caption {
    text-align: left;
    color: var(--ink-soft);
    font-size: 12px;
    padding-bottom: 6px;
}
th, td {
    border: 1px solid var(--line);
    padding: 8px 12px;
    text-align: center;
    vertical-align: middle;
}
th {
    background: var(--accent-soft);
    font-weight: 600;
    white-space: nowrap;
}
tbody tr:hover { background: #fafbfc; }
/* ---------- 表单控件 ---------- */
label { display: inline-block; }
.field { margin-bottom: 12px; }
.field__label {
    display: block;
```

```
    margin-bottom: 4px;
    color: var(--ink-soft);
    font-size: 13px;
}
input[type="text"], input[type="number"],
input[type="password"],
select, textarea {
    padding: 6px 9px;
    font-family: inherit;
    font-size: 14px;
    color: var(--ink);
    background: #fff;
    border: 1px solid #c6ccd4;
    border-radius: 4px;
    max-width: 100%;
}
input[type="number"] {
    width: 104px;
    text-align: center;
    font-variant-numeric: tabular-nums;
}
textarea { width: 100%; min-height: 64px; resize:
vertical; }
input:focus-visible, select:focus-visible,
textarea:focus-visible {
    outline: 2px solid var(--accent);
    outline-offset: 1px;
    border-color: var(--accent);
}
.checkbox {
    display: inline-flex;
    align-items: center;
    gap: 7px;
    cursor: pointer;
    user-select: none;
}
.checkbox input { width: 16px; height: 16px; margin: 0;
cursor: pointer; }
/* ---------- 按钮 ---------- */
.btn {
    display: inline-flex;
    align-items: center;
    justify-content: center;
    gap: 6px;
    padding: 8px 18px;
    font-family: inherit;
    font-size: 14px;
    line-height: 1.4;
    color: #fff;
    background: var(--accent);
```

```
    border: 1px solid transparent;                                    2751
    border-radius: 4px;                                               2752
    cursor: pointer;                                                  2753
    text-decoration: none;                                            2754
}                                                                     2755
.btn:hover { background: var(--accent-hover); }                       2756
.btn:focus-visible { outline: 2px solid var(--accent);                2757
outline-offset: 2px; }                                                2758
.btn--ghost {                                                         2759
    color: var(--ink);                                                2760
    background: #fff;                                                 2761
    border-color: #c6ccd4;                                            2762
}                                                                     2763
.btn--ghost:hover { background: #f1f3f6; }                            2764
.btn--danger {                                                        2765
    color: var(--danger);                                             2766
    background: #fff;                                                 2767
    border-color: #e6b4b0;                                            2768
}                                                                     2769
.btn--danger:hover { background: var(--danger-soft); }                2770
.btn--sm { padding: 4px 12px; font-size: 13px; }                      2771
.btn:disabled { opacity: .5; cursor: not-allowed; }                   2772
.actions { display: flex; flex-wrap: wrap; align-items:               2773
center; gap: 8px; }                                                   2774
.actions form { display: inline; }                                    2775
/* ---------- 提示条 ---------- */                                    2776
.flash {                                                              2777
    padding: 10px 14px;                                               2778
    margin-bottom: 14px;                                              2779
    border-radius: var(--radius);                                     2780
    border-left: 4px solid;                                           2781
}                                                                     2782
.flash.error {                                                        2783
    color: var(--danger);                                             2784
    background: var(--danger-soft);                                   2785
    border-left-color: var(--danger);                                 2786
}                                                                     2787
.flash.ok {                                                           2788
    color: var(--ok);                                                 2789
    background: var(--ok-soft);                                       2790
    border-left-color: var(--ok);                                     2791
}                                                                     2792
/* ---------- 文字工具类 ---------- */                                2793
.muted { color: var(--ink-soft); font-size: 12.5px; }                 2794
.mono { font-family: Consolas, "Courier New",                         2795
monospace; font-variant-numeric: tabular-nums; }                      2796
.num { font-variant-numeric: tabular-nums; }                          2797
.warn { color: var(--danger); font-weight: 600; }                     2798
.ok { color: var(--ok); font-weight: 600; }                           2799
.nowrap { white-space: nowrap; }                                      2800
```

```
.stack { display: flex; flex-direction: column; gap:
14px; }
/* 状态标签：不依赖颜色单独表意 */
.tag {
  display: inline-block;
  padding: 1px 9px;
  font-size: 12.5px;
  border-radius: 10px;
  border: 1px solid;
  white-space: nowrap;
}
.tag--ok { color: var(--ok); background:
var(--ok-soft); border-color: #b8ddc6; }
.tag--warn { color: var(--danger); background:
var(--danger-soft); border-color: #eec4c0; }
.tag--muted { color: var(--ink-soft); background:
#f1f3f6; border-color: #dde1e6; }
/* ---------- 登录页 ---------- */
.login-wrap {
  display: flex;
  justify-content: center;
  padding: 64px 20px;
}
.login-card { width: 100%; max-width: 380px; }
.login-card .field__label { font-size: 14px; color:
var(--ink); }
/* ---------- 首页功能块 ---------- */
.tile-grid {
  display: grid;
  grid-template-columns: repeat(auto-fill,
minmax(232px, 1fr));
  gap: 12px;
}
.tile {
  display: block;
  padding: 16px;
  background: var(--surface);
  border: 1px solid var(--line);
  border-radius: var(--radius);
  color: var(--ink);
  text-decoration: none;
}
.tile:hover { border-color: var(--accent); background:
#fbfcfe; }
.tile:focus-visible { outline: 2px solid var(--accent);
outline-offset: 2px; }
.tile__title { font-weight: 600; color: var(--accent);
margin-bottom: 5px; }
.tile__desc { font-size: 12.5px; color:
var(--ink-soft); text-wrap: pretty; }
```

```
/* ---------- 结果摘要 ---------- */
.stat-row {
  display: flex;
  flex-wrap: wrap;
  gap: 28px;
  margin: 4px 0 6px;
}
.stat__label { font-size: 12.5px; color: var(--ink-soft); }
.stat__value { font-size: 20px; font-weight: 700; font-variant-numeric: tabular-nums; }
/* ---------- 流程步骤条 ---------- */
.steps { display: flex; flex-wrap: wrap; gap: 8px; margin: 4px 0 18px; }
.step {
  display: flex; align-items: center; gap: 7px;
  padding: 8px 12px; font-size: 13px;
  background: #fff; border: 1px solid var(--line);
  border-radius: var(--radius);
}
.step__no {
  display: inline-flex; align-items: center; justify-content: center;
  width: 20px; height: 20px; border-radius: 50%;
  background: #eef1f5; color: var(--ink-soft);
  font-size: 12px; font-variant-numeric: tabular-nums;
}
.step__name { font-weight: 600; }
.step__state { font-size: 12.5px; color: var(--ink-soft); }
.step--done { border-color: #b8ddc6; background: var(--ok-soft); }
.step--done .step__no { background: var(--ok); color: #fff; }
.step--done .step__state { color: var(--ok); }
.step--fail { border-color: #eec4c0; background: var(--danger-soft); }
.step--fail .step__no { background: var(--danger); color: #fff; }
.step--fail .step__state { color: var(--danger); font-weight: 600; }
.step--next { border-color: var(--accent); }
.step--next .step__no { background: var(--accent); color: #fff; }
/* ---------- 提示块 ---------- */
.notice {
  padding: 14px 16px; margin: 0 0 16px;
  border-radius: var(--radius); border-left: 4px solid;
}
.notice--warn {
```

```
        background: var(--danger-soft);                                    2901
        border-left-color: var(--danger);                                  2902
        color: #7a1c17;                                                    2903
      }                                                                    2904
      .notice__title { margin: 0 0 8px; font-weight: 600; }                2905
      .notice__list { margin: 0; padding-left: 20px;                       2906
      line-height: 1.9; }                                                  2907
      .notice__list a { color: var(--accent); }                            2908
      .notice__hint { margin: 8px 0 0; font-size: 12.5px;                  2909
      color: var(--ink-soft); }                                            2910
      /* ---------- 窄屏适配 ---------- */                                  2911
      @media (max-width: 860px) {                                          2912
        .app-body { flex-direction: column; }                              2913
        .app-nav {                                                         2914
          flex: none;                                                      2915
          width: auto;                                                     2916
          border-right: none;                                              2917
          border-bottom: 1px solid var(--line);                            2918
          padding: 8px 10px;                                               2919
        }                                                                  2920
        .app-nav ul { display: flex; flex-wrap: wrap; gap:                 2921
      4px; }                                                               2922
        .app-nav a {                                                       2923
          margin-bottom: 0;                                                2924
          border-left: none;                                               2925
          border-bottom: 3px solid transparent;                            2926
        }                                                                  2927
        .app-nav a[aria-current="page"] { border-left: none;               2928
      border-bottom-color: var(--accent); }                                2929
        .app-nav__group { display: none; }                                 2930
        .app-main { padding: 16px 14px 32px; }                             2931
      }                                                                    2932
    </style>                                                               2933
</head>                                                                    2934
<body>                                                                     2935
    <a class="skip-link" href="#main">跳到主要内容</a>                      2936
    <header class="app-header">                                            2937
      <span class="app-header__title">历史遗留废弃矿山自然恢复适宜性评价系统</spa 2938
      n>                                                                   2939
      <span class="app-header__version">V1.0</span>                        2940
      <span class="app-header__spacer"></span>                             2941
      {% if current_user.is_authenticated %}                               2942
      <span class="app-header__user">                                      2943
        <span>{{ current_user.username }} ({{                              2944
      current_user.role }})</span>                                         2945
        <a href="/logout">退出</a>                                         2946
      </span>                                                              2947
      {% endif %}                                                          2948
    </header>                                                              2949
    <div class="app-body">                                                 2950
```

```
    {% if current_user.is_authenticated %}
    <nav class="app-nav" aria-label="功能导航">
      <ul>
        <li>
          <a href="/"{% if home_active %}
    aria-current="page"{% endif %}>首页</a>
        </li>
      </ul>
      {% for g in nav_groups %}
      <p class="app-nav__group">{{ g.title }}</p>
      <ul>
        {% for label, href, active in g.links %}
        <li>
          <a href="{{ href }}"{% if active %}
    aria-current="page"{% endif %}>{{ label }}</a>
        </li>
        {% endfor %}
      </ul>
      {% endfor %}
    </nav>
    {% endif %}
    <main id="main" class="app-main">
      <div class="container">
        {% with messages = get_flashed_messages(with_categor
    ies=true) %}
          {% for category, message in messages %}
          <p class="flash {{ category }}" role="status">{{
    message }}</p>
          {% endfor %}
        {% endwith %}
        {% block content %}{% endblock %}
      </div>
    </main>
  </div>
</body>
</html>
# ==== app/templates/auth/login.html ====
{% extends "base.html" %}
{% block title %}登录 · 历史遗留废弃矿山自然恢复适宜性评价系统{% endblock %}
{% block content %}
<div class="login-wrap">
  <div class="card login-card">
    <h2>用户登录</h2>
    <form method="post" novalidate>
      {{ form.hidden_tag() }}
      <div class="field">
        <label class="field__label" for="{{
    form.username.id }}">用户名</label>
        {{ form.username(id=form.username.id,
    autocomplete="username") }}
```

```
        </div>                                                                  3001
        <div class="field">                                                     3002
          <label class="field__label" for="{{                                   3003
    form.password.id }}">密码</label>                                           3004
          {{ form.password(id=form.password.id,                                 3005
    autocomplete="current-password") }}                                         3006
        </div>                                                                  3007
        <div class="field">                                                     3008
          <label class="checkbox" for="{{ form.remember.id                      3009
    }}">                                                                        3010
            {{ form.remember(id=form.remember.id) }}                            3011
            <span>记住我（下次自动登录，一个月内有效）</span>                   3012
          </label>                                                              3013
        </div>                                                                  3014
        <p style="margin:18px 0 0;">                                            3015
          {{ form.submit(class_="btn", style="width:100%;") }}                  3016
        </p>                                                                    3017
      </form>                                                                   3018
      <p class="muted" style="margin:14px 0 0;">演示账号：admin /               3019
      admin123</p>                                                              3020
    </div>                                                                      3021
</div>                                                                          3022
{% endblock %}                                                                  3023
# ==== app/templates/auth/home.html ====                                        3024
{% extends "base.html" %}                                                       3025
{% block title %}首页{% endblock %}                                             3026
{% block content %}                                                             3027
<div class="card">                                                              3028
  <h2>系统功能</h2>                                                            3029
  <p class="muted">                                                             3030
    本系统以论文《历史遗留废弃矿山自然恢复适宜性评价指标体系建设及典型案例》  3031
    建立的评价方法为算法内核。下方按论文 4.1 节的评价流程顺序排列，建议依序使用；  3032
    系统会在每一步检查前置条件，前置步骤未完成时不会放行后续计算。           3033
  </p>                                                                          3034
</div>                                                                          3035
<div class="card">                                                              3036
  <h2>评价流程</h2>                                                            3037
  <div class="tile-grid">                                                       3038
    <a class="tile" href="/project/">                                           3039
      <div class="tile__title">项目与矿山档案</div>                             3040
      <div class="tile__desc">建立评价项目、录入矿山图斑、挂接遥感影像、设置参照生态系统<  3041
    /div>                                                                       3042
    </a>                                                                        3043
    <a class="tile" href="/standard/">                                          3044
      <div class="tile__title">政策与标准知识库</div>                           3045
      <div class="tile__desc">收录 11 项地方标准与政策文件，结构化判定规则，支持省际口  3046
    径比对</div>                                                                3047
    </a>                                                                        3048
    <a class="tile" href="/constraint/">                                        3049
      <div class="tile__title">限制性指标判定</div>                             3050
```

```
        <div class="tile__desc">政策、区位约束、土壤污染、地质灾害、耕地属性五项一票否决</
    div>
    </a>
    <a class="tile" href="/unit/">
      <div class="tile__title">微地质单元</div>
      <div class="tile__desc">划分岩质边坡、碎石边坡、平台三类单元，自动计算面积占比</d
    iv>
    </a>
    <a class="tile" href="/ahp/">
      <div class="tile__title">指标体系与权重</div>
      <div class="tile__desc">指标定义维护、判断矩阵录入、权重计算与一致性检验、多专家汇总
    </div>
    </a>
    <a class="tile" href="/indicator/">
      <div class="tile__title">指标数据采集</div>
      <div class="tile__desc">按微地质单元录入土壤、植被、地形指标，支持 CSV /
    Excel 批量导入</div>
    </a>
    <a class="tile" href="/evaluate/">
      <div class="tile__title">适宜性定量评价</div>
      <div class="tile__desc">指标标准化、分单元赋分、面积加权合成、适宜性等级判定</di
    v>
    </a>
    <a class="tile" href="/report/">
      <div class="tile__title">报告与结果输出</div>
      <div class="tile__desc">结果图表、逐层得分明细，导出 PDF 与 Word
    评价报告</div>
    </a>
  </div>
</div>
<div class="card">
  <h2>系统管理</h2>
  <div class="tile-grid">
    <a class="tile" href="/users">
      <div class="tile__title">用户与操作日志</div>
      <div class="tile__desc">用户维护与角色权限；全部写操作自动留痕，可查询追溯</div
    >
    </a>
  </div>
</div>
{% endblock %}
# ==== app/templates/auth/users.html ====
{% extends "base.html" %}
{% block title %}用户与权限管理{% endblock %}
{% block content %}
<div class="card">
  <h2>用户管理</h2>
  <p class="muted">
    系统设有管理员、评价员、查看者三种角色：管理员可维护用户与基础数据，
    评价员可建项目、录数据、执行评价，查看者仅可查阅。
```

```
</p>
<p class="actions" style="margin-bottom:12px;">
  {% if current_user.role == "admin" %}
  <a class="btn btn--sm" href="/users/add">新增用户</a>
  {% endif %}
  <a class="btn btn--sm btn--ghost"
  href="/logs">查看操作日志</a>
</p>
{% if current_user.role != "admin" %}
<p class="muted">
  当前角色为「{{ role_labels.get(current_user.role,
  current_user.role) }}」，
  仅管理员可新增、编辑或删除用户。
</p>
{% endif %}
<table>
  <caption class="muted">共 {{ users|length }}
  个用户</caption>
  <thead>
    <tr>
      <th scope="col">ID</th>
      <th scope="col">用户名</th>
      <th scope="col">角色</th>
      <th scope="col">状态</th>
      {% if current_user.role == "admin" %}<th
  scope="col">操作</th>{% endif %}
    </tr>
  </thead>
  <tbody>
    {% for u in users %}
    <tr>
      <td class="num">{{ u.id }}</td>
      <td>
        {{ u.username }}
        {% if u.id == current_user.id %}<span
  class="muted">（当前登录）</span>{% endif %}
      </td>
      <td>{{ role_labels.get(u.role, u.role) }}</td>
      <td>
        {% if u.active %}<span class="tag
  tag--ok">启用</span>
        {% else %}<span class="tag tag--muted">停用</span>{%
  endif %}
      </td>
      {% if current_user.role == "admin" %}
      <td class="nowrap">
        <a href="/users/{{ u.id }}/edit">编辑</a>
         | 
        <form method="post" action="/users/{{ u.id
  }}/delete"
```

```
                    onsubmit="return confirm('确定删除用户「{{ 3151
      u.username }}」吗？\n该操作无法恢复。');"> 3152
                {{ form.hidden_tag() }} 3153
                <button type="submit" class="btn btn--sm 3154
      btn--danger" 3155
                        {% if u.id == current_user.id %}disabled 3156
                        title="不能删除当前登录的账号"{% endif 3157
      %}>删除</button> 3158
              </form> 3159
            </td> 3160
            {% endif %} 3161
          </tr> 3162
          {% endfor %} 3163
        </tbody> 3164
      </table> 3165
</div> 3166
{% endblock %} 3167
# ==== app/templates/auth/add_user.html ==== 3168
{% extends "base.html" %} 3169
{% block title %}新增用户{% endblock %} 3170
{% block content %} 3171
<div class="card" style="max-width:560px;"> 3172
  <h2>新增用户</h2> 3173
  <form method="post"> 3174
    {{ form.hidden_tag() }} 3175
    <div class="field"> 3176
      <label class="field__label" for="username">用户名</label> 3177
      <input id="username" type="text" name="username" 3178
    required 3179
             autocomplete="off" style="width:280px;"> 3180
    </div> 3181
    <div class="field"> 3182
      <label class="field__label" for="password">密码</label> 3183
      <input id="password" type="password" name="password" 3184
    required 3185
             autocomplete="new-password" 3186
    style="width:280px;"> 3187
    </div> 3188
    <div class="field"> 3189
      <label class="field__label" for="role">角色</label> 3190
      <select id="role" name="role"> 3191
        <option value="viewer">查看者（仅可查阅）</option> 3192
        <option value="evaluator">评价员（可建项目、录数据、执行评价）</option 3193
    > 3194
        <option value="admin">管理员（可维护用户与基础数据）</option> 3195
      </select> 3196
    </div> 3197
    <p class="actions" style="margin-top:18px;"> 3198
      <button type="submit" class="btn">保存</button> 3199
      <a class="btn btn--ghost" href="/users">返回</a> 3200
```

```
      </p>                                                                    3201
    </form>                                                                   3202
</div>                                                                        3203
{% endblock %}                                                                3204
# ==== app/templates/project/index.html ====                                  3205
{% extends "base.html" %}                                                     3206
{% block title %}项目与矿山档案{% endblock %}                                  3207
{% block content %}                                                           3208
<div class="card">                                                            3209
  <h2>评价项目</h2>                                                           3210
  <p class="actions" style="margin-bottom:12px;">                             3211
    <a class="btn btn--sm" href="/project/add">新建项目</a>                   3212
  </p>                                                                        3213
  {% if not projects %}                                                       3214
  <p class="muted">                                                           3215
    暂无项目。点击上方「新建项目」开始建立评价项目，                          3216
    或运行 <span class="mono">python seed_demo.py</span>                      3217
    载入论文案例。                                                            3218
  </p>                                                                        3219
  {% else %}                                                                  3220
  <table>                                                                     3221
    <caption class="muted">共 {{ projects|length }}                           3222
    个项目</caption>                                                          3223
    <thead>                                                                   3224
      <tr>                                                                    3225
        <th scope="col">项目名称</th>                                         3226
        <th scope="col">编号</th>                                             3227
        <th scope="col">状态</th>                                             3228
        <th scope="col">图斑数</th>                                           3229
        <th scope="col">操作</th>                                             3230
      </tr>                                                                   3231
    </thead>                                                                  3232
    <tbody>                                                                   3233
      {% for p in projects %}                                                 3234
      <tr>                                                                    3235
        <td style="text-align:left;">{{ p.name }}</td>                        3236
        <td class="mono">{{ p.code or "—" }}</td>                             3237
        <td><span class="tag tag--muted">{{ p.status                          3238
    }}</span></td>                                                            3239
        <td class="num">{{ p.patches.count() }}</td>                          3240
        <td class="nowrap">                                                   3241
          <a href="/project/{{ p.id }}/patches">矿山图斑</a>                  3242
           | <a href="/project/{{ p.id                              3243
    }}/edit">编辑</a>                                                         3244
           |                                                        3245
          <form method="post" action="/project/{{ p.id                        3246
    }}/delete"                                                                3247
                onsubmit="return confirm('确定删除项目「{{ p.name             3248
    }}」吗？\n该项目下的图斑、微地质单元与评价数据将一并删除，且无法恢复。');">  3249
            {{ form.hidden_tag() }}                                           3250
```

```
                <button type="submit" class="btn btn--sm
    btn--danger">删除</button>
              </form>
            </td>
          </tr>
          {% endfor %}
        </tbody>
      </table>
      {% endif %}
</div>
{% endblock %}
# ==== app/templates/project/edit.html ====
{% extends "base.html" %}
{% block title %}{{ "编辑项目" if project else "新建项目" }}{%
    endblock %}
{% block content %}
<div class="card" style="max-width:640px;">
  <h2>{{ "编辑项目" if project else "新建项目" }}</h2>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="name">项目名称</label>
      <input id="name" type="text" name="name"
    style="width:100%;" required
             value="{{ project.name if project else '' }}">
    </div>
    <div class="field">
      <label class="field__label" for="code">项目编号</label>
      <input id="code" type="text" name="code"
    style="width:240px;"
             value="{{ project.code or '' if project else
    '' }}">
    </div>
    {% if project %}
    <div class="field">
      <label class="field__label" for="status">状态</label>
      <select id="status" name="status">
        {% for s in ["进行中", "已完成", "已归档"] %}
        <option value="{{ s }}" {% if project.status == s
    %}selected{% endif %}>{{ s }}</option>
        {% endfor %}
      </select>
    </div>
    {% endif %}
    <div class="field">
      <label class="field__label" for="description">项目说明</la
    bel>
      <textarea id="description" name="description">{{
    project.description or '' if project else ''
    }}</textarea>
```

```
      </div>                                                                3301
      <p class="actions" style="margin-top:18px;">                          3302
        <button type="submit" class="btn">保存</button>                     3303
        <a class="btn btn--ghost" href="/project/">返回</a>                  3304
      </p>                                                                  3305
    </form>                                                                 3306
</div>                                                                      3307
{% endblock %}                                                              3308
# ==== app/templates/project/patches.html ====                              3309
{% extends "base.html" %}                                                   3310
{% block title %}矿山图斑 · {{ project.name }}{% endblock %}                3311
{% block content %}                                                         3312
<div class="card">                                                          3313
  <h2>矿山图斑 · {{ project.name }}</h2>                                    3314
  <p class="actions" style="margin-bottom:12px;">                           3315
    <a class="btn btn--sm" href="/project/{{ project.id                     3316
    }}/patch/add">新建图斑</a>                                              3317
    <a class="btn btn--sm btn--ghost"                                       3318
    href="/project/">返回项目列表</a>                                       3319
  </p>                                                                      3320
  {% if not project.patches.count() %}                                      3321
  <p class="muted">该项目下暂无图斑。点击上方「新建图斑」建立第一个矿山图斑。</p>          3322
  {% else %}                                                                3323
  <table>                                                                   3324
    <caption class="muted">共 {{ project.patches.count() }}                 3325
    个图斑</caption>                                                        3326
    <thead>                                                                 3327
      <tr>                                                                  3328
        <th scope="col">图斑名称</th>                                       3329
        <th scope="col">行政区</th>                                         3330
        <th scope="col">面积（公顷）</th>                                   3331
        <th scope="col">开采矿种</th>                                       3332
        <th scope="col">微地质单元</th>                                     3333
        <th scope="col">操作</th>                                           3334
      </tr>                                                                 3335
    </thead>                                                                3336
    <tbody>                                                                 3337
      {% for p in project.patches %}                                        3338
      <tr>                                                                  3339
        <td style="text-align:left;">{{ p.name }}</td>                      3340
        <td>{{ p.admin_region or "—" }}</td>                                3341
        <td class="num">{{ p.area_ha if p.area_ha is not                    3342
    none else "—" }}</td>                                                   3343
        <td>{{ p.mineral_type or "—" }}</td>                                3344
        <td class="num">{{ p.units.count() }}</td>                          3345
        <td class="nowrap">                                                 3346
          <a href="/unit/{{ p.id }}">微地质单元</a>                         3347
           | <a href="/constraint/{{ p.id                         3348
    }}">限制性判定</a>                                                     3349
           | <a href="/project/patch/{{ p.id                      3350
```

```
      }}/edit">编辑</a>                                                    3351
               |                                                 3352
              <form method="post" action="/project/patch/{{               3353
      p.id }}/delete"                                                      3354
                    onsubmit="return confirm('确定删除图斑「{{ p.name      3355
      }}」吗？\n该图斑的微地质单元、指标数据与评价结果将一并删除，且无法恢复。');">  3356
                {{ form.hidden_tag() }}                                    3357
                <button type="submit" class="btn btn--sm                   3358
      btn--danger">删除</button>                                           3359
              </form>                                                      3360
            </td>                                                          3361
          </tr>                                                            3362
          {% endfor %}                                                     3363
        </tbody>                                                           3364
    </table>                                                               3365
    {% endif %}                                                            3366
</div>                                                                     3367
{% endblock %}                                                             3368
# ==== app/templates/project/patch_edit.html ====                          3369
{% extends "base.html" %}                                                  3370
{% block title %}{{ "编辑图斑" if patch else "新建图斑" }}{%               3371
      endblock %}                                                          3372
{% block content %}                                                        3373
<div class="card" style="max-width:720px;">                                3374
    <h2>{{ "编辑图斑" if patch else "新建图斑" }} · {{ project.name        3375
      }}</h2>                                                              3376
    <form method="post">                                                   3377
      {{ form.hidden_tag() }}                                              3378
      <div class="field">                                                  3379
        <label class="field__label" for="patch-name">图斑名称</lab3380
      el>                                                                  3381
        <input id="patch-name" type="text" name="name"                     3382
      style="width:100%;" required                                         3383
               value="{{ patch.name if patch else '' }}">                  3384
      </div>                                                               3385
      <div class="field">                                                  3386
        <label class="field__label" for="admin-region">所在行政区</ 3387
      label>                                                               3388
        <input id="admin-region" type="text"                               3389
      name="admin_region" style="width:100%;"                              3390
               value="{{ patch.admin_region or '' if patch                 3391
      else '' }}">                                                         3392
      </div>                                                               3393
      <div class="field">                                                  3394
        <label class="field__label" for="area-ha">图斑面积（公顷）</la3395
      bel>                                                                 3396
        <input id="area-ha" type="number" step="any"                       3397
      name="area_ha"                                                       3398
               value="{{ patch.area_ha if patch and                        3399
      patch.area_ha is not none else '' }}">                               3400
```

```
</div>
<div class="field">
  <label class="field__label" for="mineral-type">开采矿种</label>
  <input id="mineral-type" type="text" name="mineral_type" style="width:240px;"
         value="{{ patch.mineral_type or '' if patch else '' }}">
</div>
<div class="field">
  <label class="field__label" for="mining-method">开采方式</label>
  <input id="mining-method" type="text" name="mining_method" style="width:240px;"
         value="{{ patch.mining_method or '' if patch else '' }}">
</div>
<div class="field">
  <label class="field__label" for="stop-years">停产年限</label>
  <input id="stop-years" type="number" step="any" name="stop_years"
         value="{{ patch.stop_years if patch and patch.stop_years is not none else '' }}">
</div>
<div class="field">
  <label class="checkbox" for="responsible-missing">
    <input id="responsible-missing" type="checkbox" name="responsible_entity_missing" value="1"
           {% if not patch or patch.responsible_entity_missing %}checked{% endif %}>
    <span>责任主体缺失</span>
  </label>
</div>
<h3>参照生态系统（指标标准化基准）</h3>
<p class="muted">
  论文 4.2.2 节取值：植被覆盖度 75%、Shannon-Wiener 多样性指数 0.7。
  该基准与图斑绑定管理，可按不同参照生态系统分别设定。
</p>
<div class="field">
  <label class="field__label" for="coverage-benchmark">植被覆盖度基准（%）</label>
  <input id="coverage-benchmark" type="number" step="any" name="coverage_benchmark"
         value="{{ reference.coverage_benchmark if reference else 75 }}">
</div>
<div class="field">
  <label class="field__label" for="shannon-benchmark">Shannon-Wiener 多样性指数基准</label>
```

```
        <input id="shannon-benchmark" type="number"
      step="any" name="shannon_benchmark"
                value="{{ reference.shannon_benchmark if
      reference else 0.7 }}">
      </div>
      <div class="field">
        <label class="field__label" for="dominant-species">建群种
      </label>
        <input id="dominant-species" type="text"
      name="dominant_species" style="width:100%;"
                value="{{ reference.dominant_species or '' if
      reference else '' }}">
      </div>
      <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/project/{{
      project.id }}/patches">返回</a>
      </p>
    </form>
</div>
{% if patch %}
<div class="card" style="max-width:720px;">
    <h2>遥感影像挂接</h2>
    <p class="muted">
      挂接遥感底图或无人机正射影像，用于辅助微地质单元划分与植被覆盖度判读。
      支持 PNG、JPG、GIF、WebP、BMP、TIFF，单个文件不超过 32 MB。
    </p>
    {% if patch.image_path %}
    <figure style="margin:0 0 16px;">
      <img src="{{ url_for('static', filename=patch.image_path
      ) }}"
           alt="图斑「{{ patch.name }}」已挂接的遥感影像"
           style="max-width:100%;border:1px solid
      var(--line);border-radius:6px;">
      <figcaption class="muted" style="margin-top:6px;">
        {{ patch.image_note or "遥感影像" }}
        {% if patch.image_uploaded_at %}
        · 上传于 {{ patch.image_uploaded_at.strftime("%Y-%m-%d
      %H:%M") }}
        {% endif %}
      </figcaption>
    </figure>
    <form method="post" action="/project/patch/{{ patch.id
      }}/image/delete"
          onsubmit="return confirm('确定移除该遥感影像吗？');">
      {{ form.hidden_tag() }}
      <button type="submit" class="btn btn--sm
      btn--danger">移除影像</button>
    </form>
    <hr style="margin:16px 0;border:none;border-top:1px solid
```

```
      var(--line);">                                                    3501
    {% endif %}                                                         3502
    <form method="post" action="/project/patch/{{ patch.id              3503
      }}/image"                                                         3504
          enctype="multipart/form-data">                                3505
      {{ form.hidden_tag() }}                                           3506
      <div class="field">                                               3507
        <label class="field__label" for="image-file">                   3508
          {{ "替换影像文件" if patch.image_path else "选择影像文件" }}      3509
        </label>                                                        3510
        <input id="image-file" type="file" name="image"                 3511
               accept=".png,.jpg,.jpeg,.gif,.webp,.bmp,.tif,.t          3512
      iff" required>                                                    3513
      </div>                                                            3514
      <div class="field">                                               3515
        <label class="field__label" for="image-note">影像说明（如来源、   3516
      时相）</label>                                                     3517
        <input id="image-note" type="text" name="image_note"            3518
      style="width:100%;"                                               3519
               placeholder="如：2025 年 6 月无人机正射影像"               3520
               value="{{ patch.image_note or '' }}">                    3521
      </div>                                                            3522
      <p class="actions">                                               3523
        <button type="submit" class="btn">{{ "替换影像" if              3524
      patch.image_path else "上传并挂接" }}</button>                    3525
      </p>                                                              3526
    </form>                                                             3527
</div>                                                                  3528
{% endif %}                                                             3529
{% endblock %}                                                          3530
# ==== app/templates/unit/index.html ====                               3531
{% extends "base.html" %}                                               3532
{% block title %}微地质单元 · {{ patch.name }}{% endblock %}            3533
{% block content %}                                                     3534
<div class="card">                                                      3535
    <h2>微地质单元 · {{ patch.name }}</h2>                              3536
    <p class="muted">                                                   3537
      按论文 4.1 节将矿山图斑划分为岩质边坡、碎石边坡、平台三类微地质单元。   3538
      面积占比由面积自动计算，即论文式（9） 中的权重系数。               3539
    </p>                                                               3540
    {% if units %}                                                     3541
    <table>                                                            3542
      <caption class="muted">                                          3543
        共 {{ units|length }} 个单元；                                  3544
        面积合计 <span class="num">{{ "%.4f"|format(total_area)         3545
      }}</span>；                                                       3546
        占比合计 <span class="num">{{ "%.2f%%"|format(ratio_sum *       3547
      100) }}</span>                                                    3548
        {% if ratio_sum and (ratio_sum < 0.999 or ratio_sum >           3549
      1.001) %}                                                         3550
```

```
      <span class="warn">（占比合计不等于 100%，请检查面积录入）</span>
      {% endif %}
    </caption>
    <thead>
      <tr>
        <th scope="col">微地质单元</th>
        <th scope="col">面积</th>
        <th scope="col">面积占比</th>
        <th scope="col">说明</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for u in units %}
      <tr>
        <td>{{ u.unit_type }}</td>
        <td class="num">{{ u.area if u.area is not none
    else "—" }}</td>
        <td class="num">
          {% if u.ratio is not none %}{{
    "%.2f%%"|format(u.ratio * 100) }}{% else %}—{% endif %}
        </td>
        <td style="text-align:left;">{{ u.description or
    "—" }}</td>
        <td class="nowrap">
          <a href="/unit/edit/{{ u.id }}">编辑</a>
           | 
          <form method="post" action="/unit/delete/{{ u.id
    }}"
                onsubmit="return confirm('确定删除微地质单元「{{
    u.unit_type }}」吗？\n该单元已录入的指标数据将一并删除，且无法恢复。');">
            {{ form.hidden_tag() }}
            <button type="submit" class="btn btn--sm
    btn--danger">删除</button>
          </form>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% else %}
  <p class="muted">尚未划分微地质单元。请在下方添加。</p>
  {% endif %}
  <h3>添加微地质单元</h3>
  <form method="post" action="/unit/{{ patch.id }}/add">
    {{ form.hidden_tag() }}
    <div class="actions">
      <label class="field__label" for="add-unit-type"
    style="margin:0;">类型</label>
      <select id="add-unit-type" name="unit_type">
```

```
            {% for t in unit_types %}<option value="{{ t }}">{{ t }}</option>{% endfor %}
          </select>
          <label class="field__label" for="add-unit-area" style="margin:0;">面积</label>
          <input id="add-unit-area" type="number" step="any" name="area">
          <label class="field__label" for="add-unit-desc" style="margin:0;">说明</label>
          <input id="add-unit-desc" type="text" name="description" style="width:220px;">
          <button type="submit" class="btn btn--sm">添加</button>
        </div>
      </form>
      <p class="actions" style="margin-top:20px;">
        <a class="btn btn--sm btn--ghost" href="/project/{{ patch.project_id }}/patches">返回图斑列表</a>
        <a class="btn btn--sm btn--ghost" href="/indicator/">去录入指标数据</a>
      </p>
    </div>
    {% endblock %}
    # ==== app/templates/unit/edit.html ====
    {% extends "base.html" %}
    {% block title %}编辑微地质单元{% endblock %}
    {% block content %}
    <div class="card">
      <h2>编辑微地质单元 · {{ patch.name }}</h2>
      <p class="muted">修改面积后，系统会自动重算本图斑各单元的面积占比。</p>
      <form method="post">
        {{ form.hidden_tag() }}
        <div class="field">
          <label class="field__label" for="unit-type">微地质单元类型</label>
          <select id="unit-type" name="unit_type">
            {% for t in unit_types %}
              <option value="{{ t }}" {% if unit.unit_type == t %}selected{% endif %}>{{ t }}</option>
            {% endfor %}
          </select>
        </div>
        <div class="field">
          <label class="field__label" for="unit-area">面积</label>
          <input id="unit-area" type="number" step="any" name="area"
                 value="{{ unit.area if unit.area is not none else '' }}">
        </div>
        <div class="field">
          <label class="field__label" for="unit-desc">说明</label>
```

```
        <input id="unit-desc" type="text" name="description"
      style="width:420px;"
             value="{{ unit.description or '' }}">
      </div>
      <p class="actions" style="margin-top:16px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/unit/{{
      unit.patch_id }}">返回</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/standard/index.html ====
{% extends "base.html" %}
{% block title %}政策与标准知识库{% endblock %}
{% block content %}
<div class="card">
  <h2>政策与标准知识库</h2>
  <p class="muted">
    收录论文表 1 梳理的地方标准与政策文件，并将其判定条件结构化为可执行规则，
    支持同一指标在不同省份的阈值横向比对。
  </p>
  <p class="actions" style="margin-bottom:14px;">
    <a class="btn btn--sm" href="/standard/add">添加标准</a>
    <a class="btn btn--sm btn--ghost"
    href="/standard/compare">省际口径比对</a>
  </p>
  <h3>地方标准</h3>
  {% if local %}
  <table>
    <thead>
      <tr>
        <th scope="col">省份</th>
        <th scope="col">名称</th>
        <th scope="col">编号</th>
        <th scope="col">规则数</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for s in local %}
      <tr>
        <td>{{ s.province }}</td>
        <td style="text-align:left;">{{ s.name }}</td>
        <td class="mono">{{ s.doc_number or "—" }}</td>
        <td class="num">{{ s.rules|length }}</td>
        <td class="nowrap">
          <a href="/standard/{{ s.id }}/rules">规则</a>
           | <a href="/standard/{{ s.id
    }}/edit">编辑</a>
```

```
           |                                                     3701
          <form method="post" action="/standard/{{ s.id                   3702
    }}/delete"                                                            3703
                onsubmit="return confirm('确定删除「{{                    3704
    s.province }} {{ s.name }}」吗？\n该标准下的 {{ s.rules|length        3705
    }} 条规则将一并删除，且无法恢复。');">                                3706
            {{ form.hidden_tag() }}                                       3707
            <button type="submit" class="btn btn--sm                      3708
    btn--danger">删除</button>                                            3709
          </form>                                                         3710
        </td>                                                             3711
      </tr>                                                               3712
      {% endfor %}                                                        3713
    </tbody>                                                              3714
</table>                                                                  3715
{% else %}                                                                3716
<p class="muted">暂无地方标准。运行 <span class="mono">python            3717
    seed_demo.py</span> 可载入论文表 1 的 11 项标准与政策文件。</p>       3718
{% endif %}                                                               3719
<h3>政策文件</h3>                                                         3720
{% if policy %}                                                           3721
<table>                                                                   3722
    <thead>                                                               3723
      <tr>                                                                3724
        <th scope="col">省份</th>                                         3725
        <th scope="col">名称</th>                                         3726
        <th scope="col">文号</th>                                         3727
        <th scope="col">规则数</th>                                       3728
        <th scope="col">操作</th>                                         3729
      </tr>                                                               3730
    </thead>                                                              3731
    <tbody>                                                               3732
      {% for s in policy %}                                               3733
      <tr>                                                                3734
        <td>{{ s.province }}</td>                                         3735
        <td style="text-align:left;">{{ s.name }}</td>                    3736
        <td class="mono">{{ s.doc_number or "—" }}</td>                   3737
        <td class="num">{{ s.rules|length }}</td>                         3738
        <td class="nowrap">                                               3739
          <a href="/standard/{{ s.id }}/rules">规则</a>                   3740
           | <a href="/standard/{{ s.id                         3741
    }}/edit">编辑</a>                                                     3742
           |                                                    3743
          <form method="post" action="/standard/{{ s.id                   3744
    }}/delete"                                                            3745
                onsubmit="return confirm('确定删除「{{                    3746
    s.province }} {{ s.name }}」吗？\n该文件下的 {{ s.rules|length        3747
    }} 条规则将一并删除，且无法恢复。');">                                3748
            {{ form.hidden_tag() }}                                       3749
            <button type="submit" class="btn btn--sm                      3750
```

```
      btn--danger">删除</button>                                    3751
            </form>                                                 3752
         </td>                                                      3753
      </tr>                                                         3754
      {% endfor %}                                                  3755
      </tbody>                                                      3756
   </table>                                                         3757
   {% else %}                                                       3758
   <p class="muted">暂无政策文件。</p>                               3759
   {% endif %}                                                      3760
</div>                                                              3761
{% endblock %}                                                      3762
# ==== app/templates/standard/edit.html ====                        3763
{% extends "base.html" %}                                           3764
{% block title %}{{ "编辑标准" if standard else "添加标准" }}{%      3765
   endblock %}                                                      3766
{% block content %}                                                 3767
<div class="card" style="max-width:800px;">                         3768
   <h2>{{ "编辑标准" if standard else "添加标准" }}</h2>             3769
   <form method="post">                                             3770
      {{ form.hidden_tag() }}                                       3771
      <div class="field">                                           3772
         <label class="field__label" for="province">省份</label>    3773
         <input id="province" type="text" name="province"           3774
      style="width:160px;" required                                 3775
             value="{{ standard.province if standard else           3776
      '' }}">                                                       3777
      </div>                                                        3778
      <div class="field">                                           3779
         <label class="field__label" for="std-name">名称</label>    3780
         <input id="std-name" type="text" name="name"               3781
      style="width:100%;" required                                  3782
             value="{{ standard.name if standard else ''            3783
      }}">                                                          3784
      </div>                                                        3785
      <div class="field">                                           3786
         <label class="field__label" for="doc-number">标准编号 /    3787
      文号</label>                                                  3788
         <input id="doc-number" type="text" name="doc_number"       3789
      style="width:320px;"                                          3790
             value="{{ standard.doc_number or '' if                 3791
      standard else '' }}">                                         3792
      </div>                                                        3793
      <div class="field">                                           3794
         <label class="field__label" for="category">类别</label>    3795
         <select id="category" name="category">                     3796
            {% for c in ["地方标准", "政策文件"] %}                 3797
            <option value="{{ c }}" {% if standard and              3798
      standard.category == c %}selected{% endif %}>{{ c             3799
      }}</option>                                                   3800
```

```
        {% endfor %}
      </select>
    </div>
    <div class="field">
      <label class="field__label" for="publish-date">发布时间</label>
      <input id="publish-date" type="text"
    name="publish_date" style="width:200px;"
             placeholder="如 2023-03-30"
             value="{{ standard.publish_date or '' if
    standard else '' }}">
    </div>
    <div class="field">
      <label class="field__label" for="implement-date">实施时间</
    label>
      <input id="implement-date" type="text"
    name="implement_date" style="width:200px;"
             placeholder="如 2023-04-30（未明确可留空）"
             value="{{ standard.implement_date or '' if
    standard else '' }}">
    </div>
    <div class="field">
      <label class="field__label" for="summary">特点 /
    判定标准</label>
      <textarea id="summary" name="summary"
    style="min-height:96px;">{{ standard.summary or '' if
    standard else '' }}</textarea>
    </div>
    <p class="actions" style="margin-top:18px;">
      <button type="submit" class="btn">保存</button>
      {% if standard %}
      <form method="post" action="/standard/{{ standard.id
    }}/delete" style="display:inline;"
            onsubmit="return confirm('确定删除「{{
    standard.province }} {{ standard.name
    }}」吗？\n该标准下的规则将一并删除，且无法恢复。');">
        {{ form.hidden_tag() }}
        <button type="submit" class="btn
    btn--danger">删除该标准</button>
      </form>
      {% endif %}
      <a class="btn btn--ghost" href="/standard/">返回</a>
    </p>
  </form>
</div>
{% endblock %}
#  ==== app/templates/standard/rules.html ====
{% extends "base.html" %}
{% block title %}结构化规则 · {{ standard.province }}{% endblock
    %}
```

```
{% block content %}
<div class="card">
  <h2>结构化规则 · {{ standard.province }} {{ standard.name }}</h2>
  <p class="muted">
    将标准条文拆解为「判定项 + 比较符 + 阈值 + 条件类型」的可执行规则，
    使同一指标在各省的阈值差异可自动比对。
  </p>
  {% if rules %}
  <table>
    <caption class="muted">共 {{ rules|length }} 条规则</caption>
    <thead>
      <tr>
        <th scope="col">判定项</th>
        <th scope="col">条件</th>
        <th scope="col">条件类型</th>
        <th scope="col">适用分区</th>
        <th scope="col">备注</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for r in rules %}
      <tr>
        <td style="text-align:left;">{{ r.item }}</td>
        <td class="mono nowrap">{{ r.operator }} {{ r.threshold or "" }}</td>
        <td>{{ r.condition_type }}</td>
        <td>{{ r.region or "—" }}</td>
        <td style="text-align:left;">{{ r.note or "—" }}</td>
        <td>
          <form method="post" action="/standard/rule/{{ r.id }}/delete"
                onsubmit="return confirm('确定删除规则「{{ r.item }} {{ r.operator }} {{ r.threshold or '' }}」吗？');">
            {{ form.hidden_tag() }}
            <button type="submit" class="btn btn--sm btn--danger">删除</button>
          </form>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% else %}
  <p class="muted">该标准暂无规则。请在下方添加第一条规则。</p>
  {% endif %}

  <h3>添加规则</h3>
```

```
<form method="post">
  {{ form.hidden_tag() }}
  <div class="actions">
    <div>
      <label class="field__label" for="r-item">判定项</label>
      <input id="r-item" type="text" name="item"
  placeholder="如：植被覆盖度" required>
    </div>
    <div>
      <label class="field__label" for="r-op">比较符</label>
      <select id="r-op" name="operator">
        {% for o in [">=", ">", "<=", "<", "=="]
  %}<option value="{{ o }}">{{ o }}</option>{% endfor %}
      </select>
    </div>
    <div>
      <label class="field__label" for="r-thr">阈值</label>
      <input id="r-thr" type="text" name="threshold"
  placeholder="如：50" style="width:110px;">
    </div>
    <div>
      <label class="field__label" for="r-cond">条件类型</label
  >
      <select id="r-cond" name="condition_type">
        <option value="必要条件">必要条件</option>
        <option value="充分条件">充分条件</option>
      </select>
    </div>
    <div>
      <label class="field__label" for="r-region">适用分区</lab
  el>
      <input id="r-region" type="text" name="region"
  placeholder="如宁夏分区" style="width:130px;">
    </div>
    <div>
      <label class="field__label" for="r-note">备注</label>
      <input id="r-note" type="text" name="note"
  style="width:220px;">
    </div>
    <div>
      <button type="submit" class="btn">添加规则</button>
    </div>
  </div>
</form>
<p class="actions" style="margin-top:20px;">
  <a class="btn btn--sm btn--ghost"
  href="/standard/">返回标准列表</a>
  <a class="btn btn--sm btn--ghost"
  href="/standard/compare">省际口径比对</a>
</p>
```

```
</div>
{% endblock %}
# ==== app/templates/standard/compare.html ====
{% extends "base.html" %}
{% block title %}省际口径比对{% endblock %}
{% block content %}
<div class="card">
  <h2>省际口径比对</h2>
  <p class="muted">
    同一判定项在不同省份的阈值横向对比，避免人工翻阅多份标准文件。
  </p>
  {% if not groups %}
  <p class="muted">
    暂无规则数据。请先在「标准知识库」中为标准录入规则，
    或运行 <span class="mono">python seed_demo.py</span> 载入论文表 1 数据。
  </p>
  {% else %}
  {% for item, rs in groups.items() %}
  <h3>{{ item }}</h3>
  <table>
    <thead>
      <tr>
        <th scope="col">省份</th>
        <th scope="col">标准 / 文件</th>
        <th scope="col">条件</th>
        <th scope="col">条件类型</th>
        <th scope="col">备注</th>
      </tr>
    </thead>
    <tbody>
      {% for r in rs %}
      <tr>
        <td>{{ r.standard.province }}</td>
        <td style="text-align:left;">{{ r.standard.name
    }}</td>
        <td class="mono nowrap">{{ r.operator }} {{
    r.threshold or "" }}</td>
        <td>{{ r.condition_type }}</td>
        <td style="text-align:left;">{{ r.note or "—"
    }}</td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% endfor %}
  {% endif %}
  <p class="actions" style="margin-top:16px;">
    <a class="btn btn--ghost" href="/standard/">返回标准列表</a>
  </p>
```

```
</div>
{% endblock %}
# ==== app/templates/constraint/index.html ====
{% extends "base.html" %}
{% block title %}限制性指标判定{% endblock %}
{% block content %}
<div class="card">
  <h2>限制性指标判定</h2>
  <p class="muted">
    按论文表 2 与 4.1 节流程，对每项限制性条件逐项判定。
    任一项不通过即判定为「人工辅助或生态重建」，不再进入定量评价。
  </p>
  {% if not patches %}
  <p class="muted">暂无矿山图斑。请先在「项目与档案」中建立图斑。</p>
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">矿山图斑</th>
        <th scope="col">判定结果</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for p in patches %}
      {% set r = latest.get(p.id) %}
      <tr>
        <td style="text-align:left;">{{ p.name }}</td>
        <td style="text-align:left;">
          {% if r %}
            {% if r.passed %}
            <span class="tag tag--ok">通过</span>
            {% else %}
            <span class="tag tag--warn">不通过</span>
            {% endif %}
            <span class="muted">{{ r.reason }}</span>
          {% else %}
          <span class="tag tag--muted">未判定</span>
          {% endif %}
        </td>
        <td><a href="/constraint/{{ p.id }}">执行判定</a></td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% endif %}
</div>
{% endblock %}
# ==== app/templates/constraint/check.html ====
{% extends "base.html" %}
```

```
{% block title %}限制性条件判定 · {{ patch.name }}{% endblock %}
{% block content %}
<div class="card" style="max-width:760px;">
  <h2>限制性条件判定 · {{ patch.name }}</h2>
  <p class="muted">
    五项条件全部通过方可进入定量评价；任一项不通过即判定为「人工辅助或生态重建」，
    不再进行后续计算。
  </p>
  <form method="post">
    {{ form.hidden_tag() }}
    {% for field, name, val in field_states %}
    <fieldset style="border:1px solid
    var(--line);border-radius:6px;padding:10px
    14px;margin:0 0 10px;">
      <legend style="padding:0 6px;font-weight:600;">{{
    name }}</legend>
      <div class="actions">
        <label class="checkbox">
          <input type="radio" name="{{ field }}" value="1"
                 {% if val is none or val %}checked{% endif
    %}>
          <span>通过</span>
        </label>
        <label class="checkbox">
          <input type="radio" name="{{ field }}" value="0"
                 {% if val is not none and not val
    %}checked{% endif %}>
          <span>不通过</span>
        </label>
      </div>
    </fieldset>
    {% endfor %}
    <p class="actions" style="margin-top:16px;">
      <button type="submit" class="btn">保存判定</button>
      <a class="btn btn--ghost" href="/constraint/">返回</a>
    </p>
  </form>
  {% if record %}
  <div style="margin-top:18px;padding:14px;border-radius:6px
    ;border-left:4px solid;
                {% if record.passed %}background:var(--ok-soft
    );border-left-color:var(--ok);
                {% else %}background:var(--danger-soft);border
    -left-color:var(--danger);{% endif %}">
    <p style="margin:0 0 4px;font-weight:600;">
      判定结论：
      {% if record.passed %}
      <span class="ok">通过，可进入定量评价</span>
      {% else %}
      <span class="warn">不通过 —— 建议生态重建或人工辅助</span>
```

```
        {% endif %}
      </p>
      <p class="muted" style="margin:0;">{{ record.reason }}</p>
    </div>
    {% endif %}
</div>
{% endblock %}
# ==== app/templates/ahp/index.html ====
{% extends "base.html" %}
{% block title %}AHP 权重计算{% endblock %}
{% block content %}
<div class="card" style="max-width:720px;">
    <h2>AHP 权重计算 · 选择指标层级</h2>
    <p class="muted">
      选择微地质单元类型与指标层级后，系统按论文表 3 确定判断矩阵的规模。
      岩质边坡不设土壤指标、平台不设地形指标，系统会相应调整矩阵所包含的指标。
    </p>
    <form method="post">
      {{ form.hidden_tag() }}
      <div class="field">
        <label class="field__label" for="{{ form.unit_type.id }}">微地质单元类型</label>
        {{ form.unit_type(id=form.unit_type.id) }}
      </div>
      <div class="field">
        <label class="field__label" for="{{ form.level.id }}">指标层级</label>
        {{ form.level(id=form.level.id) }}
      </div>
      <div class="field">
        <label class="field__label" for="{{ form.l1.id }}">
          一级指标（层级选「二级指标」时需选择）
        </label>
        {{ form.l1(id=form.l1.id) }}
      </div>
      <p class="actions" style="margin-top:16px;">
        {{ form.submit(class_="btn") }}
        <a class="btn btn--ghost" href="/ahp/indicators">指标定义维护</a>
        <a class="btn btn--ghost" href="/ahp/experts">多专家矩阵</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/ahp/matrix.html ====
{% extends "base.html" %}
{% block title %}判断矩阵录入{% endblock %}
{% block content %}
```

```
<div class="card">
  <h2>
    判断矩阵录入
    · {{ unit_type }}
    · {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{% endif %}
  </h2>
  <p class="muted">
    采用 1–9 标度：1 同等重要、3 稍微重要、5 明显重要、7 强烈重要、9 极端重要，
    2、4、6、8 为相邻判断的中间值。只需录入上三角，对角线为 1，下三角自动取倒数。
    矩阵已按论文表 3 的权重预填。
  </p>
  <form method="post">
    {{ form.hidden_tag() }}
    <table style="width:auto;">
      <thead>
        <tr>
          <th></th>
          {% for name in names %}<th scope="col">{{ name }}</th>{% endfor %}
        </tr>
      </thead>
      <tbody>
        {% for i in range(names|length) %}
        <tr>
          <th scope="row">{{ names[i] }}</th>
          {% for j in range(names|length) %}
            {% if i == j %}
            <td class="muted num">1</td>
            {% elif i < j %}
            <td>
              <label class="muted" style="position:absolute;left:-9999px;"
                     for="m-{{ i }}-{{ j }}">{{ names[i] }} 相对 {{ names[j] }} 的重要性</label>
              <input id="m-{{ i }}-{{ j }}" type="number" step="0.01" min="0.01"
                     name="m_{{ i }}_{{ j }}" value="{{ default[i][j] }}">
            </td>
            {% else %}
            <td class="muted num">{{ default[i][j] }}</td>
            {% endif %}
          {% endfor %}
        </tr>
        {% endfor %}
      </tbody>
    </table>
    <p class="actions">
      <button type="submit" class="btn">计算权重</button>
```

```
      <a class="btn btn--ghost" href="/ahp/">重新选择</a>
    </p>
  </form>
  {% if result %}
  <hr style="margin:22px 0;border:none;border-top:1px solid
    var(--line);">
  <h3>计算结果</h3>
  <table style="width:auto;">
    <thead>
      <tr>
        <th scope="col">指标</th>
        {% for name in names %}<th scope="col">{{ name
    }}</th>{% endfor %}
      </tr>
    </thead>
    <tbody>
      <tr>
        <th scope="row">权重</th>
        {% for w in result.weights %}<td class="mono">{{ w
    }}</td>{% endfor %}
      </tr>
    </tbody>
  </table>
  <div class="stat-row">
    <div>
      <div class="stat__label">最大特征值 λmax</div>
      <div class="stat__value mono">{{ result.lambda_max
    }}</div>
    </div>
    <div>
      <div class="stat__label">一致性指标 CI</div>
      <div class="stat__value mono">{{
    "%.4f"|format(result.ci) }}</div>
    </div>
    <div>
      <div class="stat__label">一致性比例 CR</div>
      <div class="stat__value mono">{{
    "%.4f"|format(result.cr) }}</div>
    </div>
    <div>
      <div class="stat__label">一致性检验（要求 CR &lt; 0.1）</div>
      <div class="stat__value">
        {% if result.consistent %}<span class="ok">通过</span>
        {% else %}<span class="warn">不通过</span>{% endif %}
      </div>
    </div>
  </div>
  {% if not result.consistent and result.most_inconsistent
    %}
  <p style="margin-top:14px;padding:12px;border-radius:6px;b
```

```
      ackground:var(--danger-soft);                                          4251
                border-left:4px solid var(--danger);">                       4252
      <strong class="warn">建议优先调整：</strong>                             4253
      第 {{ result.most_inconsistent.i + 1 }} 行、第 {{                      4254
      result.most_inconsistent.j + 1 }} 列的比较                             4255
      （当前 <span class="mono num">{{ result.most_inconsistent.v            4256
      alue|round(3) }}</span>，                                              4257
      理想约 <span class="mono num">{{ result.most_inconsistent.e            4258
      xpected|round(3) }}</span>）                                           4259
    </p>                                                                     4260
    {% endif %}                                                              4261
    {% endif %}                                                              4262
</div>                                                                       4263
{% endblock %}                                                               4264
# ==== app/templates/indicator/index.html ====                               4265
{% extends "base.html" %}                                                    4266
{% block title %}指标数据采集{% endblock %}                                  4267
{% block content %}                                                          4268
<div class="card">                                                           4269
    <h2>指标数据采集与管理</h2>                                             4270
    <p class="muted">                                                        4271
      按微地质单元组织录入。系统按单元类型自动决定应录入的指标：            4272
      岩质边坡不启用土壤指标、平台不启用地形指标。                          4273
    </p>                                                                     4274
    <p class="actions" style="margin-bottom:12px;">                          4275
      <a class="btn btn--sm" href="/indicator/import">批量导入（CSV          4276
      / Excel）</a>                                                          4277
    </p>                                                                     4278
    {% if not patches %}                                                     4279
    <p class="muted">                                                        4280
      暂无矿山图斑。请先在「项目与档案」中建立图斑，                        4281
      或运行 <span class="mono">python seed_demo.py</span>                   4282
      载入论文案例。                                                         4283
    </p>                                                                     4284
    {% else %}                                                               4285
    {% for p in patches %}                                                   4286
    <h3>{{ p.name }}</h3>                                                    4287
    {% if not p.units.count() %}                                             4288
    <p class="muted">该图斑尚未划分微地质单元，请先到「微地质单元」页面划分。</p>  4289
    {% else %}                                                               4290
    <table>                                                                  4291
      <thead>                                                                4292
        <tr>                                                                 4293
          <th scope="col">微地质单元</th>                                    4294
          <th scope="col">面积占比</th>                                      4295
          <th scope="col">数据状态</th>                                      4296
          <th scope="col">操作</th>                                          4297
        </tr>                                                                4298
      </thead>                                                               4299
      <tbody>                                                                4300
```

```
        {% for u in p.units %}                                                  4301
        <tr>                                                                    4302
          <td>{{ u.unit_type }}</td>                                            4303
          <td class="num">                                                      4304
            {% if u.ratio is not none %}{{                                      4305
      "%.2f%%"|format(u.ratio * 100) }}{% else %}—{% endif %}                   4306
          </td>                                                                 4307
          <td>                                                                  4308
            {% if u.indicator_data %}                                           4309
            <span class="tag tag--ok">已录入</span>                             4310
            {% else %}                                                          4311
            <span class="tag tag--muted">未录入</span>                          4312
            {% endif %}                                                         4313
          </td>                                                                 4314
          <td><a href="/indicator/{{ u.id }}">录入 / 编辑</a></td>              4315
        </tr>                                                                   4316
        {% endfor %}                                                            4317
      </tbody>                                                                  4318
    </table>                                                                    4319
    {% endif %}                                                                 4320
    {% endfor %}                                                                4321
    {% endif %}                                                                 4322
</div>                                                                          4323
{% endblock %}                                                                  4324
# ==== app/templates/indicator/edit.html ====                                   4325
{% extends "base.html" %}                                                       4326
{% block title %}指标数据录入 · {{ unit.unit_type }}{% endblock %}              4327
{% block content %}                                                             4328
<div class="card" style="max-width:820px;">                                     4329
  <h2>指标数据录入 · {{ unit.unit_type }}</h2>                                  4330
  <p class="muted">                                                             4331
    输入框留空表示该指标「不适用」，与「实测值为 0」含义不同，计算时会被跳过。  4332
    岩质边坡不设土壤指标、平台不设地形指标，系统已按单元类型隐藏不相关字段。    4333
  </p>                                                                          4334
  <form method="post">                                                          4335
    {{ form.hidden_tag() }}                                                     4336
    {% for group_name, fields, has_hm in groups %}                              4337
    <h3>{{ group_name }}</h3>                                                   4338
    {% for fname, cname, unit_ in fields %}                                     4339
    <div class="field">                                                         4340
      <label class="field__label" for="f-{{ fname }}">                          4341
        {{ cname }}{% if unit_ %} ({{ unit_ }}) {% endif %}                     4342
      </label>                                                                  4343
      {% if fname == "aspect" %}                                                4344
      <select id="f-{{ fname }}" name="aspect">                                 4345
        <option value="">— 不适用 —</option>                                    4346
        {% for a in aspect_types %}                                             4347
        <option value="{{ a }}" {% if current.get("aspect")                     4348
    == a %}selected{% endif %}>{{ a }}</option>                                 4349
        {% endfor %}                                                            4350
```

```
        </select>
        {% else %}
        <input id="f-{{ fname }}" type="number" step="any"
    name="{{ fname }}"
               value="{{ current.get(fname) if
    current.get(fname) is not none else '' }}">
        {% endif %}
      </div>
      {% endfor %}
      {% endfor %}
      {% for group_name, fields, has_hm in groups %}
        {% if has_hm %}
        <h3>重金属含量（mg/kg）</h3>
        <p class="muted">未检测的金属可留空，系统按论文注释视为未受污染处理。</p>
        <table style="width:auto;">
          <thead>
            <tr>
              {% for m in heavy_metals %}<th scope="col">{{ m
    }}</th>{% endfor %}
            </tr>
          </thead>
          <tbody>
            <tr>
              {% for m in heavy_metals %}
              <td>
                <label class="muted" style="position:absolute;
    left:-9999px;"
                       for="hm-{{ m }}">{{ m }} 含量</label>
                <input id="hm-{{ m }}" type="number"
    step="any" name="hm_{{ m }}"
                       value="{{ current.get('heavy',
    {}).get(m) if current.get('heavy', {}).get(m) is not
    none else '' }}">
              </td>
              {% endfor %}
            </tr>
          </tbody>
        </table>
        {% endif %}
      {% endfor %}
      <p class="actions" style="margin-top:18px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/indicator/">返回</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/evaluate/index.html ====
{% extends "base.html" %}
{% block title %}适宜性定量评价{% endblock %}
```

```
{% block content %}
<div class="card">
  <h2>适宜性定量评价</h2>
  <p class="muted">
    系统按论文式（８） 对各项指标做标准化，按式（９） 分单元赋分并按面积占比加权合成，
    最后按论文表 4 判定适宜性等级。未满足前置条件的图斑不能执行评价，
    下表会列出所缺步骤。
  </p>
  {% if not patches %}
  <p class="muted">暂无矿山图斑。请先到「项目与矿山档案」建立图斑。</p>
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">矿山图斑</th>
        <th scope="col">前置条件</th>
        <th scope="col">最近评价结果</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for p in patches %}
      {% set pr = progress.get(p.id) %}
      <tr>
        <td style="text-align:left;">{{ p.name }}</td>
        <td style="text-align:left;">
          {% if pr and pr.ready %}
          <span class="tag tag--ok">已具备</span>
          {% elif pr %}
          <span class="tag tag--warn">缺：{{
    pr.blocking[0].label }}</span>
          <span class="muted">{{ pr.blocking[0].reason
    }}</span>
          {% endif %}
        </td>
        <td>
          {% if latest.get(p.id) %}
          F = <span class="mono num">{{
    latest[p.id].final_score }}</span>，
          <span class="tag tag--ok">{{ latest[p.id].grade
    }}</span>
          {% else %}
          <span class="tag tag--muted">未评价</span>
          {% endif %}
        </td>
        <td class="nowrap">
          <a href="/evaluate/{{ p.id }}">执行评价</a>
          {% if latest.get(p.id) %}
           | <a href="/evaluate/result/{{
    latest[p.id].id }}">查看结果</a>
```

```
            {% endif %}                                                    4451
          </td>                                                            4452
        </tr>                                                              4453
        {% endfor %}                                                       4454
      </tbody>                                                             4455
    </table>                                                               4456
    {% endif %}                                                            4457
</div>                                                                     4458
{% endblock %}                                                             4459
# ==== app/templates/evaluate/run.html ====                                4460
{% extends "base.html" %}                                                  4461
{% import "_macros.html" as ui %}                                          4462
{% block title %}执行评价 · {{ patch.name }}{% endblock %}                 4463
{% block content %}                                                        4464
<div class="card" style="max-width:880px;">                                4465
  <h2>执行适宜性评价 · {{ patch.name }}</h2>                               4466
  <p class="muted">                                                        4467
    系统按论文式（8） 对各项指标做标准化，按式（9） 分单元赋分并按面积占比加权合成，  4468
    最后按论文表 4 判定适宜性等级。评价结果会保存，可随时回看。           4469
  </p>                                                                     4470
  {{ ui.workflow_steps(progress) }}                                        4471
  {{ ui.workflow_blocking(progress) }}                                     4472
  {% if progress.ready %}                                                  4473
  <form method="post">                                                     4474
    {{ form.hidden_tag() }}                                                4475
    <p class="actions">                                                    4476
      <button type="submit" class="btn">开始评价</button>                  4477
      <a class="btn btn--ghost" href="/evaluate/">返回</a>                 4478
    </p>                                                                   4479
  </form>                                                                  4480
  {% else %}                                                               4481
  <p class="actions">                                                      4482
    <button type="button" class="btn" disabled>开始评价</button>           4483
    <a class="btn btn--ghost" href="/evaluate/">返回</a>                   4484
  </p>                                                                     4485
  <p class="muted">前置步骤全部完成后，该按钮才可用。</p>                  4486
  {% endif %}                                                              4487
</div>                                                                     4488
<div class="card" style="max-width:880px;">                                4489
  <h2>当前流程状态</h2>                                                   4490
  <table>                                                                  4491
    <thead>                                                                4492
      <tr><th scope="col">步骤</th><th scope="col">状态</th><th          4493
    scope="col">说明</th></tr>                                             4494
    </thead>                                                               4495
    <tbody>                                                                4496
      <tr>                                                                 4497
        <td>划分微地质单元</td>                                            4498
        <td>                                                               4499
          {% if progress.units.done %}<span class="tag                     4500
```

```
      tag--ok">已完成</span>                                                4501
            {% else %}<span class="tag tag--muted">待完成</span>{ 4502
      % endif %}                                                            4503
        </td>                                                               4504
        <td style="text-align:left;">{{                                    4505
      progress.units.detail }}</td>                                         4506
      </tr>                                                                 4507
      <tr>                                                                  4508
        <td>限制性条件判定</td>                                             4509
        <td>                                                               4510
          {% if progress.constraint.passed %}<span                          4511
      class="tag tag--ok">已通过</span>                                     4512
          {% elif progress.constraint.done %}<span                          4513
      class="tag tag--warn">未通过</span>                                   4514
          {% else %}<span class="tag tag--muted">待完成</span>{ 4515
      % endif %}                                                            4516
        </td>                                                               4517
        <td style="text-align:left;">{{                                    4518
      progress.constraint.detail }}</td>                                    4519
      </tr>                                                                 4520
      <tr>                                                                  4521
        <td>录入指标数据</td>                                               4522
        <td>                                                               4523
          {% if progress.data.ready %}<span class="tag                      4524
      tag--ok">已完成</span>                                                4525
          {% else %}<span class="tag tag--muted">待完成</span>{ 4526
      % endif %}                                                            4527
        </td>                                                               4528
        <td style="text-align:left;">{{                                    4529
      progress.data.detail }}</td>                                          4530
      </tr>                                                                 4531
    </tbody>                                                                4532
  </table>                                                                  4533
</div>                                                                      4534
{% endblock %}                                                              4535
# ==== app/templates/evaluate/result.html ====                              4536
{% extends "base.html" %}                                                   4537
{% block title %}评价结果{% endblock %}                                     4538
{% block content %}                                                         4539
<div class="card">                                                          4540
  <h2>适宜性评价结果 · {{ patch.name if patch else "" }}</h2>               4541
  <div class="stat-row">                                                    4542
    <div>                                                                   4543
      <div class="stat__label">最终评价因子 F</div>                         4544
      <div class="stat__value mono">{{ r.final_score                        4545
    }}</div>                                                                4546
    </div>                                                                  4547
    <div>                                                                   4548
      <div class="stat__label">适宜性等级</div>                             4549
      <div class="stat__value ok">{{ r.grade }}</div>                       4550
```

```
      </div>                                                                    4551
      <div style="flex:1;min-width:240px;">                                     4552
        <div class="stat__label">推荐措施</div>                                 4553
        <div style="font-size:15px;">{{ r.measures }}</div>                     4554
      </div>                                                                    4555
    </div>                                                                      4556
    {% if unit_scores %}                                                        4557
    <h3>各微地质单元得分</h3>                                                   4558
    <table style="max-width:720px;">                                            4559
      <thead>                                                                   4560
        <tr>                                                                    4561
          <th scope="col">微地质单元</th>                                       4562
          <th scope="col">面积占比</th>                                         4563
          <th scope="col">单元评价因子 F′</th>                                  4564
          <th scope="col">加权贡献</th>                                         4565
        </tr>                                                                   4566
      </thead>                                                                  4567
      <tbody>                                                                   4568
        {% for u in unit_scores %}                                              4569
        <tr>                                                                    4570
          <td>{{ u.unit_type }}</td>                                            4571
          <td class="num">{{ "%.2f%%"|format(u.ratio * 100)                     4572
      }}</td>                                                                   4573
          <td class="mono">{{ u.unit_score }}</td>                              4574
          <td class="mono">{{ "%.4f"|format(u.ratio *                           4575
      u.unit_score) }}</td>                                                     4576
        </tr>                                                                   4577
        {% endfor %}                                                            4578
      </tbody>                                                                  4579
    </table>                                                                    4580
    {% endif %}                                                                 4581
    <p class="actions" style="margin-top:18px;">                                4582
      <a class="btn btn--ghost" href="/report/{{ r.patch_id                     4583
      }}">查看得分明细</a>                                                      4584
      <a class="btn btn--ghost" href="/evaluate/">返回评价列表</a>              4585
    </p>                                                                        4586
</div>                                                                          4587
{% endblock %}                                                                  4588
#  ==== app/templates/report/index.html ====                                    4589
{% extends "base.html" %}                                                       4590
{% block title %}报告输出{% endblock %}                                         4591
{% block content %}                                                             4592
<div class="card">                                                              4593
    <h2>报告与结果输出</h2>                                                     4594
    <p class="muted">                                                           4595
      可查看逐层得分明细与结果图表，并导出 PDF 或 Word 评价报告。              4596
      报告是评价结论的呈现，因此仅在评价所依赖的前置步骤全部完成后才可生成。  4597
    </p>                                                                        4598
    {% if not patches %}                                                        4599
    <p class="muted">暂无矿山图斑。请先到「项目与矿山档案」建立图斑。</p>      4600
```

```
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">矿山图斑</th>
        <th scope="col">可生成报告</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for p in patches %}
      {% set pr = progress.get(p.id) %}
      <tr>
        <td style="text-align:left;">{{ p.name }}</td>
        <td style="text-align:left;">
          {% if pr and pr.ready %}
          <span class="tag tag--ok">可以</span>
          {% elif pr %}
          <span class="tag tag--warn">暂不可</span>
          <span class="muted">{{ pr.blocking[0].reason
    }}</span>
          {% endif %}
        </td>
        <td class="nowrap">
          {% if pr and pr.ready %}
          <a href="/report/{{ p.id }}">查看得分明细</a>
           | <a href="/report/{{ p.id }}/pdf">导出
    PDF</a>
           | <a href="/report/{{ p.id }}/word">导出
    Word</a>
          {% else %}
          <span class="muted">请先完成前置步骤</span>
          {% endif %}
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% endif %}
</div>
{% endblock %}
# ==== app/templates/report/detail.html ====
{% extends "base.html" %}
{% block title %}得分明细 · {{ patch.name }}{% endblock %}
{% block content %}
<div class="card">
  <h2>得分明细 · {{ patch.name }}</h2>
  <div class="stat-row">
    <div>
      <div class="stat__label">最终评价因子 F</div>
```

```
        <div class="stat__value mono">{{ "%.4f"|format(result.final_score) }}</div>
      </div>
      <div>
        <div class="stat__label">适宜性等级</div>
        <div class="stat__value ok">{{ result.grade }}</div>
      </div>
      <div style="flex:1;min-width:240px;">
        <div class="stat__label">推荐措施</div>
        <div style="font-size:15px;">{{ result.grade_info.measure }}</div>
      </div>
    </div>
    <p class="actions" style="margin-top:16px;">
      <a class="btn" href="/report/{{ patch.id }}/pdf">导出 PDF 报告</a>
      <a class="btn btn--ghost" href="/report/{{ patch.id }}/word">导出 Word 报告</a>
      <a class="btn btn--ghost" href="/report/">返回</a>
    </p>
  </div>
  <div class="card">
    <h2>结果图表</h2>
    <h3>各微地质单元得分对比</h3>
    <div>{{ chart_units|safe }}</div>
    <h3>各一级指标平均得分</h3>
    <p class="muted">反映各一级指标对评价结果的实际影响，便于识别关键控制项。</p>
    <div>{{ chart_weight|safe }}</div>
  </div>
  {% for u in result.unit_results %}
  <div class="card">
    <h2>{{ u.unit_type }} · 占比 {{ "%.2f%%"|format(u.ratio * 100) }} · 单元得分 {{ "%.4f"|format(u.unit_score) }}</h2>
    <table>
      <thead>
        <tr>
          <th scope="col">一级指标</th>
          <th scope="col">二级指标</th>
          <th scope="col">实测值</th>
          <th scope="col">标准化值</th>
          <th scope="col">权重</th>
          <th scope="col">贡献</th>
        </tr>
      </thead>
      <tbody>
        {% for d in u.details %}
        <tr>
          <td>{{ d.l1 }}</td>
          <td style="text-align:left;">{{ d.l2 }}</td>
          <td class="mono" style="text-align:left;">{{ "—" if
```

```
      d.raw is none else d.raw }}</td>                                  4701
           <td class="mono">{{ "—" if d.std_value is none else          4702
      "%.4f"|format(d.std_value) }}</td>                                4703
           <td class="mono">{{ "%.4f"|format(d.weight) }}</td>           4704
           <td class="mono">{{ "—" if d.contribution is none            4705
      else "%.4f"|format(d.contribution) }}</td>                        4706
         </tr>                                                          4707
         {% endfor %}                                                   4708
      </tbody>                                                          4709
   </table>                                                             4710
   <p class="muted" style="margin:0;">                                  4711
      一级指标得分：                                                    4712
      {% for k, v in u.l1_scores.items() %}{{ k }} = <span             4713
      class="num">{{ "%.4f"|format(v) }}</span>{% if not               4714
      loop.last %};{% endif %}{% endfor %}                              4715
   </p>                                                                 4716
</div>                                                                 4717
{% endfor %}                                                           4718
{% endblock %}                                                         4719
#  ==== seed_demo.py ====                                              4720
#  -*- coding: utf-8 -*-                                               4721
"""内置玉田县案例：把论文第 5 节案例数据写入数据库。                   4722
用于功能验证与操作演示。数据逐字取自论文表 5 与第 5 节。               4723
运行：python seed_demo.py（幂等：已存在则先清空重建）                  4724
"""                                                                    4725
import json                                                            4726
import sys                                                             4727
sys.stdout.reconfigure(encoding="utf-8")                               4728
from app import create_app                                             4729
from app.models import (db, User, Project, MinePatch,                  4730
      ReferenceEcosystem,                                              4731
                              MicroUnit, IndicatorData, Standard,      4732
      StandardRule,                                                    4733
                              ConstraintCheck)                         4734
app = create_app()                                                     4735
#  论文表 1：11 项地方标准与政策文件                                   4736
STANDARDS = [                                                          4737
      dict(category="地方标准", province="辽宁",                       4738
      name="矿山废弃地植被自然恢复技术规程",                           4739
            doc_number="DB21/T 3702—2023",                             4740
      publish_date="2023-03-30",                                       4741
            implement_date="2023-04-30",                               4742
            summary="首个正式发布的矿山自然恢复地方标准；提出必要条件和充分条件",  4743
            rules=[("植被覆盖度", ">=", "50", "必要条件", None),       4744
                   ("无地质灾害", "==", "是", "必要条件", None),       4745
                   ("无水土污染", "==", "是", "必要条件", None),       4746
                   ("与参照系差距", "<", "20", "充分条件", "%")]),     4747
      dict(category="地方标准", province="湖北",                       4748
      name="废弃露天矿山生态环境自然恢复标准",                         4749
            doc_number="DB42/T 2421—2025",                             4750
```

```
publish_date="2025-08-20",
        implement_date="2025-10-20",
        summary="结合矿山生态需求进行自然恢复适宜性分级判断",
        rules=[("自然恢复适宜性分级", "==", "是", "必要条件", None)]),
dict(category="地方标准", province="内蒙古",
name="历史遗留废弃矿山自然恢复判定规范",
        doc_number="DB15/T 4280—2026",
publish_date="2026-01-30",
        implement_date="2026-03-01",
        summary="调查内容包含参照生态系统调查",
        rules=[("地质灾害发育程度", "<=", "弱发育", "必要条件", None),
               ("地形地貌与周边协调", "==", "是", "必要条件", None),
               ("有利于植被恢复的表层土壤", "==", "是", "必要条件", None),
               ("已形成稳定植被覆盖", "==", "是", "必要条件", None)]),
dict(category="地方标准", province="江西",
name="废弃矿山自然恢复技术规范",
        doc_number="DB36/T 2221—2026",
publish_date="2026-02-02",
        implement_date="2026-08-01",
        summary="对参照生态系统进行了要求；提出自然恢复适宜性指数",
        rules=[("参照生态系统调查", "==", "是", "必要条件", None),
               ("自然恢复适宜性指数", ">=", "阈值", "充分条件", None)]),
dict(category="地方标准", province="宁夏", name="矿山自然恢复认定规程",
        doc_number=None, publish_date="2026-06-18",
implement_date=None,
        summary="分区差异化：三个一级生态修复分区及 22 个二级分区",
        rules=[("无地质灾害风险", "==", "是", "必要条件", None),
               ("植被覆盖度趋势", "==", "稳定或正增长", "必要条件", None),
               ("与参照系差距", "<", "20", "充分条件", "%")]),
dict(category="政策文件", province="安徽",
        name="安徽省自然资源厅关于规范废弃露天矿山采取自然恢复方式进行治理的通知",
        doc_number="皖自然资修函[2020]34号",
publish_date="2020-03-03", implement_date=None,
        summary="自然恢复条件：责任主体为政府、无地质灾害、无水土污染、不在三线三边、水土条件良好",
        rules=[("责任主体", "==", "政府", "必要条件", None),
               ("无地质灾害", "==", "是", "必要条件", None),
               ("无水土污染", "==", "是", "必要条件", None)]),
dict(category="政策文件", province="山东",
        name="山东省自然资源厅关于进一步规范历史遗留废弃露天开采矿山采取自然恢复方式进行治理的通知",
        doc_number="鲁自然资字[2022]104号",
publish_date="2022-08-02", implement_date=None,
        summary="条件与安徽省相同，另增加自然恢复认定标准",
        rules=[("无地质灾害", "==", "是", "必要条件", None),
               ("无水土污染", "==", "是", "必要条件", None)]),
dict(category="政策文件", province="湖南",
        name="关于印发《湖南省历史遗留矿山自然恢复指导意见（试行）》的通知",
        doc_number="湘自资办发[2023]58号",
publish_date="2023-07-20", implement_date=None,
        summary="自然恢复认定标准：无地质灾害或无威胁对象、无水土污染、设备拆除、符合规划、植被覆盖度
>75%、连续3年正向增长",
```

```
            rules=[("植被覆盖度", ">", "75", "必要条件", "%"),
                   ("植被覆盖率趋势", "==", "连续3年正向增长", "必要条件", None),
                   ("无地质灾害", "==", "是", "必要条件", None),
                   ("符合国土空间规划", "==", "是", "必要条件", None)]),
    dict(category="政策文件", province="新疆",
            name="自治区历史遗留废弃矿山自然恢复方式销号图斑工作指南（试行）",
            doc_number="新自然资源办函〔2024〕219号",
    publish_date="2024-12-20", implement_date=None,
            summary="荒漠区沙地/裸土地/裸岩石砾地满足无地质灾害、地形协调、无水土污染即可判定",
            rules=[("植被覆盖度", ">=", "周边30%", "必要条件", None),
                   ("无地质灾害", "==", "是", "必要条件", None),
                   ("地形平缓地貌稳定", "==", "是", "必要条件", None)]),
    dict(category="政策文件", province="广西",
            name="广西壮族自治区历史遗留矿山自然恢复技术指南",
            doc_number="桂自然资办〔2024〕250号",
    publish_date="2024-12-30", implement_date=None,
            summary="除高陡边坡和裸露岩体外，水土水热条件良好、植被自然恢复趋势明显",
            rules=[("无地质灾害", "==", "是", "必要条件", None),
                   ("植被自然恢复趋势", "==", "明显", "必要条件", None),
                   ("生产设备拆除完毕", "==", "是", "必要条件", None)]),
    dict(category="政策文件", province="西藏",
            name="西藏自治区历史遗留废弃矿山生态自然恢复认定指导意见（试行）",
            doc_number="藏自然资发〔2025〕19号",
    publish_date="2025-09-24", implement_date=None,
            summary="含修复成本高、生态修复效益不显著等适用条件",
            rules=[("责任主体", "==", "政府", "必要条件", None),
                   ("地形地貌完整与周边协调", "==", "是", "必要条件", None),
                   ("无地质灾害", "==", "是", "必要条件", None)]),
]
CASE = {
    "name": "玉田县典型历史遗留废弃矿山案例",
    "patch_name": "玉田县孤树镇白云岩采石场",
    "area_ha": 5.6,
    "mineral_type": "白云岩",
    "units": [
        {
            "unit_type": "岩质边坡", "ratio": 0.1386,
            "data": {"coverage": 68.88, "shannon": 0.00,
                     "slope": 63.33, "aspect": "阴阳坡"},
        },
        {
            "unit_type": "碎石边坡", "ratio": 0.6295,
            "data": {"soil_ratio": 0.34, "thickness": 10,
    "ph": 9.24,
                     "organic_matter": 4.89,
                     "heavy_metals": {"Hg": 0.05, "As":
    5.17, "Ni": 12.60,
                                      "Cu": 11.70, "Cd":
    0.04, "Pb": 6.50},
                     "coverage": 87.92, "shannon": 0.60,
```

```
                    "slope": 28.75, "aspect": "阳坡"},
            },
            {
                "unit_type": "平台", "ratio": 0.2319,
                "data": {"soil_ratio": 0.56, "thickness": 30,
        "ph": 8.72,
                    "organic_matter": 10.92,
                    "heavy_metals": {"Hg": 0.02, "As":
        3.13, "Ni": 12.33,
                                     "Cu": 11.14, "Cd":
        0.09, "Pb": 9.98},
                    "coverage": 58.42, "shannon": 0.72},
            },
        ],
    }
    def clean_case():
        """清空案例相关数据（幂等重跑用）。
        indicator_data / score_details 等表未设级联删除，需显式清理。
        """
        from app.models import EvaluationResult, ScoreDetail,
        ConstraintCheck
        ScoreDetail.query.delete()
        EvaluationResult.query.delete()
        ConstraintCheck.query.delete()
        IndicatorData.query.delete()
        MicroUnit.query.delete()
        ReferenceEcosystem.query.delete()
        MinePatch.query.delete()
        Project.query.delete()
        db.session.commit()
    def seed():
        with app.app_context():
            db.create_all()        # 全新环境下先建表，保证本脚本可独立运行
            clean_case()
            admin = User.query.filter_by(username="admin").first
        ()
            if not admin:
                from werkzeug.security import
        generate_password_hash
                admin = User(username="admin",
                             password_hash=generate_password_has
        h("admin123"),
                             role="admin")
                db.session.add(admin)
                db.session.commit()
            project = Project(name=CASE["name"],
        code="DEMO-001",
                              description="论文第 5 节典型案例",
        created_by=admin.id)
            db.session.add(project)
```

```
        db.session.flush()
        patch = MinePatch(project_id=project.id,
name=CASE["patch_name"],
                          admin_region="河北省唐山市玉田县",
                          area_ha=CASE["area_ha"],
                          mineral_type=CASE["mineral_type"],
                          mining_method="露天开采",
stop_years=None,
                          responsible_entity_missing=True)
        db.session.add(patch)
        db.session.flush()
        ref = ReferenceEcosystem(patch_id=patch.id,
                                 coverage_benchmark=75.0,
shannon_benchmark=0.7)
        db.session.add(ref)
        for u in CASE["units"]:
            unit = MicroUnit(patch_id=patch.id,
unit_type=u["unit_type"],
                             ratio=u["ratio"], area=None)
            db.session.add(unit)
            db.session.flush()
            d = u["data"]
            ind = IndicatorData(
                unit_id=unit.id,
                soil_ratio=d.get("soil_ratio"),
                thickness=d.get("thickness"),
                ph=d.get("ph"),
                organic_matter=d.get("organic_matter"),
                heavy_metals=json.dumps(d["heavy_metals"],
ensure_ascii=False)
                             if d.get("heavy_metals") else
None,
                coverage=d.get("coverage"),
                shannon=d.get("shannon"),
                slope=d.get("slope"),
                aspect=d.get("aspect"),
            )
            db.session.add(ind)
        #  限制性条件判定记录：演示案例按论文第 5 节描述属「无地质灾害隐患点」、
        #  具备自然恢复条件，故五项全部通过。缺此记录会使后续评价被流程门禁拦住。
        db.session.add(ConstraintCheck(
            patch_id=patch.id,
            policy_pass=True, location_pass=True,
pollution_pass=True,
            geo_pass=True, farmland_pass=True,
            passed=True, reason="全部通过"))
        db.session.commit()
        print("已写入案例：", CASE["name"])
        print("  图斑：", CASE["patch_name"])
        print("  微地质单元：岩质边坡 13.86% / 碎石边坡 62.95% / 平台
```

```
    23.19%")
        print("  管理员账号：admin / admin123")
        seed_standards()
        # 指标定义（M6 指标体系维护）
        from app.services.indicator_service import
    seed_indicators
        n = seed_indicators()
        print("已写入指标定义：%d 项" % n)
def seed_standards():
    """写入论文表 1 的 11 项标准与政策文件及其结构化规则。"""
    StandardRule.query.delete()
    Standard.query.delete()
    db.session.commit()
    for s in STANDARDS:
        std = Standard(category=s["category"],
    province=s["province"],
                       name=s["name"],
    doc_number=s["doc_number"],
                       publish_date=s["publish_date"],
                       implement_date=s["implement_date"],
                       summary=s["summary"])
        db.session.add(std)
        db.session.flush()
        for item, op, thr, ctype, note in s["rules"]:
            db.session.add(StandardRule(standard_id=std.id,
    item=item,
                                        operator=op,
    threshold=thr,

    condition_type=ctype, note=note))
    db.session.commit()
    print("已写入标准/政策文件：%d 项，规则 %d 条"
          % (len(STANDARDS), sum(len(s["rules"]) for s in
    STANDARDS)))
if __name__ == "__main__":
    seed()
# ==== test_ahp.py ====
# -*- coding: utf-8 -*-
"""地基自检：验证 AHP 引擎与应用可启动。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
import numpy as np
from app.services import ahp_service as ahp
LINE = "=" * 66
# ------------------------------------------------------------
    ----------
# 1） 完全一致矩阵 → 权重应精确还原，CR=0
# ------------------------------------------------------------
    ----------
print(LINE)
```

```
print("测试 1  完全一致判断矩阵的权重还原与一致性")
print(LINE)
# 由权重构造完全一致矩阵：a_ij = w_i / w_j
w_true = np.array([0.44, 0.31, 0.25])
A = np.outer(w_true, 1.0 / w_true)
r = ahp.weights_and_consistency(A)
print("  判断矩阵 A：")
for row in A:
    print("    " + "  ".join(f"{v:8.4f}" for v in row))
print()
print("  计算权重 :", "  ".join(f"{v:.4f}" for v in
    r["weights"]))
print("  真实权重 :", "  ".join(f"{v:.4f}" for v in w_true))
print("  λmax = %.6f   CI = %.3e   CR = %.3e   一致: %s"
      % (r["lambda_max"], r["ci"], r["cr"],
    r["consistent"]))
assert np.allclose(r["weights"], w_true, atol=1e-6),
    "权重未精确还原"
assert r["consistent"], "完全一致矩阵应通过检验"
# ------------------------------------------------------------
    ----------------
# 2） 上三角输入 → 完整矩阵 → 一致性定位
# ------------------------------------------------------------
    ----------------
print()
print(LINE)
print("测试 2  上三角录入与最不一致项定位")
print(LINE)
# 一个不太一致的 3 阶矩阵：上三角 [2, 5, 3]（a12=2, a13=5, a23=3）
M = ahp.matrix_from_upper([2, 5, 3])
r2 = ahp.weights_and_consistency(M)
bad = ahp.find_most_inconsistent(M)
print("  由 [a12=2, a13=5, a23=3] 构造矩阵：")
for row in M:
    print("    " + "  ".join(f"{v:8.4f}" for v in row))
print("  权重:", "  ".join(f"{v:.4f}" for v in r2["weights"]),
      "  CR = %.3f  一致: %s" % (r2["cr"], r2["consistent"]))
print("  最不一致比较项: 第(%d,%d) 项，当前 %.3f，理想 %.3f"
      % (bad["i"] + 1, bad["j"] + 1, bad["value"],
    bad["expected"]))
# ------------------------------------------------------------
    ----------------
# 3） 多专家聚合（几何平均）
# ------------------------------------------------------------
    ----------------
print()
print(LINE)
print("测试 3  多专家判断矩阵聚合（几何平均）")
print(LINE)
m1 = ahp.matrix_from_upper([2, 4, 3])
```

```
m2 = ahp.matrix_from_upper([3, 5, 2])
agg = ahp.aggregate_matrices([m1, m2])
print("  专家1 上三角 [2,4,3],专家2 上三角 [3,5,2]")
print("  聚合后上三角: a12=%.4f a13=%.4f a23=%.4f"
      % (agg[0, 1], agg[0, 2], agg[1, 2]))
assert abs(agg[0, 1] - np.sqrt(2 * 3)) < 1e-9, "几何平均错误"
# ------------------------------------------------------------------
    ------------------
# 4) 论文表 3 权重可还原性
# ------------------------------------------------------------------
    ------------------
print()
print(LINE)
print("测试 4  论文表 3 碎石边坡一级权重 (0.44, 0.31, 0.25)")
print(LINE)
j = ahp.judge_from_weights([0.44, 0.31, 0.25])
print("  λmax = %.6f  CR = %.3e  一致: %s" %
    (j["lambda_max"], j["cr"], j["consistent"]))
assert j["consistent"], "论文权重应能作为完全一致矩阵主特征向量"
# ------------------------------------------------------------------
    ------------------
# 5) 应用可启动 + 建表
# ------------------------------------------------------------------
    ------------------
print()
print(LINE)
print("测试 5  Flask 应用启动与建表")
print(LINE)
from app import create_app
from app.models import db
app = create_app()
with app.app_context():
    db.create_all()
    tables = sorted(db.metadata.tables.keys())
print("  已创建 %d 张表:" % len(tables))
for t in tables:
    print("    -", t)
print()
print(LINE)
print("全部测试通过 √")
print(LINE)
# ==== test_evaluate.py ====
# -*- coding: utf-8 -*-
"""评价引擎自检:用论文玉田县案例跑通完整计算链。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
from app.services import evaluation_service as ev
from app.services import standardization as std
LINE = "=" * 70
# ------------------------------------------------------------------
```

```
    ------------------
# 论文第 5 节案例数据（表 5，逐字抄录）
# ------------------------------------------------------------
    ------------------
# 三类单元面积占比：岩质边坡 １３．８６%、碎石边坡 ６２．９５%、平台 ２３．１９%
# 注：岩质边坡无土壤指标、平台无地形指标，对应"不适用"。
case = [
    {
        "unit_type": "岩质边坡",
        "ratio": 0.1386,
        "data": {
            "coverage": 68.88, "shannon": 0.00,
            "slope": 63.33, "aspect": "阴阳坡",
        },
    },
    {
        "unit_type": "碎石边坡",
        "ratio": 0.6295,
        "data": {
            "soil_ratio": 0.34, "thickness": 10, "ph":
    9.24, "organic_matter": 4.89,
            "heavy_metals": {"Hg": 0.05, "As": 5.17, "Ni":
    12.60,
                             "Cu": 11.70, "Cd": 0.04, "Pb":
    6.50},
            "coverage": 87.92, "shannon": 0.60,
            "slope": 28.75, "aspect": "阳坡",
        },
    },
    {
        "unit_type": "平台",
        "ratio": 0.2319,
        "data": {
            "soil_ratio": 0.56, "thickness": 30, "ph":
    8.72, "organic_matter": 10.92,
            "heavy_metals": {"Hg": 0.02, "As": 3.13, "Ni":
    12.33,
                             "Cu": 11.14, "Cd": 0.09, "Pb":
    9.98},
            "coverage": 58.42, "shannon": 0.72,
        },
    },
]
# ------------------------------------------------------------
    ------------------
# 逐单元输出标准化与得分
# ------------------------------------------------------------
    ------------------
print(LINE)
print("玉田县案例 · 评价引擎完整计算")
```

```
print(LINE)
for u in case:
    r = ev.score_unit(u["unit_type"], u["data"])
    print()
    print("  【%s】  (面积占比 %.2f%%)" % (u["unit_type"],
    u["ratio"] * 100))
    print("  " + "-" * 60)
    for l1, score in r["l1_scores"].items():
        print("    %s 得分 = %.4f" % (l1, score))
    print("    单元评价因子 F' = %.4f" % r["unit_score"])
print()
print(LINE)
print("面积加权合成（式(9)）")
print(LINE)
res = ev.score_mine(case)
total = res["final_score"]
print()
for ur in res["unit_results"]:
    print("  %s  %.2f%% × %.4f = %.4f"
          % (ur["unit_type"], ur["ratio"] * 100,
    ur["unit_score"], ur["weighted"]))
print("  " + "-" * 60)
print("  最终评价因子 F = %.4f" % total)
print()
print("  适宜性等级：%s" % res["grade"])
print("  判定依据  ：%s" % res["grade_info"]["meaning"])
print("  推荐措施  ：%s" % res["grade_info"]["measure"])
print()
print(LINE)
print("结论")
print(LINE)
print("  - 评价引擎按论文式(8)(9) 与表 3 权重完整跑通")
print("  - 坡度标准化采用论文 25° 分界的线性折减默认值（见 standardization.py）")
print("  - 最终 F ≈ %.2f，判定为「%s」，与论文「最终适宜自然恢复」的结论一致"
      % (total, res["grade"]))
print("  - 论文表 5 所载最终得分为 0.76；差异集中在坡度、坡向的标准化口径，")
print("    该口径在 standardization.std_slope / aspect_score
    中集中定义、可配置")
print(LINE)
# ==== test_web_ahp.py ====
# -*- coding: utf-8 -*-
"""M6 AHP 网页界面自检：走通选择→录入→计算 全流程。"""
import sys
sys.stdout.reconfigure(encoding="utf-8")
from app import create_app
LINE = "=" * 66
# 测试配置：关闭 CSRF 以简化测试客户端（真实运行仍开启 CSRF）
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///:memory:"
```

```
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 建立表并创建测试用户，登录后再访问业务页面
with app.app_context():
    from app.models import db, User
    from werkzeug.security import generate_password_hash
    db.create_all()
    if not User.query.filter_by(username="admin").first():
        db.session.add(User(username="admin",
                            password_hash=generate_password_
    hash("admin123"),
                            role="admin"))
        db.session.commit()
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
print("测试 1  首页与 AHP 入口")
print(LINE)
r = client.get("/")
print("  首页状态码：", r.status_code, "（应 200）")
assert r.status_code == 200
r = client.get("/ahp/")
print("  AHP 入口状态码：", r.status_code, "（应 200）")
assert r.status_code == 200
assert "微地质单元类型" in r.get_data(as_text=True)
print()
print(LINE)
print("测试 2  选择碎石边坡一级指标 → 跳转矩阵页")
print(LINE)
r = client.post("/ahp/", data={
    "unit_type": "碎石边坡", "level": "1", "submit": "生成判断矩阵",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转后页面含「判断矩阵录入」：", "判断矩阵录入" in body)
print("  含三个一级指标（土壤/植被/地形）：", all(n in body for n in
    ("土壤指标", "植被指标", "地形指标")))
assert "判断矩阵录入" in body
assert all(n in body for n in ("土壤指标", "植被指标", "地形指标"))
print()
print(LINE)
print("测试 3  提交论文表 3 一致矩阵 → 还原权重")
print(LINE)
# 碎石边坡一级权重 (0.44, 0.31, 0.25) 对应的完全一致矩阵上三角：
# a12 = 0.44/0.31 = 1.4194, a13 = 0.44/0.25 = 1.76, a23 =
    0.31/0.25 = 1.24
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24",
```

```
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  结果页含「计算结果」:", "计算结果" in body)
print("  含权重 0.44:", "0.44" in body, " 含 0.31:", "0.31" in
    body, " 含 0.25:", "0.25" in body)
print("  含一致性检验通过:", "通过" in body)
assert "计算结果" in body
assert "0.44" in body and "0.31" in body and "0.25" in body
print()
print(LINE)
print("测试 4  不一致矩阵应提示不通过并定位")
print(LINE)
# 明显不一致：a12=9, a13=1, a23=9
r = client.post("/ahp/matrix?unit_type=碎石边坡&level=1", data={
    "m_0_1": "9", "m_0_2": "1", "m_1_2": "9",
}, follow_redirects=True)
body = r.get_data(as_text=True)
print("  含「不通过」:", "不通过" in body)
print("  含「建议优先调整」:", "建议优先调整" in body)
assert "不通过" in body
assert "建议优先调整" in body
print()
print(LINE)
print("M6 网页界面全部测试通过 √")
print(LINE)
# ==== test_web_modules.py ====
# -*- coding: utf-8 -*-
"""M1–M5 模块自检：登录、项目档案、微地质单元、标准库、限制性判定。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        os.path.dirname(__file__), "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
print(LINE)
print("测试 1  未登录应被重定向到登录页")
print(LINE)
r = client.get("/project/")
print("  /project/ 状态码:", r.status_code, " 重定向到:",
    r.headers.get("Location"))
assert r.status_code == 302 and "login" in
    r.headers.get("Location", "")
```

```
print()
print(LINE)
print("测试 2  登录")
print(LINE)
r = client.post("/login", data={"username": "admin",
    "password": "admin123"},
                follow_redirects=True)
body = r.get_data(as_text=True)
print("  登录后含「系统功能」:", "系统功能" in body)
print("  含用户名 admin:", "admin" in body)
assert "系统功能" in body
print()
print(LINE)
print("测试 3  M2 项目与矿山档案")
print(LINE)
r = client.get("/project/")
body = r.get_data(as_text=True)
print("  项目列表状态码:", r.status_code)
print("  含案例项目:", "玉田县典型历史遗留废弃矿山案例" in body)
assert "玉田县典型历史遗留废弃矿山案例" in body
# 新建项目
client.post("/project/add", data={"name": "测试项目A", "code":
    "T-001"})
r = client.get("/project/")
print("  新建项目后含「测试项目A」:", "测试项目A" in r.get_data(as_text=True
    ))
assert "测试项目A" in r.get_data(as_text=True)
# 清理测试数据，避免污染演示库
with app.app_context():
    from app.models import db, Project
    for p in Project.query.filter_by(name="测试项目A").all():
        db.session.delete(p)
    db.session.commit()
print("  已清理测试项目")
print()
print(LINE)
print("测试 4  M5 微地质单元（面积占比自动计算）")
print(LINE)
from app.models import MinePatch
with app.app_context():
    patch = MinePatch.query.filter_by(name="玉田县孤树镇白云岩采石场").f
    irst()
    pid = patch.id
r = client.get("/unit/%d" % pid)
body = r.get_data(as_text=True)
print("  单元列表含三类单元:", all(t in body for t in ("岩质边坡",
    "碎石边坡", "平台")))
print("  含占比合计:", "占比合计" in body)
assert "岩质边坡" in body and "占比合计" in body
print()
```

```
print(LINE)
print("测试 5  M3 标准知识库与省际口径比对")
print(LINE)
r = client.get("/standard/")
body = r.get_data(as_text=True)
print("  含辽宁/湖南/新疆标准:", all(p in body for p in ("辽宁", "湖南",
    "新疆")))
assert "辽宁" in body
r = client.get("/standard/compare")
body = r.get_data(as_text=True)
print("  口径比对页含「植被覆盖度」分组:", "植被覆盖度" in body)
print("  含多省份并列:", "湖南" in body and "新疆" in body)
assert "植被覆盖度" in body
print()
print(LINE)
print("测试 6  M4 限制性指标判定（一票否决）")
print(LINE)
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("  含五项条件:", all(x in body for x in ("政策约束", "区位约束",
    "土壤污染", "地质灾害", "耕地属性")))
assert all(x in body for x in ("政策约束", "区位约束", "土壤污染",
    "地质灾害", "耕地属性"))
# 全部通过
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "1",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("  全通过后显示「通过，可进入定量评价」:", "可进入定量评价" in body)
assert "可进入定量评价" in body
# 一项不通过
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "0",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
body = r.get_data(as_text=True)
print("  一项不通过后显示「不通过」:", "不通过" in body)
print("  含失败项「区位约束」:", "区位约束" in body)
assert "不通过" in body
# 恢复为全部通过，避免把演示数据留在「不通过」状态影响后续步骤
client.post("/constraint/%d" % pid, data={
    "policy_pass": "1", "location_pass": "1",
    "pollution_pass": "1",
    "geo_pass": "1", "farmland_pass": "1"})
r = client.get("/constraint/%d" % pid)
print("  已恢复为全部通过:", "可进入定量评价" in r.get_data(as_text=True))
print()
```

```
print(LINE)
print("M1–M5 全部测试通过 √")
print(LINE)
# ==== test_web_evaluate.py ====
# -*- coding: utf-8 -*-
"""M7 + M8 端到端自检：录数据 → 执行评价 → 查看结果。"""
import os
import sys
sys.stdout.reconfigure(encoding="utf-8")
from app import create_app
LINE = "=" * 66
# 与真实配置共用同一个 app.db，但关闭 CSRF 便于测试客户端
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        os.path.dirname(__file__), "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
print(LINE)
print("测试 1  M7 指标数据列表")
print(LINE)
r = client.get("/indicator/")
body = r.get_data(as_text=True)
print("  状态码：", r.status_code)
print("  含案例图斑：", "玉田县孤树镇白云岩采石场" in body)
print("  含三个单元：", all(n in body for n in ("岩质边坡", "碎石边坡",
    "平台")))
assert r.status_code == 200
assert "岩质边坡" in body and "碎石边坡" in body and "平台" in body
print()
print(LINE)
print("测试 2  M7 指标数据录入页（按单元类型显示字段）")
print(LINE)
from app.models import MicroUnit, MinePatch
with app.app_context():
    unit_rock = MicroUnit.query.filter_by(unit_type="岩质边坡").
    first()
    unit_gravel = MicroUnit.query.filter_by(unit_type="碎石边坡"
    ).first()
r = client.get("/indicator/%d" % unit_rock.id)
b_rock = r.get_data(as_text=True)
print("  岩质边坡页：无「土石比」字段：", "土石比" not in b_rock, "（应 True）")
print("  岩质边坡页：含「植被覆盖度」：", "植被覆盖度" in b_rock)
assert "土石比" not in b_rock and "植被覆盖度" in b_rock
```

```
r = client.get("/indicator/%d" % unit_gravel.id)
b_gravel = r.get_data(as_text=True)
print("  碎石边坡页：含「土石比」字段：", "土石比" in b_gravel)
print("  碎石边坡页：含 8 种重金属：", all(m in b_gravel for m in
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn")))
assert "土石比" in b_gravel and all(m in b_gravel for m in
    ("Cd", "Hg", "Pb", "Cr", "As", "Ni", "Cu", "Zn"))
print()
print(LINE)
print("测试 3  M8 执行评价")
print(LINE)
with app.app_context():
    patch = MinePatch.query.first()
r = client.post("/evaluate/%d" % patch.id, data={},
    follow_redirects=True)
body = r.get_data(as_text=True)
print("  跳转至结果页：", "适宜性评价结果" in body)
print("  含最终评价因子 F：", "最终评价因子 F" in body)
print("  含等级「适宜」：", "适宜" in body)
print("  含推荐措施「自然恢复」：", "自然恢复" in body)
assert "最终评价因子 F" in body and "自然恢复" in body
print()
print(LINE)
print("测试 4  结果持久化")
print(LINE)
from app.models import EvaluationResult
with app.app_context():
    res = EvaluationResult.query.order_by(EvaluationResult.i
    d.desc()).first()
    print("  最终得分 F =", res.final_score)
    print("  等级 =", res.grade)
    assert res.grade == "适宜"
    assert 0.7 <= res.final_score < 1.0
print()
print(LINE)
print("M7 + M8 端到端全部测试通过 √")
print(LINE)
# ==== test_web_report.py ====
# -*- coding: utf-8 -*-
"""M9 报告输出自检：得分明细页 + Word 导出。"""
import os
import sys
import io
sys.stdout.reconfigure(encoding="utf-8")
from app import create_app
LINE = "=" * 66
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        os.path.dirname(__file__), "app", "app.db")
```

```
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
app = create_app(TestConfig)
client = app.test_client()
# 登录（案例种子已创建 admin 用户）
client.post("/login", data={"username": "admin",
    "password": "admin123"})
from app.models import MinePatch
with app.app_context():
    patch = MinePatch.query.first()
    pid = patch.id
print(LINE)
print("测试 1  报告列表")
print(LINE)
r = client.get("/report/")
print("  状态码：", r.status_code)
assert r.status_code == 200
print()
print(LINE)
print("测试 2  得分明细页")
print(LINE)
r = client.get("/report/%d" % pid)
body = r.get_data(as_text=True)
print("  状态码：", r.status_code)
print("  含「得分明细」：", "得分明细" in body)
print("  含「最终评价因子 F」：", "最终评价因子 F" in body)
print("  含二级指标明细（植被覆盖度/坡度/坡向）：", all(n in body for n in
    ("植被覆盖度", "坡度", "坡向")))
print("  含权重/标准化值列：", "标准化值" in body and "贡献" in body)
assert r.status_code == 200
assert "植被覆盖度" in body and "标准化值" in body
print()
print(LINE)
print("测试 3  Word 报告导出")
print(LINE)
r = client.get("/report/%d/word" % pid)
print("  状态码：", r.status_code)
print("  Content-Type:", r.headers.get("Content-Type",
    "")[:60])
data = r.data
print("  文件大小：", len(data), "字节")
print("  是否为有效 docx（PK 头）：", data[:2] == b"PK")
assert r.status_code == 200 and data[:2] == b"PK"
# 用 python-docx 打开验证内容
import docx
d = docx.Document(io.BytesIO(data))
text = "\n".join(p.text for p in d.paragraphs)
print("  含报告标题：", "自然恢复适宜性评价报告" in text)
print("  含评价结论章节：", "评价结论" in text)
```

```
print("  含得分明细章节：", "得分明细" in text)
print("  段落数：", len(d.paragraphs), " 表格数：", len(d.tables))
assert "自然恢复适宜性评价报告" in text and len(d.tables) >= 3
print()
print(LINE)
print("M9 报告输出全部测试通过 √")
print(LINE)
# ==== test_ui_fixes.py ====
# -*- coding: utf-8 -*-
"""界面修复自检：「记住我」与各模块删除功能。"""
import os
import re
import sys
from app import create_app
sys.stdout.reconfigure(encoding="utf-8")
LINE = "=" * 70
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" + os.path.join(
        os.path.dirname(__file__), "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    REMEMBER_COOKIE_DURATION = __import__("datetime").timede
    lta(days=30)
app = create_app(TestConfig)
client = app.test_client()
# ------------------------------------------------------------
    ----------------
print(LINE)
print("一、「记住我」")
print(LINE)
html = client.get("/login").get_data(as_text=True)
# 1) 复选框必须有可点击关联的 label（这是原先失效的根因）
wrapped = re.search(r'<label[^>]*class="checkbox"[^>]*>\s*<i
    nput[^>]*name="remember"',
                        html) is not None
print("  %s 「记住我」文字与复选框包在同一 label 内" % ("[符合]" if wrapped
    else "[不符合]"))
assert wrapped, "复选框未与文字关联，点击文字不会勾选"
has_for = 'for="remember"' in html
print("  %s label 带 for 属性指向复选框" % ("[符合]" if has_for else
    "[不符合]"))
# 2) 勾选后应下发带过期时间的 remember_token
c = app.test_client()
r = c.post("/login", data={"username": "admin", "password":
    "admin123",
                           "remember": "y"},
    follow_redirects=False)
cookies = r.headers.getlist("Set-Cookie")
```

```
rt = [h for h in cookies if h.startswith("remember_token=")]
print("  %s 勾选后下发 remember_token" % ("[符合]" if rt else
    "[不符合]"))
assert rt, "未下发 remember_token"
exp = re.search(r"Expires=([^;]+)", rt[0])
print("  %s 该 cookie 带过期时间：%s"
      % ("[符合]" if exp else "[不符合]", exp.group(1) if exp
    else "无（会话 cookie）"))
assert exp, "remember_token 无过期时间，关浏览器即失效"
# 3) 不勾选时不应下发 remember_token
c2 = app.test_client()
r2 = c2.post("/login", data={"username": "admin",
    "password": "admin123"},
              follow_redirects=False)
rt2 = [h for h in r2.headers.getlist("Set-Cookie") if
    h.startswith("remember_token=")]
print("  %s 未勾选时不下发 remember_token" % ("[符合]" if not rt2
    else "[不符合]"))
assert not rt2
# 4) 携带 remember_token 的新会话应自动登录
c3 = app.test_client()
c3.set_cookie("remember_token",
              rt[0].split(";")[0].split("=", 1)[1],
    domain="localhost")
r3 = c3.get("/", follow_redirects=False)
print("  %s 仅凭 remember_token 访问首页可自动登录（状态码 %d）"
      % ("[符合]" if r3.status_code == 200 else "[不符合]",
    r3.status_code))
assert r3.status_code == 200
# ------------------------------------------------------------
    ----------------
print()
print(LINE)
print("二、删除功能")
print(LINE)
client.post("/login", data={"username": "admin",
    "password": "admin123"})
# 项目：列表页须有删除按钮
html = client.get("/project/").get_data(as_text=True)
has_del = "/delete" in html and "btn--danger" in html
print("  %s 项目列表含删除按钮" % ("[符合]" if has_del else "[不符合]"))
assert has_del, "项目列表缺少删除按钮"
print("  %s 删除前有二次确认" % ("[符合]" if "confirm(" in html else
    "[不符合]"))
assert "confirm(" in html
# 图斑列表
with app.app_context():
    from app.models import MinePatch, Project
    patch = MinePatch.query.first()
    patch_id = patch.id
```

```
html = client.get("/project/%d/patches" %
    patch_id).get_data(as_text=True)
print("  %s 图斑列表含删除按钮" % ("[符合]" if "/project/patch/" in
    html and "btn--danger" in html else "[不符合]"))
assert "btn--danger" in html
# 微地质单元
html = client.get("/unit/%d" % patch_id).get_data(as_text=Tr
    ue)
print("  %s 微单元列表含删除按钮" % ("[符合]" if "/unit/delete/" in
    html else "[不符合]"))
assert "/unit/delete/" in html
print("  %s 微单元编辑为独立页面（表单不再嵌在表格行内）"
      % ("[符合]" if "/unit/edit/" in html else "[不符合]"))
# 标准
html = client.get("/standard/").get_data(as_text=True)
print("  %s 标准列表含删除按钮" % ("[符合]" if "btn--danger" in html
    else "[不符合]"))
assert "btn--danger" in html
# 实际执行一次删除：新建项目 → 删除 → 确认消失
client.post("/project/add", data={"name": "待删除测试项目",
    "code": "DEL-1"})
with app.app_context():
    from app.models import db, Project
    tmp = Project.query.filter_by(name="待删除测试项目").first()
    tmp_id = tmp.id
html = client.get("/project/").get_data(as_text=True)
print("  %s 新建项目成功" % ("[符合]" if "待删除测试项目" in html else
    "[不符合]"))
r = client.post("/project/%d/delete" % tmp_id,
    follow_redirects=True)
html = r.get_data(as_text=True)
gone = "待删除测试项目" not in html
print("  %s 删除项目后列表中不再出现" % ("[符合]" if gone else "[不符合]"))
assert gone
print("  %s 有删除成功提示" % ("[符合]" if "已删除" in html else
    "[不符合]"))
print()
print(LINE)
print("界面修复自检全部通过 √")
print(LINE)
# ==== test_new_features.py ====
# -*- coding: utf-8 -*-
"""补齐模块自检：M1 操作日志与用户维护、M2 遥感影像、M6 指标与多专家矩阵、
M7 批量导入、M9 图表与 PDF 导出。
对应计划书功能清单与系统实现的差异核对结论。
"""
import io
import json
import os
import re
```

```
import sys
sys.stdout.reconfigure(encoding="utf-8")
import fitz
from app import create_app
from app.models import (db, User, Project, MinePatch,
    MicroUnit, IndicatorData,
                        Indicator, ExpertMatrix,
    OperationLog)
from app.utils.schema import ensure_schema
from app.utils import browser, charts
LINE = "=" * 72
BASE = os.path.dirname(os.path.abspath(__file__))
class TestConfig:
    SECRET_KEY = "test"
    SQLALCHEMY_DATABASE_URI = "sqlite:///" +
    os.path.join(BASE, "app", "app.db")
    SQLALCHEMY_TRACK_MODIFICATIONS = False
    WTF_CSRF_ENABLED = False
    TESTING = True
    UPLOAD_FOLDER = os.path.join(BASE, "app", "static",
    "uploads")
    MAX_CONTENT_LENGTH = 32 * 1024 * 1024
    ALLOWED_IMAGE_EXT = {"png", "jpg", "jpeg", "gif",
    "webp", "bmp", "tif", "tiff"}
app = create_app(TestConfig)
with app.app_context():
    db.create_all()
    ensure_schema(db)
    ExpertMatrix.query.delete()
    db.session.commit()
    patch_id = MinePatch.query.first().id
c = app.test_client()
c.post("/login", data={"username": "admin", "password":
    "admin123"})
def ok(flag, label, extra=""):
    print("  %s %-44s %s" % ("[符合]" if flag else "[不符合]",
    label, extra))
    assert flag, label
# ----------------------------------------------------------------
    ----------------
print(LINE)
print("M1 操作日志与用户维护")
print(LINE)
before = None
with app.app_context():
    before = OperationLog.query.count()
c.post("/project/add", data={"name": "日志核对项目", "code":
    "LOG-T"})
with app.app_context():
    after = OperationLog.query.count()
```

```
    last = OperationLog.query.order_by(OperationLog.id.desc(
    )).first()
ok(after == before + 1, "写操作被写入操作日志", "%d -> %d" % (before,
    after))
ok(last.action == "新建项目", "日志记录了可读动作名", last.action)
ok("project" in (last.target_type or ""), "日志记录了对象类型",
    last.target_type)
r = c.get("/logs")
body = r.get_data(as_text=True)
ok(r.status_code == 200 and "操作日志" in body, "操作日志查看页可访问")
ok("新建项目" in body, "日志页显示操作记录")
# 用户编辑 / 删除
with app.app_context():
    u = User(username="tmp_user", role="viewer",
             password_hash=User.query.first().password_hash)
    db.session.add(u); db.session.commit()
    uid = u.id
r = c.get("/users/%d/edit" % uid)
ok(r.status_code == 200, "用户编辑页可访问")
c.post("/users/%d/edit" % uid, data={"role": "evaluator",
    "active": "1"})
with app.app_context():
    ok(User.query.get(uid).role == "evaluator", "角色已修改并保存")
# 最后一个管理员保护
with app.app_context():
    admin_id = User.query.filter_by(role="admin").first().id
c.post("/users/%d/edit" % admin_id, data={"role": "viewer",
    "active": "1"})
with app.app_context():
    ok(User.query.get(admin_id).role == "admin",
    "拒绝把最后一个管理员降级")
c.post("/users/%d/delete" % uid, follow_redirects=True)
with app.app_context():
    ok(User.query.get(uid) is None, "用户删除生效")
# 清理日志核对用的项目
with app.app_context():
    for p in Project.query.filter_by(name="日志核对项目").all():
        db.session.delete(p)
    db.session.commit()
# ------------------------------------------------------------------
    ------------------
print()
print(LINE)
print("M2 遥感影像挂接")
print(LINE)
doc = fitz.open()
pg = doc.new_page(width=320, height=200)
pg.insert_text((20, 60), "RS image", fontname="china-s",
    fontsize=16)
png = doc[0].get_pixmap(dpi=100).tobytes("png")
```

```
doc.close()
r = c.post("/project/patch/%d/image" % patch_id,
           data={"image": (io.BytesIO(png), "影像.png"),
                 "image_note": "2025年6月无人机正射影像"},
           content_type="multipart/form-data",
    follow_redirects=True)
body = r.get_data(as_text=True)
ok("已挂接" in body and "<img" in body, "影像上传并显示")
with app.app_context():
    rel = MinePatch.query.get(patch_id).image_path
ok(bool(rel), "数据库记录影像路径", rel)
abs_path = os.path.join(TestConfig.UPLOAD_FOLDER,
    rel.replace("uploads/", "", 1))
ok(os.path.isfile(abs_path), "影像文件落盘")
ok(c.get("/static/" + rel).status_code == 200, "影像可经 URL
    访问")
r = c.post("/project/patch/%d/image" % patch_id,
           data={"image": (io.BytesIO(b"x"), "bad.txt")},
           content_type="multipart/form-data",
    follow_redirects=True)
ok("不支持" in r.get_data(as_text=True), "拒绝非图片格式")
c.post("/project/patch/%d/image/delete" % patch_id,
    follow_redirects=True)
with app.app_context():
    ok(MinePatch.query.get(patch_id).image_path is None,
    "影像移除后字段清空")
ok(not os.path.isfile(abs_path), "影像文件已删除")
# ------------------------------------------------------------
    ----------------
print()
print(LINE)
print("M6 指标定义维护与多专家矩阵")
print(LINE)
r = c.get("/ahp/indicators")
body = r.get_data(as_text=True)
with app.app_context():
    n_ind = Indicator.query.count()
ok(r.status_code == 200 and n_ind > 0, "指标定义已入库并可查看", "%d
    项" % n_ind)
ok(all(k in body for k in ("一级指标", "二级指标", "三级指标")),
   "指标树按层级展示")
ok("岩质边坡" in body and "平台" in body, "显示各指标启用的微地质单元")
c.post("/ahp/indicators/coverage/edit",
       data={"name": "植被覆盖度", "unit": "%", "ideal_type":
    "benefit",
             "ideal_value": "75", "enabled_units": ["岩质边坡"],
             "description": "自检修改"})
with app.app_context():
    ok(Indicator.query.filter_by(code="coverage").first().de
    scription == "自检修改",
```

```
        "指标定义可编辑")
c.post("/ahp/indicators/reset")
with app.app_context():
    ok("自检修改" not in (Indicator.query.filter_by(code="covera
    ge").first().description or ""),
        "可按论文取值重置")
base = {"m_0_1": "1.4194", "m_0_2": "1.76", "m_1_2": "1.24"}
for name in ("张工", "李工", "王工"):
    d = dict(base); d.update({"action": "add", "unit_type":
    "碎石边坡",
                              "level": "1", "expert_name":
    name})
    c.post("/ahp/experts", data=d)
with app.app_context():
    ok(ExpertMatrix.query.filter_by(unit_type="碎石边坡",
    level=1).count() == 3,
        "录入 3 位专家的判断矩阵")
r = c.get("/ahp/experts/aggregate?unit_type=碎石边坡&level=1")
body = r.get_data(as_text=True)
ok("汇总后权重" in body and "各专家矩阵的一致性" in body,
   "多专家矩阵几何平均汇总可用")
ws = re.search(r"<th scope=\"row\">权重</th>\s*(.*?)</tr>",
    body, re.S)
vals = re.findall(r">([0-9.]+)</td>", ws.group(1)) if ws
    else []
ok(len(vals) == 3 and abs(sum(map(float, vals)) - 1.0) <
    0.01,
   "汇总权重归一", "、".join(vals))
ok(abs(float(vals[0]) - 0.44) < 0.02, "汇总权重接近论文表 3 取值",
    vals[0] if vals else "")
# ------------------------------------------------------------
    ------------------
print()
print(LINE)
print("M7 批量导入")
print(LINE)
r = c.get("/indicator/import/template")
tpl = r.data.decode("utf-8-sig")
ok(r.status_code == 200 and "单元类型" in tpl, "导入模板可下载")
ok("土石比" in tpl and "植被覆盖度" in tpl, "模板包含指标列")
csv_text = (
    "单元类型,土石比,土层厚度,pH,有机质,Hg,As,Ni,Cu,Cd,Pb,植被覆盖度,Shannon,坡度
    ,坡向\n"
    "碎石边坡,0.34,10,9.24,4.89,0.05,5.17,12.60,11.70,0.04,6.50,
    87.92,0.60,28.75,阳坡\n"
    "平台,0.56,30,8.72,10.92,0.02,3.13,12.33,11.14,0.09,9.98,5
    8.42,0.72,,,\n"
    "不存在单元,1,1,7,1,,,,,,,,,\n"
)
r = c.post("/indicator/import",
```

```
                data={"patch_id": str(patch_id),
                      "datafile": (io.BytesIO(csv_text.encode("ut
    f-8")), "d.csv")},
                content_type="multipart/form-data",
    follow_redirects=True)
body = r.get_data(as_text=True)
ok("导入结果" in body, "导入结果页展示")
ok("不存在单元" in body, "错误行被报告")
with app.app_context():
    u = MicroUnit.query.filter_by(patch_id=patch_id,
    unit_type="碎石边坡").first()
    ind = IndicatorData.query.filter_by(unit_id=u.id).first(
    )
    ok(abs(ind.soil_ratio - 0.34) < 1e-6 and
    abs(ind.coverage - 87.92) < 1e-6,
        "导入数据已写入数据库")
r = c.post("/indicator/import",
                data={"patch_id": str(patch_id),
                      "datafile": (io.BytesIO(b"x"), "d.pdf")},
                content_type="multipart/form-data",
    follow_redirects=True)
ok("仅支持" in r.get_data(as_text=True), "拒绝不支持的导入格式")
# ------------------------------------------------------------
    ----------------
print()
print(LINE)
print("M9 图表与 PDF 导出")
print(LINE)
svg = charts.hbars([("岩质边坡", 0.54, "0.5401"), ("平台", 0.84,
    "0.8376")])
ok("<svg" in svg and "0.8376" in svg, "条形图生成且带数值标签")
svg2 = charts.donut([("植被指标", 0.6, "0.6000"), ("土壤指标", 0.4,
    "0.4000")])
ok("<svg" in svg2 and "占比" in svg2, "环形图生成且带图例")
r = c.get("/report/%d" % patch_id)
body = r.get_data(as_text=True)
ok(body.count("<svg") >= 2, "得分明细页含图表", "%d 个 SVG" %
    body.count("<svg"))
exe = browser.browser_path()
if exe is None:
    print("  [跳过] 未找到浏览器，PDF 导出无法验证")
else:
    r = c.get("/report/%d/pdf" % patch_id)
    data = r.data
    ok(data[:4] == b"%PDF", "PDF 报告导出成功", "%d 字节" %
    len(data))
    d = fitz.open(stream=data, filetype="pdf")
    txt = "".join(d[i].get_text() for i in
    range(d.page_count))
    ok("自然恢复适宜性评价报告" in txt, "PDF 含报告标题")
```

```
    ok("评价结论" in txt and "得分明细" in txt, "PDF 含结论与明细章节")
    ok(d.page_count >= 2, "PDF 页数合理", "%d 页" % d.page_count)
    d.close()
print()
print(LINE)
print("补齐模块自检全部通过 √")
print(LINE)
# ==== app/services/__init__.py ====
# -*- coding: utf-8 -*-
# ==== app/services/audit.py ====
# -*- coding: utf-8 -*-
"""操作审计（M1）。
在应用工厂中以 after_request 钩子统一埋点：凡是成功的写操作（POST/PUT/DELETE）
且已登录，即写入 operation_logs 表，无需在每个视图里手工调用。
动作名称由「端点 → 中文动作」映射表给出，未收录的端点回退为方法 + 路径。
"""
from flask import request
from flask_login import current_user
from app.models import db, OperationLog
# 端点 → 可读动作名
ACTION_NAMES = {
    "auth.login": "登录系统",
    "auth.logout": "退出登录",
    "auth.add_user": "新增用户",
    "auth.edit_user": "编辑用户",
    "auth.delete_user": "删除用户",
    "auth.reset_password": "重置用户密码",
    "project.add": "新建项目",
    "project.edit": "编辑项目",
    "project.delete": "删除项目",
    "project.patch_add": "新建矿山图斑",
    "project.patch_edit": "编辑矿山图斑",
    "project.patch_delete": "删除矿山图斑",
    "project.upload_image": "上传遥感影像",
    "unit.add": "添加微地质单元",
    "unit.edit": "编辑微地质单元",
    "unit.delete": "删除微地质单元",
    "standard.add": "添加标准/政策文件",
    "standard.edit": "编辑标准/政策文件",
    "standard.delete": "删除标准/政策文件",
    "standard.rules": "维护标准规则",
    "standard.rule_delete": "删除标准规则",
    "constraint.check": "执行限制性条件判定",
    "ahp.save_expert": "录入专家判断矩阵",
    "ahp.delete_expert": "删除专家判断矩阵",
    "ahp.aggregate": "汇总多专家矩阵",
    "indicator.edit": "保存指标数据",
    "indicator.import_data": "批量导入指标数据",
    "indicator.update_meta": "维护指标定义",
    "evaluate.run": "执行适宜性评价",
```

```
}
#  无需记录的操作（查看类或纯导航）
SKIP_ENDPOINTS = {"static", "auth.index"}
def action_name(endpoint):
    if endpoint in ACTION_NAMES:
        return ACTION_NAMES[endpoint]
    return "写操作 %s %s" % (request.method, request.path)
def extract_target_id():
    """从视图参数中取出被操作对象的主键，取不到则返回 None。"""
    args = request.view_args or {}
    for key in ("pid", "patch_id", "unit_id", "sid", "rid",
    "result_id", "user_id"):
        if key in args:
            try:
                return int(args[key])
            except (TypeError, ValueError):
                return None
    return None
def record(endpoint, status_code):
    """记录一次写操作。失败（4xx/5xx）也记录，便于追溯。"""
    if endpoint in SKIP_ENDPOINTS:
        return
    try:
        detail = "路径 %s" % request.path
        if status_code >= 400:
            detail += "；结果 HTTP %d" % status_code
        db.session.add(OperationLog(
            user_id=current_user.id if
    current_user.is_authenticated else None,
            action=action_name(endpoint),
            target_type=(endpoint or "").split(".")[0] or
    None,
            target_id=extract_target_id(),
            detail=detail,
        ))
        db.session.commit()
    except Exception:
        #  审计失败不应影响业务响应
        db.session.rollback()
#  ==== app/services/import_service.py ====
#  -*- coding: utf-8 -*-
"""指标数据批量导入（M7）。
支持 CSV 与 Excel（.xlsx/.xls）。一行对应一个微地质单元，
按「单元类型」匹配图斑内已有的微地质单元后写入指标数据。
表头约定（缺失列按不适用处理）：
    单元类型， 土石比， 土层厚度， pH， 有机质，
    Hg， As， Ni， Cu， Cd， Pb， Cr， Zn，
    植被覆盖度， Shannon， 坡度， 坡向
"""
import csv
```

```
import io
import json
from app.utils.constants import HEAVY_METALS, ASPECT_TYPES,
    UNIT_TYPES
#  表头（顺序即导出模板顺序）
HEADERS = (["单元类型", "土石比", "土层厚度", "pH", "有机质"]
           + list(HEAVY_METALS)
           + ["植被覆盖度", "Shannon", "坡度", "坡向"])
FIELD_ALIASES = {
    "单元类型": "unit_type",
    "土石比": "soil_ratio",
    "土层厚度": "thickness",
    "pH": "ph", "PH": "ph", "ph": "ph",
    "有机质": "organic_matter",
    "植被覆盖度": "coverage",
    "Shannon": "shannon", "shannon": "shannon", "生物多样性":
    "shannon",
    "坡度": "slope",
    "坡向": "aspect",
}
for _m in HEAVY_METALS:
    FIELD_ALIASES[_m] = "hm_" + _m
def build_template_csv():
    """生成导入模板（含表头与示例行）。"""
    buf = io.StringIO()
    w = csv.writer(buf)
    w.writerow(HEADERS)
    #  示例：碎石边坡一行，说明每列含义
    row = {h: "" for h in HEADERS}
    row.update({"单元类型": "碎石边坡", "土石比": "0.34", "土层厚度": "10",
                "pH": "9.24", "有机质": "4.89", "Hg": "0.05",
    "As": "5.17",
                "Ni": "12.60", "Cu": "11.70", "Cd": "0.04",
    "Pb": "6.50",
                "植被覆盖度": "87.92", "Shannon": "0.60",
                "坡度": "28.75", "坡向": "阳坡"})
    w.writerow([row[h] for h in HEADERS])
    return buf.getvalue()
def _read_rows(filename, stream):
    """读取上传文件为「表头 -> 值」的字典列表。"""
    ext = filename.rsplit(".", 1)[-1].lower()
    if ext == "csv":
        text = stream.read().decode("utf-8-sig",
    errors="replace")
        reader = csv.DictReader(io.StringIO(text))
        return [dict(r) for r in reader]
    if ext in ("xlsx", "xls"):
        from openpyxl import load_workbook
        wb = load_workbook(io.BytesIO(stream.read()),
    data_only=True)
```

```
        ws = wb.active
        rows = list(ws.iter_rows(values_only=True))
        if not rows:
            return []
        header = [str(c).strip() if c is not None else ""
    for c in rows[0]]
        out = []
        for r in rows[1:]:
            if all(c is None or str(c).strip() == "" for c
    in r):
                continue
            out.append({header[i]: (r[i] if i < len(r) else
    None)
                        for i in range(len(header)) if
    header[i]})
        return out
    raise ValueError("不支持的文件类型：.%s" % ext)
def _num(value):
    if value is None:
        return None, None
    s = str(value).strip()
    if s == "":
        return None, None
    try:
        return float(s), None
    except ValueError:
        return None, "「%s」不是有效数值" % s
def parse_upload(filename, stream, unit_type_map):
    """解析上传文件。
    参数
    ----
    unit_type_map : dict, {单元类型: MicroUnit 对象}
    返回
    ----
    (parsed, errors)
      parsed : list[dict], 每项 {unit_type, fields{...},
    heavy_metals{}}
      errors : list[str], 逐行问题说明
    """
    rows = _read_rows(filename, stream)
    parsed, errors = [], []
    if not rows:
        return [], ["文件为空或没有数据行"]
    # 表头校验
    unknown = [h for h in (rows[0].keys() if rows else [])
    if h not in FIELD_ALIASES]
    if unknown:
        errors.append("存在无法识别的列，将被忽略：%s" %
    "、".join(map(str, unknown)))
    for idx, raw in enumerate(rows, start=2):       # 第 1 行是表头
```

```
        rec = {}
        for k, v in raw.items():
            key = FIELD_ALIASES.get(str(k).strip())
            if key:
                rec[key] = v
        unit_type = str(rec.get("unit_type") or "").strip()
        if not unit_type:
            errors.append("第 %d 行：缺少「单元类型」" % idx)
            continue
        if unit_type not in unit_type_map:
            errors.append("第 %d 行：图斑内不存在单元类型「%s」（现有：%s）"
                          % (idx, unit_type,
"、".join(unit_type_map) or "无"))
            continue
        if unit_type not in UNIT_TYPES:
            errors.append("第 %d 行：单元类型「%s」不是系统支持的三种之一"
                          % (idx, unit_type))
            continue
        fields, heavy = {}, {}
        ok = True
        for key, val in rec.items():
            if key in ("unit_type",):
                continue
            if key.startswith("hm_"):
                n, err = _num(val)
                if err:
                    errors.append("第 %d 行：%s %s" % (idx,
key[3:], err))
                    ok = False
                elif n is not None:
                    heavy[key[3:]] = n
                continue
            if key == "aspect":
                s = str(val).strip() if val is not None
else ""
                if s == "":
                    fields["aspect"] = None
                elif s in ASPECT_TYPES:
                    fields["aspect"] = s
                else:
                    errors.append("第 %d 行：坡向「%s」无效（应为 %s）"
                                  % (idx, s,
"、".join(ASPECT_TYPES)))
                    ok = False
                continue
            n, err = _num(val)
            if err:
                errors.append("第 %d 行：%s %s" % (idx, key,
err))
                ok = False
```

```
                else:                                                    6201
                    fields[key] = n                                      6202
            if ok:                                                       6203
                parsed.append({"unit_type": unit_type,                   6204
    "fields": fields,                                                    6205
                               "heavy_metals": heavy})                   6206
    return parsed, errors                                                6207
# ==== app/services/indicator_service.py ====                            6208
# -*- coding: utf-8 -*-                                                  6209
"""指标定义维护（M6）。                                                  6210
把论文表 2/表 3 的指标体系写入 indicators 表，使指标名称、单位、          6211
理想值与取值依据可在界面上查看维护，而不再只存在于代码常量中。          6212
代码常量（app.utils.constants）仍是计算时的**权威取值来源**；           6213
本表用于展示、核对与记录取值依据。二者由 seed_indicators() 保持同步。   6214
"""                                                                      6215
from app.models import db, Indicator                                     6216
from app.utils.constants import (                                        6217
    UNIT_TYPES, UNIT_ENABLED_INDICATORS, IDEAL_SOIL_RATIO,               6218
    IDEAL_THICKNESS,                                                     6219
    IDEAL_PH_RANGE, IDEAL_ORGANIC, IDEAL_COVERAGE,                       6220
    IDEAL_SHANNON,                                                       6221
    HEAVY_METAL_LIMIT, SLOPE_BREAK, ASPECT_SCORES,                       6222
)                                                                        6223
ALL_UNITS = ",".join(UNIT_TYPES)                                         6224
# (code, name, level, parent, unit, ideal_value,                         6225
    ideal_type, enabled_units, desc, is_constraint)                      6226
TREE = [                                                                 6227
    # ---- 一级指标 ----                                                6228
    ("soil", "土壤指标", 1, None, None, None, None, "碎石边坡,平台",     6229
    "论文表 2 评价指标之一级指标", False),                               6230
    ("veg", "植被指标", 1, None, None, None, None, ALL_UNITS,            6231
    "论文表 2 评价指标之一级指标", False),                               6232
    ("terrain", "地形指标", 1, None, None, None, None,                   6233
    "岩质边坡,碎石边坡", "论文表 2 评价指标之一级指标", False),         6234
    # ---- 二级指标：土壤 ----                                          6235
    ("soil_ratio", "土石比", 2, "soil", "—",                             6236
    str(IDEAL_SOIL_RATIO), "benefit", "碎石边坡,平台",                   6237
     "论文 4.2.2 节：理想值 0.8，根据矿山修复实际情况确定", False),     6238
    ("thickness", "土层厚度", 2, "soil", "cm",                           6239
    str(IDEAL_THICKNESS), "benefit", "碎石边坡,平台",                    6240
     "论文 4.2.2 节：理想值 20 cm", False),                              6241
    ("ph", "pH", 2, "soil", "—", "%g~%g" % IDEAL_PH_RANGE,               6242
    "mid", "碎石边坡,平台",                                              6243
     "论文 4.2.2 节：pH 用适中指标的标准化方法，理想区间 %g~%g" %        6244
    IDEAL_PH_RANGE, False),                                              6245
    ("organic_matter", "有机质", 2, "soil", "g/kg",                      6246
    str(IDEAL_ORGANIC), "benefit", "碎石边坡,平台",                      6247
     "《全国第二次土壤普查养分等级划分标准》二等（较丰富）最低值 30 g/kg", False),  6248
    ("pollution", "污染指标", 2, "soil", "mg/kg", "见三级指标",          6249
    "cost", "碎石边坡,平台",                                             6250
```

```
        "同时作为限制性指标与评价指标；8 种重金属等权合成（各 0.125）", True),
    # ---- 二级指标：植被 ----
    ("coverage", "植被覆盖度", 2, "veg", "%",
    str(IDEAL_COVERAGE), "benefit", ALL_UNITS,
        "论文 4.2.2 节：参照生态系统取值 75%", False),
    ("shannon", "生物多样性", 2, "veg", "—", str(IDEAL_SHANNON),
    "benefit", ALL_UNITS,
        "Shannon-Wiener 多样性指数，参照生态系统取值 0.7", False),
    # ---- 二级指标：地形 ----
    ("slope", "坡度", 2, "terrain", "°", "%g° 分界" %
    SLOPE_BREAK, "cost", "岩质边坡,碎石边坡",
        "论文 4.2.2 节引《新一轮退耕还林还草总体方案》发改西部〔2014〕1772 号：25°
    为陡坡重要分界", False),
    ("aspect", "坡向", 2, "terrain", "—",
    "阳坡0.2/阴阳坡0.6/阴坡1.0", "discrete", "岩质边坡,碎石边坡",
        "论文 4.2.2 节：阴坡及半阴坡水分条件更优，故赋值高于阳坡", False),
]
# 污染指标下的三级指标（8 种重金属）
for _m, _limit in HEAVY_METAL_LIMIT.items():
    TREE.append((
        "hm_" + _m, _m, 3, "pollution", "mg/kg",
    str(_limit), "cost",
        "碎石边坡,平台",
        "《绿化种植土壤》CJ/T 340—2016 土壤重金属含量 IV 级最高值 %g mg/kg" %
    _limit,
        False,
    ))
def seed_indicators():
    """写入/更新指标定义（幂等）。返回写入条数。"""
    code_map = {
        "benefit": "越大越优", "cost": "越小越优",
        "mid": "适中型", "discrete": "离散赋值",
    }
    n = 0
    for (code, name, level, parent, unit, ideal, itype,
    units, desc, is_con) in TREE:
        ind = Indicator.query.filter_by(code=code).first()
        if ind is None:
            ind = Indicator(code=code)
            db.session.add(ind)
        ind.name = name
        ind.level = level
        ind.parent_code = parent
        ind.unit = unit
        ind.ideal_value = ideal
        ind.ideal_type = itype
        ind.enabled_units = units
        ind.description = desc
        ind.is_constraint = is_con
        ind.is_evaluative = (level in (2, 3))
```

```
            n += 1                                                    6301
        db.session.commit()                                           6302
        return n                                                      6303
def ideal_type_label(code):                                           6304
        return {"benefit": "越大越优", "cost": "越小越优",              6305
                "mid": "适中型", "discrete": "离散赋值"}.get(code,      6306
        code or "—")                                                  6307
#  ==== app/services/workflow.py ====                                 6308
#  -*- coding: utf-8 -*-                                              6309
"""评价流程前置条件（M8/M9 的流程门禁）。                                6310
论文 4.1 节确定的评价流程存在严格的先后关系：                            6311
        建立图斑 → 划分微地质单元 → 限制性条件判定（通过）               6312
        → 录入指标数据 → 计算评价因子并判定等级 → 输出报告               6313
本模块把该顺序固化为可检查的状态。后续模块在入口处调用它，                6314
前置步骤未完成时直接拦截并给出「缺什么、去哪补」的提示，                  6315
避免在条件不具备时产出看似合理的评价结论。                              6316
"""                                                                   6317
from app.models import (db, MicroUnit, IndicatorData,                 6318
        ConstraintCheck)                                              6319
#  指标数据非空字段（用于判断某单元是否已录入数据）                       6320
DATA_FIELDS = ("soil_ratio", "thickness", "ph",                       6321
        "organic_matter",                                             6322
                "coverage", "shannon", "slope", "aspect")             6323
def _has_data(ind):                                                   6324
        if ind is None:                                               6325
            return False                                              6326
        if any(getattr(ind, f, None) is not None for f in             6327
        DATA_FIELDS):                                                 6328
            return True                                               6329
        return bool(ind.heavy_metals_dict())                          6330
def patch_progress(patch_id):                                         6331
        """检查某图斑的评价流程完成情况。                               6332
        返回 dict：                                                   6333
            units          {done, count, ratio_sum, detail}           6334
            constraint     {done, passed, detail}                     6335
            data           {done, ready, missing, detail}             6336
            blocking       未满足的前置步骤列表（按流程顺序）           6337
            ready          是否可进入定量评价                          6338
        """                                                           6339
        units = MicroUnit.query.filter_by(patch_id=patch_id).all      6340
        ()                                                            6341
        ratio_sum = sum(u.ratio or 0.0 for u in units)                6342
        #  ---- 步骤一：微地质单元 ----                                6343
        units_step = {                                                6344
            "done": bool(units),                                      6345
            "count": len(units),                                      6346
            "ratio_sum": ratio_sum,                                   6347
            "detail": ("已划分 %d 个微地质单元，面积占比合计 %.2f%%"      6348
                       % (len(units), ratio_sum * 100))               6349
                      if units else "尚未划分微地质单元",               6350
```

```
    }
    # ---- 步骤二：限制性条件判定 ----
    cc = ConstraintCheck.query.filter_by(patch_id=patch_id)\
        .order_by(ConstraintCheck.id.desc()).first()
    if cc is None:
        constraint_step = {"done": False, "passed": False,
                           "detail": "尚未执行限制性条件判定"}
    elif cc.passed:
        constraint_step = {"done": True, "passed": True,
                           "detail": "限制性条件判定已通过"}
    else:
        constraint_step = {"done": True, "passed": False,
                           "detail": "限制性条件判定未通过（%s）" %
    (cc.reason or "")}
    # ---- 步骤三：指标数据 ----
    missing = []
    for u in units:
        ind = IndicatorData.query.filter_by(unit_id=u.id).fi
    rst()
        if not _has_data(ind):
            missing.append(u.unit_type)
    if not units:
        data_step = {"done": False, "ready": False,
    "missing": [],
                     "detail": "尚无可录入数据的微地质单元"}
    elif missing:
        data_step = {"done": False, "ready": False,
    "missing": missing,
                     "detail": "以下微地质单元尚未录入指标数据：%s"
                               % "、".join(missing)}
    else:
        data_step = {"done": True, "ready": True,
    "missing": [],
                     "detail": "全部 %d 个微地质单元的指标数据已录入" %
    len(units)}
    ratio_ok = ratio_sum > 0.999 if units else False
    if units and not ratio_ok:
        units_step["detail"] += "（占比合计不等于 100%，请检查面积录入）"
    # ---- 汇总未满足的前置条件（按流程顺序） ----
    blocking = []
    if not units_step["done"]:
        blocking.append({
            "step": "units", "label": "划分微地质单元",
            "reason": "该图斑尚未划分微地质单元，无法确定各单元的面积权重。",
            "fix_url": "/unit/%d" % patch_id, "fix_label":
    "去划分微地质单元",
        })
    elif not ratio_ok:
        blocking.append({
```

```
                "step": "units", "label": "补齐单元面积",
                "reason": "各微地质单元面积占比合计不等于 100%，加权结果会失真。",
                "fix_url": "/unit/%d" % patch_id, "fix_label":
    "去核对单元面积",
            })
        if not constraint_step["done"]:
            blocking.append({
                "step": "constraint", "label": "执行限制性条件判定",
                "reason": "尚未执行限制性条件判定。论文流程要求先判定政策、区位约束、"
                          "土壤污染、地质灾害与耕地属性，通过后方可进入定量评价。",
                "fix_url": "/constraint/%d" % patch_id,
    "fix_label": "去执行限制性判定",
            })
        elif not constraint_step["passed"]:
            blocking.append({
                "step": "constraint", "label": "限制性条件判定未通过",
                "reason": "限制性条件判定未通过。按论文流程，此类矿山应直接判定为"
                          "「人工辅助或生态重建」，不进入定量评价。",
                "fix_url": "/constraint/%d" % patch_id,
    "fix_label": "去复核判定结果",
            })
        if units_step["done"] and not data_step["ready"]:
            blocking.append({
                "step": "data", "label": "录入指标数据",
                "reason": data_step["detail"] +
    "。缺少数据时计算会把空值按不适用跳过，"
                          "得出的结论不能反映实际情况。",
                "fix_url": "/indicator/", "fix_label":
    "去录入指标数据",
            })
        return {
            "units": units_step,
            "constraint": constraint_step,
            "data": data_step,
            "blocking": blocking,
            "ready": not blocking,
        }
# ==== app/templates/_macros.html ====
{# 共用宏：流程状态面板 #}
{# 评价流程步骤条：显示各前置步骤的完成情况 #}
{% macro workflow_steps(progress) %}
<div class="steps">
  <div class="step {% if progress.units.done %}step--done{%
    else %}step--todo{% endif %}">
    <span class="step__no">1</span>
    <span class="step__name">划分微地质单元</span>
    <span class="step__state">
      {% if progress.units.done %}已完成{% else %}待完成{% endif
    %}
    </span>
```

```
    </div>
    <div class="step {% if progress.constraint.passed
      %}step--done{% elif progress.constraint.done
      %}step--fail{% else %}step--todo{% endif %}">
      <span class="step__no">2</span>
      <span class="step__name">限制性条件判定</span>
      <span class="step__state">
        {% if progress.constraint.passed %}已通过
        {% elif progress.constraint.done %}未通过
        {% else %}待完成{% endif %}
      </span>
    </div>
    <div class="step {% if progress.data.ready %}step--done{%
      else %}step--todo{% endif %}">
      <span class="step__no">3</span>
      <span class="step__name">录入指标数据</span>
      <span class="step__state">
        {% if progress.data.ready %}已完成{% else %}待完成{% endif
      %}
      </span>
    </div>
    <div class="step step--next">
      <span class="step__no">4</span>
      <span class="step__name">执行适宜性评价</span>
      <span class="step__state">当前步骤</span>
    </div>
  </div>
{% endmacro %}
{# 前置条件未满足时的提示块，逐条给出原因与去处理入口 #}
{% macro workflow_blocking(progress) %}
{% if progress.blocking %}
<div class="notice notice--warn">
  <p class="notice__title">尚不能执行本步骤，请先完成以下前置工作：</p>
  <ol class="notice__list">
    {% for b in progress.blocking %}
    <li>
      <strong>{{ b.label }}</strong> —— {{ b.reason }}
      <a href="{{ b.fix_url }}">{{ b.fix_label }}</a>
    </li>
    {% endfor %}
  </ol>
  <p class="notice__hint">
    说明：本系统按论文 4.1 节的评价流程设置前置检查，
    避免在前置条件不具备时产出不合理的评价结论。
  </p>
</div>
{% endif %}
{% endmacro %}
{# 一行式状态摘要，用于列表页 #}
{% macro workflow_badge(progress) %}
```

```
{% if progress.ready %}
<span class="tag tag--ok">可执行评价</span>
{% elif progress.blocking %}
<span class="tag tag--warn">{{ progress.blocking[0].label }}</span>
<span class="muted">{{ progress.blocking[0].reason }}</span>
{% endif %}
{% endmacro %}
# ==== app/templates/ahp/experts.html ====
{% extends "base.html" %}
{% block title %}多专家判断矩阵{% endblock %}
{% block content %}
<div class="card">
  <h2>多专家判断矩阵 · 录入与汇总</h2>
  <p class="muted">
    论文明确权重由「组织 １０ 位专家对各项指标权重进行打分」确定。
    此页支持多位专家分别录入判断矩阵，再用几何平均汇总，
    保持矩阵互反性，并重新做一致性检验。
  </p>
  <form method="get" class="actions" style="margin-bottom:14px;">
    <div>
      <label class="field__label" for="unit-type">微地质单元类型</label>
      <select id="unit-type" name="unit_type"
              onchange="this.form.submit()">
        {% for t in ["岩质边坡", "碎石边坡", "平台"] %}
          <option value="{{ t }}" {% if unit_type == t %}selected{% endif %}>{{ t }}</option>
        {% endfor %}
      </select>
    </div>
    <div>
      <label class="field__label" for="level">指标层级</label>
      <select id="level" name="level"
              onchange="this.form.submit()">
        <option value="1" {% if level == 1 %}selected{% endif %}>一级指标</option>
        <option value="2" {% if level == 2 %}selected{% endif %}>二级指标</option>
      </select>
    </div>
  </form>
  <p class="muted">
    当前范围：<strong>{{ unit_type }}</strong> ·
    {% if level == 1 %}一级指标{% else %}二级指标（{{ l1 }}）{% endif %}。
    若选二级指标，请先到
    <a href="/ahp/">权重计算</a> 选择一级指标后进入本页。
  </p>
```

```
{% if rows %}
<table>
  <caption class="muted">已录入 {{ rows|length }} 位专家的矩阵</caption>
  <thead>
    <tr>
      <th scope="col">专家</th>
      <th scope="col">矩阵规模</th>
      <th scope="col">该矩阵 CR</th>
      <th scope="col">一致性</th>
      <th scope="col">录入时间</th>
      <th scope="col">操作</th>
    </tr>
  </thead>
  <tbody>
    {% for r in rows %}
    <tr>
      <td>{{ r.expert_name }}</td>
      <td class="num">{{ n }} × {{ n }}</td>
      <td class="mono">{{ "%.4f"|format(r.cr) if r.cr is not none else "—" }}</td>
      <td>
        {% if r.cr is not none and r.cr < 0.1 %}<span class="tag tag--ok">通过</span>{% else %}<span class="tag tag--warn">不通过</span>{% endif %}
      </td>
      <td class="mono nowrap">{{ r.created_at.strftime("%Y-%m-%d %H:%M") if r.created_at else "—" }}</td>
      <td>
        <form method="post" action="/ahp/experts/{{ r.id }}/delete"
              onsubmit="return confirm('确定删除专家「{{ r.expert_name }}」的矩阵吗？');">
          {{ form.hidden_tag() }}
          <button type="submit" class="btn btn--sm btn--danger">删除</button>
        </form>
      </td>
    </tr>
    {% endfor %}
  </tbody>
</table>
<p class="actions" style="margin-bottom:10px;">
  <a class="btn" href="/ahp/experts/aggregate?unit_type={{ unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif %}">
    汇总计算（几何平均）
  </a>
</p>
```

```
{% else %}
<p class="muted">尚未录入任何专家矩阵。请在下方录入第一位专家的判断矩阵。</p>
{% endif %}
<h3>录入一位专家的判断矩阵</h3>
<form method="post">
  {{ form.hidden_tag() }}
  <input type="hidden" name="action" value="add">
  <input type="hidden" name="unit_type" value="{{ unit_type }}">
  <input type="hidden" name="level" value="{{ level }}">
  {% if l1 %}<input type="hidden" name="l1" value="{{ l1 }}">{% endif %}
  <div class="field">
    <label class="field__label" for="expert-name">专家姓名</label>
    <input id="expert-name" type="text" name="expert_name"
           style="width:220px;" placeholder="如：张工">
  </div>
  <table style="width:auto;">
    <thead>
      <tr>
        <th></th>
        {% for name in names %}<th scope="col">{{ name }}</th>{% endfor %}
      </tr>
    </thead>
    <tbody>
      {% for i in range(n) %}
      <tr>
        <th scope="row">{{ names[i] }}</th>
        {% for j in range(n) %}
          {% if i == j %}
          <td class="muted num">1</td>
          {% elif i < j %}
          <td>
            <label class="muted" style="position:absolute;left:-9999px;"
                   for="e-{{ i }}-{{ j }}">{{ names[i] }} 相对 {{ names[j] }} 的重要性</label>
            <input id="e-{{ i }}-{{ j }}" type="number" step="0.01" min="0.01"
                   name="m_{{ i }}_{{ j }}" value="{{ default[i][j] }}">
          </td>
          {% else %}
          <td class="muted num">{{ default[i][j] }}</td>
          {% endif %}
        {% endfor %}
      </tr>
      {% endfor %}
```

```
        </tbody>                                                    6651
      </table>                                                      6652
      <p class="actions" style="margin-top:12px;">                  6653
        <button type="submit" class="btn">录入该专家矩阵</button>    6654
        <a class="btn btn--ghost" href="/ahp/indicators">指标维护<  6655
      /a>                                                           6656
        <a class="btn btn--ghost" href="/ahp/">返回权重计算</a>     6657
      </p>                                                          6658
    </form>                                                         6659
</div>                                                              6660
{% endblock %}                                                      6661
# ==== app/templates/ahp/experts_aggregate.html ====                6662
{% extends "base.html" %}                                           6663
{% block title %}多专家矩阵汇总{% endblock %}                       6664
{% block content %}                                                 6665
<div class="card">                                                  6666
  <h2>多专家矩阵汇总 · {{ unit_type }} · {% if level == 1 %}一级指标{% 6667
    else %}二级指标（{{ l1 }}）{% endif %}</h2>                     6668
  <p class="muted">                                                 6669
    对各位专家的判断矩阵按对应元素取几何平均：                      6670
    a<sub>ij</sub> = (a<sub>ij</sub><sup>(1)</sup> × … ×            6671
    a<sub>ij</sub><sup>(k)</sup>)<sup>1/k</sup>。                   6672
    该方式可保持矩阵的互反性，汇总后重新计算权重与一致性比例。     6673
  </p>                                                              6674
  {% if not result %}                                               6675
  <p class="muted">                                                 6676
    该范围内尚未录入专家矩阵。                                      6677
    <a href="/ahp/experts?unit_type={{ unit_type                    6678
    }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif             6679
    %}">去录入</a>                                                  6680
  </p>                                                              6681
  {% else %}                                                        6682
  <div class="stat-row">                                            6683
    <div>                                                           6684
      <div class="stat__label">参与汇总的专家</div>                 6685
      <div class="stat__value">{{ result.count }} 位</div>          6686
    </div>                                                          6687
    <div>                                                           6688
      <div class="stat__label">λmax</div>                           6689
      <div class="stat__value mono">{{ result.lambda_max            6690
    }}</div>                                                        6691
    </div>                                                          6692
    <div>                                                           6693
      <div class="stat__label">CI</div>                             6694
      <div class="stat__value mono">{{                              6695
    "%.4f"|format(result.ci) }}</div>                               6696
    </div>                                                          6697
    <div>                                                           6698
      <div class="stat__label">CR（要求 &lt; 0.1）</div>            6699
      <div class="stat__value mono">{{                              6700
```

```
    "%.4f"|format(result.cr) }}</div>                                  6701
    </div>                                                             6702
    <div>                                                              6703
      <div class="stat__label">一致性检验</div>                        6704
      <div class="stat__value">                                        6705
        {% if result.consistent %}<span class="ok">通过</span>         6706
        {% else %}<span class="warn">不通过</span>{% endif %}          6707
      </div>                                                           6708
    </div>                                                             6709
  </div>                                                               6710
  <h3>汇总后权重</h3>                                                  6711
  <table style="width:auto;">                                          6712
    <thead>                                                            6713
      <tr>                                                             6714
        <th scope="col">指标</th>                                      6715
        {% for name in names %}<th scope="col">{{ name                 6716
    }}</th>{% endfor %}                                                6717
      </tr>                                                            6718
    </thead>                                                           6719
    <tbody>                                                            6720
      <tr>                                                             6721
        <th scope="row">权重</th>                                      6722
        {% for w in result.weights %}<td class="mono">{{ w             6723
    }}</td>{% endfor %}                                                6724
      </tr>                                                            6725
    </tbody>                                                           6726
  </table>                                                             6727
  <h3>各专家矩阵的一致性</h3>                                          6728
  <table style="width:auto;">                                          6729
    <thead>                                                            6730
      <tr><th scope="col">专家</th><th scope="col">该矩阵              6731
    CR</th><th scope="col">判定</th></tr>                              6732
    </thead>                                                           6733
    <tbody>                                                            6734
      {% for r in rows %}                                              6735
      <tr>                                                             6736
        <td>{{ r.expert_name }}</td>                                   6737
        <td class="mono">{{ "%.4f"|format(r.cr) if r.cr is             6738
    not none else "—" }}</td>                                          6739
        <td>                                                           6740
          {% if r.cr is not none and r.cr < 0.1 %}<span                6741
    class="ok">通过</span>                                             6742
          {% else %}<span class="warn">不通过</span>{% endif %}        6743
        </td>                                                          6744
      </tr>                                                            6745
      {% endfor %}                                                     6746
    </tbody>                                                           6747
  </table>                                                             6748
  {% if not result.consistent and result.most_inconsistent             6749
    %}                                                                 6750
```

```
  <p style="padding:12px;border-radius:6px;background:var(--danger-soft);
             border-left:4px solid var(--danger);">
    <strong class="warn">汇总矩阵一致性不通过，建议优先调整：</strong>
    第 {{ result.most_inconsistent.i + 1 }} 行、第 {{ result.most_inconsistent.j + 1 }} 列的比较
    （当前 <span class="mono">{{ result.most_inconsistent.value|round(3) }}</span>，
    理想约 <span class="mono">{{ result.most_inconsistent.expected|round(3) }}</span>）。
    该偏差通常来自个别专家打分，可先核对一致性较差的专家矩阵。
  </p>
  {% endif %}
  {% endif %}
  <p class="actions" style="margin-top:16px;">
    <a class="btn btn--ghost" href="/ahp/experts?unit_type={{ unit_type }}&level={{ level }}{% if l1 %}&l1={{ l1 }}{% endif %}">返回录入</a>
    <a class="btn btn--ghost" href="/ahp/">返回权重计算</a>
  </p>
</div>
{% endblock %}
# ==== app/templates/ahp/indicator_edit.html ====
{% extends "base.html" %}
{% block title %}编辑指标 · {{ ind.name }}{% endblock %}
{% block content %}
<div class="card" style="max-width:680px;">
  <h2>编辑指标 · {{ ind.name }}</h2>
  <p class="muted">
    指标编码 <span class="mono">{{ ind.code }}</span>、层级 {{ ind.level }} 级，
    由论文指标体系确定，不可修改。
  </p>
  <form method="post">
    {{ form.hidden_tag() }}
    <div class="field">
      <label class="field__label" for="name">指标名称</label>
      <input id="name" type="text" name="name"
             style="width:100%;" required
             value="{{ ind.name }}">
    </div>
    <div class="field">
      <label class="field__label" for="unit">计量单位</label>
      <input id="unit" type="text" name="unit"
             style="width:180px;"
             value="{{ ind.unit or '' }}">
    </div>
    <div class="field">
      <label class="field__label" for="ideal-type">标准化类型</label
```

```
    <select id="ideal-type" name="ideal_type">
      {% for code, label in [("benefit","越大越优"),("cost","越小越优"),("mid","适中型"),("discrete","离散赋值")] %}
      <option value="{{ code }}" {% if ind.ideal_type == code %}selected{% endif %}>{{ label }}</option>
      {% endfor %}
    </select>
  </div>
  <div class="field">
    <label class="field__label" for="ideal-value">理想取值</label>
    <input id="ideal-value" type="text" name="ideal_value" style="width:100%;"
           value="{{ ind.ideal_value or '' }}">
    <p class="muted" style="margin:4px 0 0;">
      如土石比 0.8、土层厚度 20、pH 5~9、植被覆盖度 75。
    </p>
  </div>
  <fieldset style="border:1px solid var(--line);border-radius:6px;padding:10px 14px;margin:0 0 12px;">
    <legend style="padding:0 6px;font-weight:600;">启用该指标的微地质单元</legend>
    {% for t in unit_types %}
    <label class="checkbox" style="margin-right:18px;">
      <input type="checkbox" name="enabled_units" value="{{ t }}"
             {% if t in ind.units_list() %}checked{% endif %}>
      <span>{{ t }}</span>
    </label>
    {% endfor %}
    <p class="muted" style="margin:8px 0 0;">
      论文设定：岩质边坡不启用土壤指标，平台不启用地形指标。
    </p>
  </fieldset>
  <div class="field">
    <label class="checkbox" for="is-constraint">
      <input id="is-constraint" type="checkbox" name="is_constraint" value="1"
             {% if ind.is_constraint %}checked{% endif %}>
      <span>同时作为限制性指标（参与一票否决判定）</span>
    </label>
  </div>
  <div class="field">
    <label class="field__label" for="description">取值依据说明</label>
    <textarea id="description" name="description" style="min-height:88px;">{{ ind.description or '' }}</textarea>
```

```
      </div>
      <p class="actions" style="margin-top:16px;">
        <button type="submit" class="btn">保存</button>
        <a class="btn btn--ghost" href="/ahp/indicators">返回指标列表</a>
      </p>
    </form>
</div>
{% endblock %}
# ==== app/templates/ahp/indicators.html ====
{% extends "base.html" %}
{% block title %}指标定义维护{% endblock %}
{% block content %}
<div class="card">
  <h2>指标体系维护</h2>
  <p class="muted">
    按论文表 2 与表 3 建立的指标树。此处可核对指标名称、单位、标准化类型、
    理想取值与取值依据，并按需调整启用该指标的微地质单元。
    计算时的权威取值仍以 <span class="mono">app/utils/constants.py</span> 为准，
    本页用于展示与核对。
  </p>
  <p class="actions" style="margin-bottom:14px;">
    <a class="btn btn--sm btn--ghost"
    href="/ahp/">返回权重计算</a>
    <a class="btn btn--sm btn--ghost"
    href="/ahp/experts">多专家矩阵</a>
    <form method="post" action="/ahp/indicators/reset"
    style="display:inline;"
          onsubmit="return confirm('将按论文取值重置全部指标定义，自定义修改会丢失。
    确定继续？');">
      {{ form.hidden_tag() }}
      <button type="submit" class="btn btn--sm
    btn--danger">按论文取值重置</button>
    </form>
  </p>
  {% for level, title in [(1, "一级指标"), (2, "二级指标"), (3,
    "三级指标")] %}
  {% set level_items = all_ind|selectattr("level",
    "equalto", level)|list %}
  {% if level_items %}
  <h3>{{ title }}（{{ level_items|length }} 项）</h3>
  <table>
    <thead>
      <tr>
        <th scope="col">指标</th>
        <th scope="col">上级指标</th>
        <th scope="col">单位</th>
        <th scope="col">标准化类型</th>
        <th scope="col">理想取值</th>
```

```
          <th scope="col">启用单元</th>                                    6901
          <th scope="col">限制性</th>                                      6902
          <th scope="col">操作</th>                                        6903
        </tr>                                                              6904
      </thead>                                                             6905
      <tbody>                                                              6906
        {% for i in level_items %}                                         6907
        <tr>                                                               6908
          <td style="text-align:left;">                                    6909
            <span class="mono muted">{{ i.code                             6910
      }}</span><br>{{ i.name }}                                            6911
          </td>                                                            6912
          <td>{{ (all_ind|selectattr("code", "equalto",                   6913
      i.parent_code)|list|first).name if i.parent_code else                6914
      "—" }}</td>                                                          6915
          <td>{{ i.unit or "—" }}</td>                                     6916
          <td>{{ type_label(i.ideal_type) }}</td>                          6917
          <td class="mono">{{ i.ideal_value or "—" }}</td>                 6918
          <td style="text-align:left;">{{ i.enabled_units or               6919
      "—" }}</td>                                                          6920
          <td>                                                             6921
            {% if i.is_constraint %}<span class="tag                       6922
      tag--warn">是</span>                                                 6923
            {% else %}<span class="tag tag--muted">否</span>{%             6924
      endif %}                                                             6925
          </td>                                                            6926
          <td><a href="/ahp/indicators/{{ i.code                           6927
      }}/edit">编辑</a></td>                                               6928
        </tr>                                                              6929
        {% endfor %}                                                       6930
      </tbody>                                                             6931
    </table>                                                               6932
    {% endif %}                                                            6933
    {% endfor %}                                                           6934
</div>                                                                     6935
{% endblock %}                                                             6936
# ==== app/templates/auth/edit_user.html ====                              6937
{% extends "base.html" %}                                                  6938
{% block title %}编辑用户 · {{ user.username }}{% endblock %}              6939
{% block content %}                                                        6940
<div class="card" style="max-width:560px;">                                6941
  <h2>编辑用户 · {{ user.username }}</h2>                                  6942
  <form method="post">                                                     6943
    {{ form.hidden_tag() }}                                                6944
    <div class="field">                                                    6945
      <label class="field__label" for="role">角色</label>                  6946
      <select id="role" name="role">                                       6947
        {% for code, label in role_labels.items() %}                       6948
        <option value="{{ code }}" {% if user.role == code                 6949
      %}selected{% endif %}>{{ label }}</option>                           6950
```

```
            {% endfor %}
          </select>
        </div>
        <div class="field">
          <label class="checkbox" for="active">
            <input id="active" type="checkbox" name="active"
        value="1"
                   {% if user.active %}checked{% endif %}>
            <span>启用该账号</span>
          </label>
          <p class="muted" style="margin:6px 0 0;">
            停用后该账号无法登录，但其历史操作记录会保留。
            系统至少需保留一名启用状态的管理员。
          </p>
        </div>
        <div class="field">
          <label class="field__label" for="password">重置密码（留空则不修改
        ）</label>
          <input id="password" type="password" name="password"
                 autocomplete="new-password"
        style="width:280px;"
                 placeholder="留空表示保持原密码">
        </div>
        <p class="actions" style="margin-top:18px;">
          <button type="submit" class="btn">保存</button>
          <a class="btn btn--ghost" href="/users">返回</a>
        </p>
      </form>
</div>
{% endblock %}
# ==== app/templates/auth/logs.html ====
{% extends "base.html" %}
{% block title %}操作日志{% endblock %}
{% block content %}
<div class="card">
  <h2>操作日志</h2>
  <p class="muted">
    系统自动记录所有写操作（新建、编辑、删除、导入、执行评价等），
    便于追溯数据变更。失败的操作也会记录并标注 HTTP 状态码。
  </p>
  <p class="actions" style="margin-bottom:12px;">
    <a class="btn btn--sm btn--ghost"
    href="/users">返回用户管理</a>
  </p>
  {% if not records %}
  <p class="muted">暂无操作记录。</p>
  {% else %}
  <table>
    <caption class="muted">共 {{ total }} 条记录，当前第 {{ page }}
    / {{ pages }} 页</caption>
```

```
      <thead>                                                                  7001
        <tr>                                                                   7002
          <th scope="col">时间</th>                                            7003
          <th scope="col">操作人</th>                                          7004
          <th scope="col">操作</th>                                            7005
          <th scope="col">对象</th>                                            7006
          <th scope="col">说明</th>                                            7007
        </tr>                                                                  7008
      </thead>                                                                 7009
      <tbody>                                                                  7010
        {% for r in records %}                                                 7011
        <tr>                                                                   7012
          <td class="mono nowrap">{{ r.created_at.strftime("%Y                 7013
      -%m-%d %H:%M:%S") if r.created_at else "—" }}</td>                       7014
          <td>{{ users_map.get(r.user_id, "—") }}</td>                         7015
          <td style="text-align:left;">{{ r.action }}</td>                     7016
          <td>                                                                 7017
            {% if r.target_type %}{{ r.target_type }}{% if                     7018
      r.target_id %}#{{ r.target_id }}{% endif %}                              7019
            {% else %}—{% endif %}                                             7020
          </td>                                                                7021
          <td style="text-align:left;">{{ r.detail or "—"                      7022
      }}</td>                                                                  7023
        </tr>                                                                  7024
        {% endfor %}                                                           7025
      </tbody>                                                                 7026
    </table>                                                                   7027
    {% if pages > 1 %}                                                         7028
    <nav class="actions" aria-label="日志分页">                                7029
      {% if page > 1 %}                                                        7030
      <a class="btn btn--sm btn--ghost" href="/logs?page={{                    7031
      page - 1 }}">上一页</a>                                                  7032
      {% endif %}                                                              7033
      <span class="muted">第 {{ page }} / {{ pages }} 页</span>                7034
      {% if page < pages %}                                                    7035
      <a class="btn btn--sm btn--ghost" href="/logs?page={{                    7036
      page + 1 }}">下一页</a>                                                  7037
      {% endif %}                                                              7038
    </nav>                                                                     7039
    {% endif %}                                                                7040
    {% endif %}                                                                7041
</div>                                                                         7042
{% endblock %}                                                                 7043
# ==== app/templates/error.html ====                                           7044
{% extends "base.html" %}                                                      7045
{% block title %}{{ code }} · {{ message }}{% endblock %}                      7046
{% block content %}                                                            7047
<div class="card" style="max-width:560px;">                                    7048
    <h2>{{ code }} · {{ message }}</h2>                                        7049
    <p class="muted">                                                          7050
```

```
      请返回上一页重试；若确认该操作应当可用，请检查当前账号的角色权限。
    </p>
    <p class="actions" style="margin-top:16px;">
      <a class="btn" href="/">返回首页</a>
    </p>
</div>
{% endblock %}
# ==== app/templates/indicator/import.html ====
{% extends "base.html" %}
{% block title %}批量导入指标数据{% endblock %}
{% block content %}
<div class="card" style="max-width:860px;">
    <h2>批量导入指标数据</h2>
    <p class="muted">
      支持 CSV 与 Excel（.xlsx/.xls）。文件每行对应一个微地质单元，
      系统按「单元类型」匹配图斑内已划分的单元后写入数据。
      先下载模板按格式填写，可减少出错。
    </p>
    <p class="actions" style="margin-bottom:14px;">
      <a class="btn btn--sm btn--ghost"
      href="/indicator/import/template">下载导入模板（CSV）</a>
      <a class="btn btn--sm btn--ghost"
      href="/indicator/">返回指标数据</a>
    </p>
    <form method="post" enctype="multipart/form-data">
      {{ form.hidden_tag() }}
      <div class="field">
        <label class="field__label" for="patch-id">目标矿山图斑</lab
      el>
        <select id="patch-id" name="patch_id" required>
          <option value="">— 请选择 —</option>
          {% for p in patches %}
          <option value="{{ p.id }}">{{ p.name }} ({{
      p.units.count() }} 个单元）</option>
          {% endfor %}
        </select>
      </div>
      <div class="field">
        <label class="field__label" for="datafile">数据文件</label
      >
        <input id="datafile" type="file" name="datafile"
               accept=".csv,.xlsx,.xls" required>
      </div>
      <p class="actions" style="margin-top:16px;">
        <button type="submit" class="btn">开始导入</button>
      </p>
    </form>
    <h3>模板列说明</h3>
    <p class="muted">
      共 {{ headers|length }} 列。留空表示该指标「不适用」。
```

```
      坡向须填「阳坡 / 阴阳坡 / 阴坡」之一。未检测的重金属留空即可
      （按论文注释视为未受污染）。
    </p>
    <p class="mono muted" style="line-height:1.9;">{{
      headers|join(" , ") }}</p>
    <h3>导入规则</h3>
    <ul class="muted" style="padding-left:20px;line-height:1.9
      ;">
      <li>「单元类型」必须与图斑内已划分的单元一致，否则该行会被跳过并在结果中提示。</li>
      <li>同一单元类型若在文件中出现多行，以最后一行为准。</li>
      <li>已有数据的单元会被覆盖更新；没有数据的单元将新建记录。</li>
      <li>数值列请填纯数字，不要带单位（如填 <span class="mono">10</span> 而非
      <span class="mono">10cm</span>）。</li>
    </ul>
</div>
{% if report %}
<div class="card" style="max-width:860px;">
    <h2>导入结果</h2>
    <div class="stat-row">
      <div>
        <div class="stat__label">目标图斑</div>
        <div style="font-size:15px;">{{ report.patch }}</div>
      </div>
      <div>
        <div class="stat__label">成功导入</div>
        <div class="stat__value ok">{{ report.saved }}
      个单元</div>
      </div>
      <div>
        <div class="stat__label">问题行</div>
        <div class="stat__value {% if report.errors %}warn{%
      endif %}">{{ report.errors|length }}</div>
      </div>
    </div>
    {% if report.errors %}
    <h3>未导入的行</h3>
    <ul class="muted" style="padding-left:20px;line-height:1.9
      ;">
      {% for e in report.errors %}<li>{{ e }}</li>{% endfor %}
    </ul>
    {% else %}
    <p class="ok" style="margin-top:12px;">全部数据行均已成功导入。</p>
    {% endif %}
    <p class="actions" style="margin-top:16px;">
      <a class="btn btn--ghost" href="/indicator/">查看指标数据</a>
    </p>
</div>
{% endif %}
{% endblock %}
# ==== app/templates/report/print.html ====
```

```
<!DOCTYPE html>
<html lang="zh-CN">
<head>
<meta charset="UTF-8">
<title>自然恢复适宜性评价报告 · {{ patch.name }}</title>
<style>
  @page { size: A4; margin: 18mm 16mm; }
  * { box-sizing: border-box; }
  body {
    margin: 0; color: #23262b; background: #fff;
    font-family: "Microsoft YaHei", "SimSun", sans-serif;
    font-size: 10.5pt; line-height: 1.65;
  }
  h1 { font-size: 17pt; text-align: center; margin: 0 0
    6px; }
  h2 { font-size: 13pt; color: #1f3a5f; margin: 20px 0 8px;
       border-bottom: 1px solid #dfe3e8; padding-bottom:
    4px; }
  h3 { font-size: 11.5pt; color: #1f3a5f; margin: 16px 0
    6px; }
  .sub { text-align: center; color: #6b7280; font-size:
    9.5pt; margin-bottom: 18px; }
  table { width: 100%; border-collapse: collapse; margin:
    8px 0 14px;
          font-variant-numeric: tabular-nums; }
  th, td { border: 1px solid #c9ccd1; padding: 4px 8px;
    text-align: center; }
  th { background: #eef2f7; font-weight: 600; }
  .left { text-align: left; }
  .mono { font-family: Consolas, "Courier New", monospace; }
  .kv { width: auto; }
  .kv th { width: 30%; text-align: left; }
  .kv td { text-align: left; }
  ul { margin: 6px 0 12px; padding-left: 20px; }
  .note { color: #6b7280; font-size: 9.5pt; }
  .chart { margin: 10px 0 16px; }
  .sig { margin-top: 28px; font-size: 10pt; }
  .avoid-break { break-inside: avoid; }
  svg { max-width: 100%; }
</style>
</head>
<body>
  <h1>历史遗留废弃矿山自然恢复适宜性评价报告</h1>
  <p class="sub">
    {{ patch.name }}
    {% if patch.admin_region %} · {{ patch.admin_region
    }}{% endif %}
    <br>历史遗留废弃矿山自然恢复适宜性评价系统 Ｖ１.０ ·
    生成时间 {{ generated_at }}
  </p>
```

```
<h2>一、图斑基本信息</h2>
<table class="kv">
  <tr><th>图斑名称</td><td>{{ patch.name }}</td></tr>
  <tr><th>所在行政区</td><td>{{ patch.admin_region or "—"
  }}</td></tr>
  <tr><th>图斑面积（公顷）</td><td>{{ patch.area_ha if
  patch.area_ha is not none else "—" }}</td></tr>
  <tr><th>开采矿种</td><td>{{ patch.mineral_type or "—"
  }}</td></tr>
  <tr><th>开采方式</td><td>{{ patch.mining_method or "—"
  }}</td></tr>
  <tr><th>责任主体缺失</td><td>{{ "是" if patch.responsible_entit
  y_missing else "否" }}</td></tr>
  {% if reference %}
  <tr><th>参照生态系统</td>
      <td>植被覆盖度基准 {{ reference.coverage_benchmark }}%，
          多样性指数基准 {{ reference.shannon_benchmark
  }}</td></tr>
  {% endif %}
</table>
<h2>二、评价结论</h2>
<table class="kv">
  <tr><th>最终评价因子 F</td><td class="mono"><strong>{{
  "%.4f"|format(result.final_score) }}</strong></td></tr>
  <tr><th>适宜性等级</td><td><strong>{{ result.grade
  }}</strong></td></tr>
  <tr><th>判定依据</td><td>{{ result.grade_info.meaning
  }}</td></tr>
  <tr><th>推荐措施</td><td>{{ result.grade_info.measure
  }}</td></tr>
</table>
<h2>三、各微地质单元得分</h2>
<table>
  <thead>
    <tr><th>微地质单元</th><th>面积占比</th><th>单元评价因子
  F'</th><th>加权贡献</th></tr>
  </thead>
  <tbody>
    {% for u in result.unit_results %}
    <tr>
      <td>{{ u.unit_type }}</td>
      <td class="mono">{{ "%.2f%%"|format(u.ratio * 100)
  }}</td>
      <td class="mono">{{ "%.4f"|format(u.unit_score)
  }}</td>
      <td class="mono">{{ "%.4f"|format(u.weighted)
  }}</td>
    </tr>
    {% endfor %}
  </tbody>
```

```
</table>
<div class="chart">{{ chart_units|safe }}</div>
<h2>四、一级指标权重分布</h2>
<div class="chart">{{ chart_weight|safe }}</div>
<h2>五、得分明细</h2>
{% for u in result.unit_results %}
<h3>{{ u.unit_type }}（占比 {{ "%.2f%%"|format(u.ratio *
100) }}，单元得分 {{ "%.4f"|format(u.unit_score) }}）</h3>
<table>
  <thead>
    <tr>
      <th>一级指标</th><th>二级指标</th><th>实测值</th>
      <th>标准化值</th><th>权重</th><th>贡献</th>
    </tr>
  </thead>
  <tbody>
    {% for d in u.details %}
    <tr>
      <td>{{ d.l1 }}</td>
      <td class="left">{{ d.l2 }}</td>
      <td class="mono left">{{ "—" if d.raw is none else
d.raw }}</td>
      <td class="mono">{{ "—" if d.std_value is none else
"%.4f"|format(d.std_value) }}</td>
      <td class="mono">{{ "%.4f"|format(d.weight) }}</td>
      <td class="mono">{{ "—" if d.contribution is none
else "%.4f"|format(d.contribution) }}</td>
    </tr>
    {% endfor %}
  </tbody>
</table>
<p class="note">
  一级指标得分：
  {% for k, v in u.l1_scores.items() %}{{ k }} = {{
  "%.4f"|format(v) }}{% if not loop.last %}；{% endif %}{%
  endfor %}
</p>
{% endfor %}
<h2>六、评价依据说明</h2>
<ul class="note">
  <li>指标标准化按论文式（８） 执行，按指标性质分为越大越优、越小越优、
      适中型与离散赋值四类。</li>
  <li>单元评价因子按论文式（９） 由二级指标权重加权求得；
      整矿评价因子由各微地质单元按面积占比加权合成。</li>
  <li>适宜性等级按论文表 4 划分：F&lt;０．３ 不适宜、０．３≤F≤０．７ 基本适宜、F≥０．７
  适宜。</li>
  <li>指标权重取自论文表 3，按岩质边坡、碎石土边坡、平台三类微地质单元分别确定。</li>
</ul>
<p class="sig">
  评价人（签字）：＿＿＿＿＿＿＿＿＿＿＿＿＿＿＿＿＿＿　　日期：＿＿＿＿＿＿＿＿＿＿＿＿
```

```
  </p>
</body>
</html>
# ==== app/templates/unit/list.html ====
{% extends "base.html" %}
{% block title %}微地质单元{% endblock %}
{% block content %}
<div class="card">
  <h2>微地质单元</h2>
  <p class="muted">
    按论文 4.1 节将矿山图斑划分为岩质边坡、碎石边坡、平台三类微地质单元。
    面积占比由面积自动计算，即论文式（9） 中决定各单元权重的系数。
  </p>
  {% if not rows %}
  <p class="muted">
    暂无矿山图斑。请先到「项目与矿山档案」建立图斑。
  </p>
  {% else %}
  <table>
    <thead>
      <tr>
        <th scope="col">矿山图斑</th>
        <th scope="col">已划分单元</th>
        <th scope="col">占比合计</th>
        <th scope="col">操作</th>
      </tr>
    </thead>
    <tbody>
      {% for r in rows %}
      <tr>
        <td style="text-align:left;">{{ r.patch.name }}</td>
        <td class="num">{{ r.count }} 个</td>
        <td class="num">
          {% if r.count %}
            {{ "%.2f%%"|format(r.ratio_sum * 100) }}
            {% if r.ratio_sum < 0.999 or r.ratio_sum >
    1.001 %}
              <span class="warn">（不等于 100%）</span>
            {% endif %}
          {% else %}—{% endif %}
        </td>
        <td class="nowrap">
          <a href="/unit/{{ r.patch.id }}">划分与维护</a>
           | <a href="/indicator/">录入指标数据</a>
        </td>
      </tr>
      {% endfor %}
    </tbody>
  </table>
  {% endif %}
```

```
</div>
{% endblock %}
# ==== app/utils/__init__.py ====
# -*- coding: utf-8 -*-
# ==== app/utils/browser.py ====
# -*- coding: utf-8 -*-
"""HTML 转 PDF（M9 报告导出）。
复用本机已安装的 Chrome/Edge 无头模式打印，避免引入 reportlab、
weasyprint 等额外依赖。生成过程使用系统临时目录中的 ASCII 文件名，
避免中文路径在 file:// URL 下出问题。
"""
import os
import shutil
import subprocess
import tempfile
import uuid
CANDIDATES = [
    r"C:\Program Files\Google\Chrome\Application\chrome.exe",
    r"C:\Program Files (x86)\Google\Chrome\Application\chrome.exe",
    r"C:\Program Files (x86)\Microsoft\Edge\Application\msedge.exe",
    r"C:\Program Files\Microsoft\Edge\Application\msedge.exe",
]
# 允许通过环境变量覆盖，便于在其他机器上指定浏览器
ENV_KEY = "DSH_BROWSER_PATH"
def browser_path():
    """定位可用的浏览器可执行文件，找不到返回 None。"""
    override = os.environ.get(ENV_KEY)
    if override and os.path.isfile(override):
        return override
    for p in CANDIDATES:
        if os.path.isfile(p):
            return p
    for name in ("chrome", "msedge", "chromium"):
        found = shutil.which(name)
        if found:
            return found
    return None
def html_to_pdf(html, timeout=90):
    """把 HTML 字符串渲染为 PDF，返回字节；失败返回 (None, 原因)。"""
    exe = browser_path()
    if exe is None:
        return None, ("未找到可用的浏览器（Chrome/Edge），无法生成 PDF。"
                      "可设置环境变量 %s 指向浏览器可执行文件。" % ENV_KEY)
    tmpdir = tempfile.mkdtemp(prefix="dshpdf_")
    tag = uuid.uuid4().hex[:8]
    html_path = os.path.join(tmpdir, "r_%s.html" % tag)
```

```
    pdf_path = os.path.join(tmpdir, "r_%s.pdf" % tag)
    try:
        with open(html_path, "w", encoding="utf-8") as f:
            f.write(html)
        cmd = [
            exe, "--headless=new", "--disable-gpu",
    "--no-sandbox",
            "--no-first-run", "--no-default-browser-check",
            "--no-pdf-header-footer",
            "--virtual-time-budget=4000",
            "--print-to-pdf=%s" % pdf_path,
            "file:///" + html_path.replace("\\", "/"),
        ]
        p = subprocess.run(cmd, capture_output=True,
    text=True,
                           timeout=timeout,
    encoding="utf-8", errors="replace")
        if not os.path.isfile(pdf_path) or
    os.path.getsize(pdf_path) < 800:
            return None, "浏览器未能生成 PDF（%s）" % (p.stderr or
    "").strip()[:200]
        with open(pdf_path, "rb") as f:
            return f.read(), None
    except subprocess.TimeoutExpired:
        return None, "生成 PDF 超时，请稍后重试"
    except Exception as e:
        return None, "生成 PDF 失败：%s" % e
    finally:
        shutil.rmtree(tmpdir, ignore_errors=True)
# ==== app/utils/charts.py ====
# -*- coding: utf-8 -*-
"""轻量图表（M9）。
用纯 SVG 输出，不依赖任何绘图库或前端脚本：
- 服务端渲染，页面与打印（PDF）中都能稳定显示
- 每个图元都带数值标签，黑白打印时仍可读（不单靠颜色区分）
- 单色方案，符合界面「一个视图一个强调色」的约定
"""
from html import escape
ACCENT = "#1f3a5f"
ACCENT_SOFT = "#7f93ad"
INK = "#23262b"
MUTED = "#6b7280"
LINE = "#dfe3e8"
# 灰度梯度：黑白打印时相邻扇区 / 条柱仍可区分
GRAYS = ["#1f3a5f", "#456188", "#6d86a8", "#96a9c4",
    "#bcc9da", "#dbe3ec"]
def _fmt(v, digits=4):
    try:
        return ("%%.%df" % digits) % float(v)
    except (TypeError, ValueError):
```

```
        return str(v)                                                    7451
def hbars(items, width=620, bar_h=22, gap=12, label_w=150,               7452
    value_w=76):                                                         7453
    """水平条形图。items: [(标签, 数值, 显示文本)]"""                    7454
    if not items:                                                        7455
        return '<p class="muted">暂无数据</p>'                           7456
    n = len(items)                                                       7457
    height = n * (bar_h + gap) + 10                                      7458
    max_v = max((abs(float(v)) for _, v, _ in items),                    7459
    default=1) or 1                                                      7460
    body_w = width - label_w - value_w                                   7461
    parts = ['<svg viewBox="0 0 %d %d" width="100%%"                     7462
    height="%d" '                                                        7463
             'role="img" aria-label="条形图" '                           7464
             'style="max-width:%dpx;font-family:inherit;">'              7465
    % (width, height, height, width)]                                    7466
    for i, (label, value, disp) in enumerate(items):                     7467
        y = i * (bar_h + gap) + 5                                        7468
        frac = (abs(float(value)) / max_v) if max_v else 0               7469
        w = max(2.0, body_w * frac)                                      7470
        fill = GRAYS[i % len(GRAYS)]                                     7471
        parts.append(                                                    7472
            '<text x="%d" y="%d" text-anchor="end"                       7473
    font-size="12" fill="%s">%s</text>'                                  7474
            % (label_w - 10, y + bar_h * 0.72, INK,                      7475
    escape(str(label))))                                                 7476
        parts.append(                                                    7477
            '<rect x="%d" y="%d" width="%.1f" height="%d"                7478
    rx="3" fill="%s"/>'                                                  7479
            % (label_w, y, w, bar_h, fill))                              7480
        parts.append(                                                    7481
            '<text x="%d" y="%d" font-size="12" fill="%s" '              7482
            'style="font-variant-numeric:tabular-nums;">%s</            7483
    text>'                                                               7484
            % (label_w + w + 8, y + bar_h * 0.72, MUTED,                 7485
    escape(str(disp))))                                                  7486
    parts.append('</svg>')                                               7487
    return "".join(parts)                                                7488
def donut(items, size=210, thickness=34):                                7489
    """环形图（权重分布）。items: [(标签, 数值, 显示文本)]               7490
    图例中给出数值与占比，黑白打印时靠图例与灰度深浅区分。               7491
    """                                                                  7492
    vals = [(l, float(v), d) for l, v, d in items if                     7493
    float(v) > 0]                                                        7494
    if not vals:                                                         7495
        return '<p class="muted">暂无数据</p>'                           7496
    total = sum(v for _, v, _ in vals) or 1                              7497
    r = (size - thickness) / 2                                           7498
    cx = cy = size / 2                                                   7499
    import math                                                          7500
```

```
    parts = ['<svg viewBox="0 0 %d %d" width="%d"                    7501
height="%d" role="img" '                                             7502
             'aria-label="权重分布环形图">' % (size, size, size,     7503
size)]                                                               7504
    angle = -math.pi / 2                                             7505
    for i, (label, v, disp) in enumerate(vals):                      7506
        sweep = 2 * math.pi * (v / total)                            7507
        a0, a1 = angle, angle + sweep                                7508
        x0, y0 = cx + r * math.cos(a0), cy + r *                     7509
math.sin(a0)                                                         7510
        x1, y1 = cx + r * math.cos(a1), cy + r *                     7511
math.sin(a1)                                                         7512
        large = 1 if sweep > math.pi else 0                          7513
        parts.append(                                                7514
            '<path d="M %.2f %.2f A %.2f %.2f 0 %d 1 %.2f            7515
%.2f" '                                                              7516
            'fill="none" stroke="%s" stroke-width="%d"/>'            7517
            % (x0, y0, r, r, large, x1, y1, GRAYS[i %                7518
len(GRAYS)], thickness))                                             7519
        angle = a1                                                   7520
    parts.append('<text x="%d" y="%d" text-anchor="middle"           7521
font-size="12" '                                                     7522
                 'fill="%s">合计</text>' % (cx, cy - 2,              7523
MUTED))                                                              7524
    parts.append('<text x="%d" y="%d" text-anchor="middle"           7525
font-size="15" '                                                     7526
                 'fill="%s" style="font-variant-numeric:tabu         7527
lar-nums;">%s</text>'                                                7528
                 % (cx, cy + 18, INK, _fmt(total, 4)))               7529
    parts.append('</svg>')                                           7530
    legend = ['<table style="width:auto;margin-top:10px;"><t         7531
head><tr>'                                                           7532
              '<th scope="col">项目</th><th                          7533
scope="col">数值</th>'                                               7534
              '<th scope="col">占比</th></tr></thead><tbody>']       7535
    for i, (label, v, disp) in enumerate(vals):                      7536
        legend.append(                                               7537
            '<tr><td style="text-align:left;">'                      7538
            '<span style="display:inline-block;width:10px;he         7539
ight:10px;'                                                          7540
            'border-radius:2px;background:%s;margin-right:6p         7541
x;"></span>%s</td>'                                                  7542
            '<td class="mono">%s</td><td                             7543
class="mono">%.1f%%</td></tr>'                                       7544
            % (GRAYS[i % len(GRAYS)], escape(str(label)),            7545
escape(str(disp)),                                                   7546
               100.0 * v / total))                                   7547
    legend.append('</tbody></table>')                                7548
    return ('<div style="display:flex;flex-wrap:wrap;gap:20p         7549
x;align-items:center;">'                                             7550
```

```
                + "".join(parts) + "".join(legend) + '</div>')
# ==== app/utils/schema.py ====
# -*- coding: utf-8 -*-
"""轻量结构迁移。
SQLite 下 `db.create_all()` 只建缺失的表，不会给已有表补列。
本模块在启动时检查目标列是否存在，缺失则 ALTER TABLE 补上，
使旧数据库无需重建即可用上新字段。
"""
from sqlalchemy import text
# （表名， 列名， 列定义）
REQUIRED_COLUMNS = [
    # M2 遥感影像挂接
    ("mine_patches", "image_path", "VARCHAR(256)"),
    ("mine_patches", "image_note", "VARCHAR(256)"),
    ("mine_patches", "image_uploaded_at", "DATETIME"),
    # M6 指标维护
    ("indicators", "enabled_units", "VARCHAR(128)"),
    ("indicators", "description", "TEXT"),
    # M6 多专家矩阵
    ("expert_matrices", "unit_type", "VARCHAR(32)"),
    ("expert_matrices", "l1_name", "VARCHAR(64)"),
    ("expert_matrices", "cr", "FLOAT"),
]
def ensure_schema(db):
    """为已有表补齐缺失列。返回补列清单。"""
    added = []
    try:
        with db.engine.connect() as conn:
            for table, column, ddl in REQUIRED_COLUMNS:
                rows = conn.execute(text("PRAGMA
table_info(%s)" % table)).fetchall()
                if not rows:
                    continue                                #
表还不存在，create_all 会建
                existing = {r[1] for r in rows}
                if column in existing:
                    continue
                conn.execute(text(
                    "ALTER TABLE %s ADD COLUMN %s %s" %
(table, column, ddl)))
                conn.commit()
                added.append("%s.%s" % (table, column))
    except Exception as e:                                  #
迁移失败不应阻断启动
        print("[schema] 结构检查跳过：%s" % e)
    return added
# ==== app/utils/uploads.py ====
# -*- coding: utf-8 -*-
"""上传文件处理（M2 遥感影像、M7 批量导入）。"""
import os
```

```
import re
import uuid
from datetime import datetime
from werkzeug.utils import secure_filename
def _safe_ext(filename):
    return os.path.splitext(filename)[1].lower().lstrip(".")
def ext_allowed(filename, allowed):
    return _safe_ext(filename) in allowed
def save_upload(file_storage, subdir, allowed_ext,
    upload_root):
    """保存上传文件，返回相对 static 的路径（如 uploads/patches/xxx.png）。
    文件名重命名为「时间戳_随机串.扩展名」，避免中文名与重名问题。
    """
    if file_storage is None or not file_storage.filename:
        return None, "未选择文件"
    ext = _safe_ext(file_storage.filename)
    if ext not in allowed_ext:
        return None, "不支持的文件类型：.%s" % ext
    target_dir = os.path.join(upload_root, subdir)
    os.makedirs(target_dir, exist_ok=True)
    # 仅保留安全字符，并加时间戳与随机串防重名
    base = secure_filename(os.path.splitext(file_storage.fil
    ename)[0]) or "file"
    base = re.sub(r"[^A-Za-z0-9_\-]", "", base)[:40] or
    "file"
    name = "%s_%s.%s" % (datetime.now().strftime("%Y%m%d%H%M
    %S"),
                         uuid.uuid4().hex[:6], ext)
    abs_path = os.path.join(target_dir, name)
    file_storage.save(abs_path)
    rel = os.path.join("uploads", subdir,
    name).replace("\\", "/")
    return rel, None
def remove_upload(rel_path, upload_root):
    """删除已保存的上传文件（相对 static 的路径）。"""
    if not rel_path:
        return
    abs_path = os.path.join(upload_root,
    rel_path.replace("uploads/", "", 1))
    try:
        if os.path.isfile(abs_path):
            os.remove(abs_path)
    except OSError:
        pass
# ==== app/views/__init__.py ====
# -*- coding: utf-8 -*-
```